



ORIGENE

genetic screening report

SAMPLE REPORT

Sample

Customer's Gender	Female
Customer's ID	Sample
Customer's Address	8F Jalan Teknologi 3/6 Taman Sains Selangor 1, Kota Damansara (PJU5) 47810 Petaling Jaya Selangor, MY

Sample Reference	018354
Sample Type	Saliva
Sampling Date	10/05/2023
Sample Receipt Date	10/05/2023
Report Date	24/05/2023
Doctor's Name	Sample
Doctor's Address	Malaysian Genomics Resource Centre Berhad 8F Jalan Teknologi 3/6 Taman Sains Selangor 1, Kota Damansara (PJU5) 47810 Petaling Jaya Selangor, MY

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Dear,

Sample

Thank you for choosing us to screen your DNA for genetic predispositions towards various diseases and medical conditions. You have taken an important step towards personalized healthcare.

Your genetic screening report contains several sections, which are explained below. Before you start reading them, it is important that you understand the information that each section provides.

Limitations of Genetic Screening

An explanatory note which describes the nature of genetic screening and how the information contained in this report should, and should not be used. This section also provides some background as to how the information was generated.

List of Useful Terms

A description of terms used throughout this report.

Your Genetic Risk Summary

An overall summary of your genetic screening results will be shown using bar indicators. For health conditions, it will indicate your genetic risk level compared to the general population. For traits, it will indicate your propensity to display the trait.

Your Genetic Screening Results

This section will display your genetic screening results, which include the bar indicators mentioned above. The number and type of genetic marker(s) found in your DNA will also be displayed. Other useful information on the health conditions or traits such as the nature of the condition, risk factors, typical symptoms, and recommended interventions will also be available, if applicable.

Please remember that health conditions can be caused by a combination of factors, which include infection, environmental influence, lifestyle, and genetics. While you cannot change your genes, the information available from genetic screening can help you make changes, where necessary, to the first three factors. We hope that the report you are about to read will be beneficial to you and your doctor for the long-term management of your health.

Yours sincerely,



Leong Yien Hung

Managing Director

Malaysian Genomics Resource Centre Berhad

Limitations of Genetic Screening

Please read and understand this section, which explains the limitations of genetic screening, before proceeding to view your genetic screening report. Do consult your physician, genetic counselor or trained healthcare professional for advice on the medical interpretations and on your health management decisions, actions and choices after reading this genetic screening report.

Genetic screening can indicate whether an individual has a predisposition, or is at an increased likelihood, of having an inherited disease or disorder. However, it cannot indicate if the individual will show symptoms, how severe the symptoms will be, or whether the disease or disorder will progress over time. A negative test result does not mean that an individual will not get an inherited disease or disorder because our genetic screening tests are designed to detect only highly significant genetic markers which have been documented during medical research. Environmental and lifestyle factors also play a role in the development of inherited diseases, disorders, and even traits.

The results generated from our genetic screening test, which includes all additional health related information, is meant for informational purposes only and not an indicator of an individual's current or future health status, nor a diagnosis for any disease, disorder, condition or other health issues. The information provided in the test results should be confirmed by your doctor or a healthcare professional in a clinical setting before any further action/treatment is to be taken.

We do not guarantee or warrant the recommendation, diagnosis, therapy, treatment, medication, device, action or strategy of any individuals with relation to our genetic screening test results or any part of our website content. The recommendation given in the test results is for informational purposes only and is not intended as a substitute for, nor does it replace, any professional medical advice, diagnosis, or treatment. It is advisable not to avoid, disregard or delay seeking and obtaining medical advice from qualified healthcare providers due to information which may have been obtained from any genetic screening test results.

List of Useful Terms

Cell	All organisms are made of cells. A cell is the smallest structural and functional unit of an organism. Most DNA is located in the cell nucleus.
Chromosome	A chromosome is a structure found inside the nucleus of a cell. It is made up of proteins and DNA organized into genes. Each cell normally contains 23 pairs of chromosomes.
DNA	DNA, or deoxyribonucleic acid, is a long two-stranded molecule that contains a unique genetic code. It holds the instructions for making all the proteins in an organism. It can be found in every cell and is passed down from parents to their offspring.
DNA Sequence	DNA contains four basic chemical building blocks or bases, which are adenine (A), cytosine (C), guanine (G) and thymine (T). The sequence, or order, of these bases forms the instructions in the genome.
Genes	Genes are sections of DNA that contain codes for the production of specific proteins. Genes comprise sequences of DNA and are arranged at specific locations on chromosomes inside cells.
Genetic Markers	Genetic markers are DNA sequences with known physical locations on the chromosome. Each chromosome has many genetic markers. Genetic risk markers can help associate inherited diseases with the responsible gene.
Genome	The genome is a complete set of DNA in an organism.
Genome-wide Association Study (GWAS)	A genome-wide association study involves rapidly scanning markers across the complete sets of DNA, or genomes, of many individuals to find genetic variations associated with a particular disease. When new genetic associations are identified, the information can be utilized by researchers to develop better ways to detect, treat and prevent the disease. These studies are useful in finding genetic variations that contribute to common, complex diseases.
Genotype	A genotype refers to the genetic makeup of an organism. It is the collection of genes belonging to an individual.
Protein	Proteins are complex and functional molecules that play a range of critical roles in the body, and protein synthesis is the process of creating protein molecules. Most genes contain the information needed to make proteins. Proteins carry out most of the work in cells and are needed for the function, structure, and regulation of tissues and organs in the body.

Your Genetic Screening Results

Derma

1. Acne Risk
2. Body Odor (Bromhidrosis)
3. Cellulite
4. Collagen Integrity
5. Dry Skin
6. Eczema (Atopic Dermatitis)
7. Folate-Folic Acid Deficiency
8. Freckles
9. Generalized Psoriasis
10. Glycation Protection
11. Keloid Risk
12. Melanin and Pigmentation
13. Omega-3 and Omega-6 Deficiency
14. Premature Wrinkles
15. Response to Antioxidants
16. Skin Barrier
17. Skin Elasticity
18. Skin Lightening Ability
19. Sunspots
20. Tanning Response
21. Varicose Veins

Wellness

22. Achilles Tendinopathy
23. Adiponectin Level
24. Ankylosing Spondylitis
25. Anterior Cruciate Ligament (ACL) Rupture
26. Caffeine Metabolism
27. Calcium Metabolism
28. Carbohydrate Metabolism
29. Carotene Level
30. CoEnzyme Q10
31. Collagen Formation
32. Concussion
33. Detoxification Capacity
34. Endurance Sports Performance
35. Energy Utilization
36. Glaucoma
37. Gluten Sensitivity
38. Hay Fever
39. HDL Cholesterol
40. High Blood Sugar (Hyperglycemia)
41. High Triglycerides
42. Homocysteine Metabolism
43. Inflammatory Response
44. Lactose Intolerance
45. LDL Cholesterol
46. Longevity
47. Maximal Oxygen Consumption (VO2 Max)
48. Osteoarthritis
49. Osteoporosis
50. Pain Tolerance
51. Polyunsaturated Fatty Acids
52. Power Sports Performance
53. Respiratory Infection - Severity
54. Respiratory Infection- Susceptibility
55. Rheumatoid Arthritis
56. Salt-Sensitivity High Blood Pressure
57. Selenium
58. Soft Tissue Injury
59. Sports Induced Sudden Cardiac Arrest
60. Steroid Hormones Metabolism
61. Stress Sensitivity
62. Total Cholesterol
63. Vision Degeneration
64. Vitamin A
65. Vitamin B2
66. Vitamin B6
67. Vitamin B12
68. Vitamin C
69. Vitamin D
70. Vitamin E

Sports & Fitness

71. Bone & Joint Health: Achilles Tendinopathy
72. Bone & Joint Health: Anterior Cruciate Ligament (ACL) Rupture
73. Bone & Joint Health: Calcium Metabolism
74. Bone & Joint Health: Osteoarthritis
75. Bone & Joint Health: Osteoporotic Fractures
76. Bone & Joint Health: Vitamin D
77. Energy Metabolism & Utilization: Dietary Choline Tendency
78. Energy Metabolism & Utilization: Phosphorus Metabolism
79. Energy Metabolism & Utilization: Vitamin B12
80. Energy Metabolism & Utilization: Vitamin B6
81. Energy Metabolism & Utilization: Vitamin B9 - Folate Tendency
82. Macronutrient Metabolism: Carbohydrate Utilization
83. Macronutrient Metabolism: Cholesterol Response to Dietary Fat
84. Macronutrient Metabolism: Fat Utilization
85. Macronutrient Metabolism: Insulin Response to Dietary Fat
86. Macronutrient Metabolism: Polyunsaturated Fat Metabolism
87. Macronutrient Metabolism: Protein Utilization
88. Macronutrient Metabolism: Vitamin E
89. Mental Ability and Performance: Impulse Control and Taste Preference
90. Mental Ability and Performance: Intrinsic Motivation to Exercise
91. Mental Ability and Performance: Sleep Duration
92. Mental Ability and Performance: Stress Sensitivity
93. Muscle Building Capacity: Testosterone
94. Muscle Building Capacity: Zinc Metabolism
95. Muscle Performance: Copper Metabolism
96. Muscle Performance: Iron Metabolism
97. Muscle Performance: Magnesium Metabolism
98. Oxidative Stress Metabolism: Antioxidants
99. Oxidative Stress Metabolism: CoEnzyme Q10
100. Oxidative Stress Metabolism: Vitamin A
101. Oxidative Stress Metabolism: Vitamin C
102. Response to Cardiovascular Training: Exercise Heart Rate Response
103. Response to Cardiovascular Training: Fat Loss Response to Cardio
104. Response to Cardiovascular Training: Fitness Response to Cardio
105. Response to Cardiovascular Training: Glucose Response to Cardio
106. Response to Cardiovascular Training: HDL Response to Cardio
107. Response to Cardiovascular Training: Insulin Sensitivity Response to Cardio
108. Response to Cardiovascular Training: Triglycerides Response to Cardio
109. Response To Resistance Training: Body composition
110. Response To Sports Drinks: Caffeine Metabolism
111. Response To Sports Drinks: Lactose Metabolism
112. Sports-Induced Injury Risk: Concussion
113. Sports-Induced Injury Risk: Soft Tissue Injury
114. Training Modes and Energy: Endurance Capacity
115. Training Modes and Energy: Energy Utilization
116. Training Modes and Energy: Exercise Intensity - Power Capacity
117. Weight Management: Weight and Satiety
118. Weight Management: Weight Loss Metabolism
119. Weight Management: Weight Loss Tendency
120. Weight Management: Weight Regain Tendency

Your Genetic Screening Results

Behavioral & Physical Traits

121.	Ability To Memorize	140.	Instant Gratification
122.	Addictive Personality	141.	Intelligence Quotient (IQ)
123.	Alcohol Dependence	142.	Lumbar Disc Disease (Lower Back Pain)
124.	Alcohol Flush	143.	Male Pattern Baldness
125.	Appetite Regulation	144.	Memory Skills
126.	Body/Volume (Hair)	145.	Monobrow
127.	Body Hair	146.	Motion Sickness
128.	Emotional Quotient (EQ)	147.	Nicotine Dependence
129.	Extroversion	148.	Optimism and Empathy
130.	Food Addiction	149.	Positive Affectivity
131.	General Cognitive Ability	150.	Reading and Spelling
132.	Hair Dryness/Brittleness	151.	Reward Dependence
133.	Hair Graying	152.	Sensitive Personality
134.	Hair Thickness	153.	Sleep Depth
135.	Hair Thinning	154.	Sleep Duration
136.	Hair Type	155.	Sleep Latency
137.	Harm Avoidance Behaviour	156.	Sleep Quality
138.	Information Processing Power	157.	Sweet Tooth
139.	Insomnia		

Cardio & Metabolic

158.	Abdominal Aortic Aneurysm	169.	Grave's Disease (Hyperthyroidism)
159.	Arrhythmogenic Right Ventricular Dysplasia	170.	Heart Attack
160.	Atrial Fibrillation	171.	Hypertension
161.	Chronic Obstructive Pulmonary Disease	172.	Hypothyroidism
162.	Coronary Heart Disease	173.	Metabolic Syndrome
163.	Deep Vein Thrombosis and Pulmonary Embolism	174.	Non-Alcoholic Fatty Liver Disease
164.	Dilated Cardiomyopathy	175.	Obesity
165.	Dyslipidemia	176.	Peripheral Arterial Disease
166.	Gallstone	177.	Type 1 Diabetes
167.	Gestational Diabetes	178.	Type 2 Diabetes
168.	Gout		

Onco

179.	Acute Lymphoblastic Leukemia	197.	Lung Cancer
180.	Basal Cell Carcinoma	198.	Melanoma
181.	Bladder Cancer	199.	Meningioma
182.	Breast Cancer (Female)	200.	Nasopharyngeal Cancer
183.	Breast Cancer (Male)	201.	Non-Hodgkin's Lymphoma
184.	Cervical Cancer (Female)	202.	Oral and Throat Cancers
185.	Cervical Cancer (Male)	203.	Osteosarcoma
186.	Chronic Lymphocytic Leukemia	204.	Ovarian Cancer (Female)
187.	Colorectal Cancer	205.	Ovarian Cancer (Male)
188.	Endometrial Cancer (Female)	206.	Pancreatic Cancer
189.	Endometrial Cancer (Male)	207.	Prostate Cancer (Female)
190.	Esophageal Cancer	208.	Prostate Cancer (Male)
191.	Gallbladder Cancer	209.	Squamous Cell Carcinoma of the Skin
192.	Glioma Development (Brain Tumor)	210.	Stomach Cancer
193.	Hepatocellular Carcinoma (Liver Cancer)	211.	Testicular Germ-Cell Tumor (Female)
194.	Hodgkin's Lymphoma	212.	Testicular Germ-Cell Tumor (Male)
195.	Kidney Cancer	213.	Thyroid Cancer
196.	Laryngeal Cancer		

Neuro

214.	Alzheimer's Disease	218.	Migraine Headache
215.	Brain Aneurysm	219.	Parkinson's Disease
216.	Depression	220.	Stroke
217.	Epilepsy		

Your Genetic Screening Results

Allergy Profile

221.

Allergy to Cats
222.

Allergy to Dust Mites
223.

Allergy to Grass
224.

Allergy to Metal (Nickel)
225.

Allergy to Milk
226.

Allergy to Peanuts
227.

Allergy to Shrimp
228.

Allergy to Wheat

Childhood-Onset Diseases

229.

Adolescent Idiopathic Scoliosis
230.

Allergic Rhinitis
231.

Asthma
232.

Attention-Deficit Hyperactivity Disorder (ADHD)
233.

Celiac Disease
234.

Childhood Epilepsy
235.

Crohn's Disease
236.

Dyslexia
237.

Early Speech Development
238.

Invasive Pneumococcal Disease
239.

Juvenile Idiopathic Arthritis
240.

Kawasaki Disease
241.

Long QT Syndrome
242.

Meningococcal Disease
243.

Obesity (Childhood Onset)
244.

Obsessive-Compulsive Disorder (OCD)
245.

Pathological Myopia
246.

Restless Legs Syndrome
247.

Teething
248.

Tourette Syndrome
249.

Ulcerative Colitis

Derma

Acne Risk



Body Odor (Bromhidrosis)



Cellulite



Collagen Integrity



Dry Skin



Eczema (Atopic Dermatitis)



Folate-Folic Acid Deficiency



Freckles



Generalized Psoriasis



Glycation Protection



Keloid Risk



Melanin & Pigmentation



Omega-3 & Omega-6 Deficiency



Premature Wrinkles



Response To Antioxidants



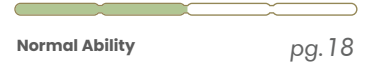
Skin Barrier



Skin Elasticity



Skin Lightening Ability



Sunspots



Tanning Response



Varicose Veins



Wellness

Achilles Tendinopathy



Adiponectin Level



Ankylosing Spondylitis



Anterior Cruciate Ligament (ACL) Rupture



Caffeine Metabolism



Calcium Metabolism



Carbohydrate Metabolism



Carotene Level



CoEnzyme Q10



Collagen Formation



Concussion



Detoxification Capacity



Endurance Sports Performance



Energy Utilization



Glaucoma



Gluten Sensitivity



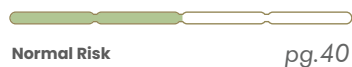
Hay Fever



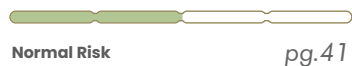
HDL Cholesterol



High Blood Sugar (Hyperglycemia)



High Triglycerides



Homocysteine Metabolism



Inflammatory Response



Lactose Intolerance



pg.44

Rheumatoid Arthritis



pg.55

Vitamin B6



pg.66

LDL Cholesterol



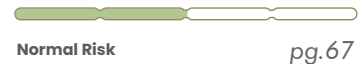
pg.45

Salt-Sensitivity High Blood Pressure



pg.56

Vitamin B12



pg.67

Longevity



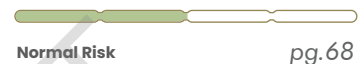
pg.46

Selenium



pg.57

Vitamin C



pg.68

Maximal Oxygen Consumption (VO2 Max)



pg.47

Soft Tissue Injury



pg.58

Vitamin D



pg.69

Osteoarthritis



pg.48

Sports Induced Sudden Cardiac Arrest



pg.59

Vitamin E



pg.70

Osteoporosis



pg.49

Steroid Hormones Metabolism



pg.60

Pain Tolerance



pg.50

Stress Sensitivity



pg.61

Polyunsaturated Fatty Acids



pg.51

Total Cholesterol



pg.62

Power Sports Performance



pg.52

Vision Degeneration



pg.63

Respiratory Infection - Severity



pg.53

Vitamin A



pg.64

Respiratory Infection - Susceptibility



pg.54

Vitamin B2



pg.65

Sports & Fitness

Bone & Joint Health: Achilles Tendinopathy

 Slightly Lower Risk pg.71

Bone & Joint Health: Anterior Cruciate Ligament (ACL) Rupture

 Slightly Lower Risk pg.72

Bone & Joint Health: Calcium Metabolism

 Normal Metabolism pg.73

Bone & Joint Health: Osteoarthritis

 Low Risk pg.74

Bone & Joint Health: Osteoporotic Fractures

 Low Risk pg.75

Bone & Joint Health: Vitamin D

 Optimal Utilization pg.76

Energy Metabolism & Utilization: Dietary Choline Tendency

 Normal Tendency pg.77

Energy Metabolism & Utilization: Phosphorous Metabolism

 Optimal Metabolism pg.78

Energy Metabolism & Utilization: Vitamin B12

 Normal Utilization pg.79

Energy Metabolism & Utilization: Vitamin B6

 Optimal Utilization pg.80

Energy Metabolism & Utilization: Vitamin B9 - Folate Tendency

 Optimal Tendency pg.81

Macronutrient Metabolism: Carbohydrate Utilization

 Low Utilization pg.82

Macronutrient Metabolism: Cholesterol Response To Dietary Fat

 Favorable Response pg.83

Macronutrient Metabolism: Fat Utilization

 Normal Utilization pg.84

Macronutrient Metabolism: Insulin Response To Dietary Fat

 Favorable Response pg.85

Macronutrient Metabolism: Polyunsaturated Fat Metabolism

 Normal Metabolism pg.86

Macronutrient Metabolism: Protein Utilization

 Optimal Utilization pg.87

Macronutrient Metabolism: Vitamin E

 Normal Metabolism pg.88

Mental Ability & Performance: Impulse Control & Taste Preference

 High Tendency pg.89

Mental Ability & Performance: Intrinsic Motivation To Exercise

 High Motivation pg.90

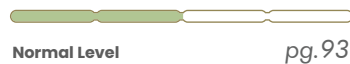
Mental Ability & Performance:
Sleep Duration



Mental Ability & Performance:
Stress Sensitivity



Muscle Building Capacity:
Testosterone



Muscle Building Capacity:
Zinc Metabolism



Muscle Performance:
Copper Metabolism



Muscle Performance:
Iron Metabolism



Muscle Performance:
Magnesium Metabolism



Oxidative Stress Metabolism:
Antioxidants



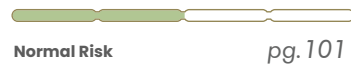
Oxidative Stress Metabolism:
Coenzyme Q10



Oxidative Stress Metabolism:
Vitamin A



Oxidative Stress Metabolism:
Vitamin C



Response To Cardiovascular Training:
Exercise Heart Rate Response



Response To Cardiovascular Training:
Fat Loss Response To Cardio



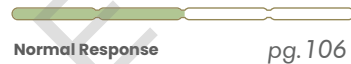
Response To Cardiovascular Training:
Fitness Response To Cardio



Response To Cardiovascular Training:
Glucose Response To Cardio



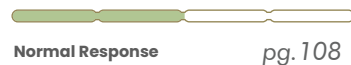
Response To Cardiovascular Training:
HDL Response To Cardio



Response To Cardiovascular Training:
Insulin Sensitivity Response To Cardio



Response To Cardiovascular Training:
Triglycerides Response To Cardio



Response To Resistance Training:
Body Composition



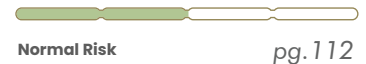
Response To Sports Drinks:
Caffeine Metabolism



Response To Sports Drinks:
Lactose Metabolism



Spots-Induced Injury Risk:
Concussion



Spots-Induced Injury Risk:
Soft Tissue Injury



Training Modes & Energy:
Endurance Capacity



Training Modes & Energy:
Energy Utilization



Training Modes & Energy:
Exercise Intensity - Power Capacity



Weight Management:
Weight & Satiety



Weight Management:
Weight Loss Metabolism



Weight Management:
Weight Loss Tendency



Weight Management:
Weight Regain Tendency



Behavioral & Physical Traits

Ability To Memorize



Addictive Personality



Alcohol Dependence



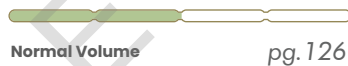
Alcohol Flush



Appetite Regulation



Body/Volume (Hair)



Body Hair



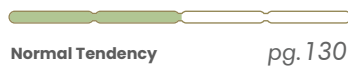
Emotional Quotient (EQ)



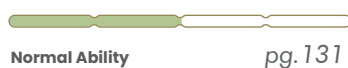
Extroversion



Food Addiction



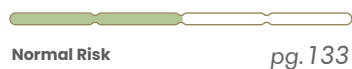
General Cognitive Ability



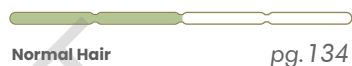
Hair Dryness/Brittleness



Hair Graying



Hair Thickness



Hair Thinning



Hair Type



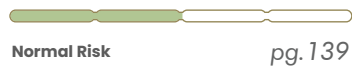
Harm Avoidance Behaviour



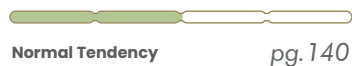
Information Processing Power



Insomnia



Instant Gratification



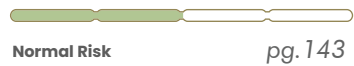
Intelligence Quotient (IQ)



Lumbar Disc Disease (Lower Back Pain)



Male Pattern Baldness



pg. 143

Sleep Duration



pg. 154

Memory Skills



pg. 144

Sleep Latency



pg. 155

Monobrow



pg. 145

Sleep Quality



pg. 156

Motion Sickness



pg. 146

Sweet Tooth



pg. 157

Nicotine Dependence



pg. 147

Optimism and Empathy



pg. 148

Positive Affectivity



pg. 149

Reading and Spelling



pg. 150

Reward Dependence



pg. 151

Sensitive Personality



pg. 152

Sleep Depth



pg. 153

Cardio & Metabolic

Abdominal Aortic Aneurysm



Arrhythmogenic Right Ventricular Dysplasia



Atrial Fibrillation



Chronic Obstructive Pulmonary Disease



Coronary Heart Disease



Deep Vein Thrombosis and Pulmonary Embolism



Dilated Cardiomyopathy



Dyslipidemia



Gallstone



Gestational Diabetes



Gout



Graves' Disease (Hyperthyroidism)



Heart Attack



Hypertension



Hypothyroidism



Metabolic Syndrome



Non-Alcoholic Fatty Liver Disease



Obesity



Peripheral Arterial Disease



Type 1 Diabetes



Type 2 Diabetes



Acute Lymphoblastic Leukemia

 Slightly Lower Risk *pg. 179*

Basal Cell Carcinoma

 Low Risk *pg. 180*

Bladder Cancer

 Low Risk *pg. 181*

Breast Cancer (Female)

 Moderate Risk *pg. 182*

Breast Cancer (Male)

 Moderate Risk *pg. 183*

Cervical Cancer (Female)

 Low Risk *pg. 184*

Cervical Cancer (Male)

 Low Risk *pg. 185*

Chronic Lymphocytic Leukemia

 Slightly Lower Risk *pg. 186*

Colorectal Cancer

 High Risk *pg. 187*

Endometrial Cancer (Female)

 Moderate Risk *pg. 188*

Endometrial Cancer (Male)

 Moderate Risk *pg. 189*

Esophageal Cancer

 Moderate Risk *pg. 190*

Gallbladder Cancer

 Low Risk *pg. 191*

Glioma Development (Brain Tumor)

 Slightly Lower Risk *pg. 192*

Hepatocellular Carcinoma (Liver Cancer)

 Slightly Lower Risk *pg. 193*

Hodgkin's Lymphoma

 Slightly Lower Risk *pg. 194*

Kidney Cancer

 Low Risk *pg. 195*

Laryngeal Cancer

 Moderate Risk *pg. 196*

Lung Cancer

 Slightly Lower Risk *pg. 197*

Melanoma

 Low Risk *pg. 198*

Meningioma

 Moderate Risk *pg. 199*

Nasopharyngeal Cancer

 Moderate Risk *pg. 200*

Non-Hodgkin's Lymphoma



Testicular Germ-Cell Tumor (Male)



Oral and Throat Cancers



Thyroid Cancer



Osteosarcoma



Ovarian Cancer (Female)



Ovarian Cancer (Male)



Pancreatic Cancer



Prostate Cancer (Female)



Prostate Cancer (Male)



Squamous Cell Carcinoma of the Skin



Stomach Cancer



Testicular Germ-Cell Tumor (Female)



Neuro

Alzheimer's Disease



Brain Aneurysm



Depression



Epilepsy



Migraine Headache



Parkinson's Disease



Stroke



Allergy

Allergy to Cats



Allergy to Dust Mites



Allergy to Grass



Allergy to Metal (Nickel)



Allergy to Milk



Allergy to Peanuts



Allergy to Shrimp



Allergy to Wheat



Childhood-Onset Diseases

Adolescent Idiopathic Scoliosis



Allergic Rhinitis



Asthma



Attention-Deficit Hyperactivity Disorder (ADHD)



Celiac Disease



Childhood Epilepsy



Crohn's Disease



Dyslexia



Early Speech Development



Invasive Pneumococcal Disease



Juvenile Idiopathic Arthritis



Kawasaki Disease



Long QT Syndrome



Meningococcal Disease



Obesity (Childhood Onset)



Obsessive-Compulsive Disorder (OCD)



Pathological Myopia



Restless Legs Syndrome



Teething



Tourette Syndrome



Ulcerative Colitis



Your Genetic Risk Summary

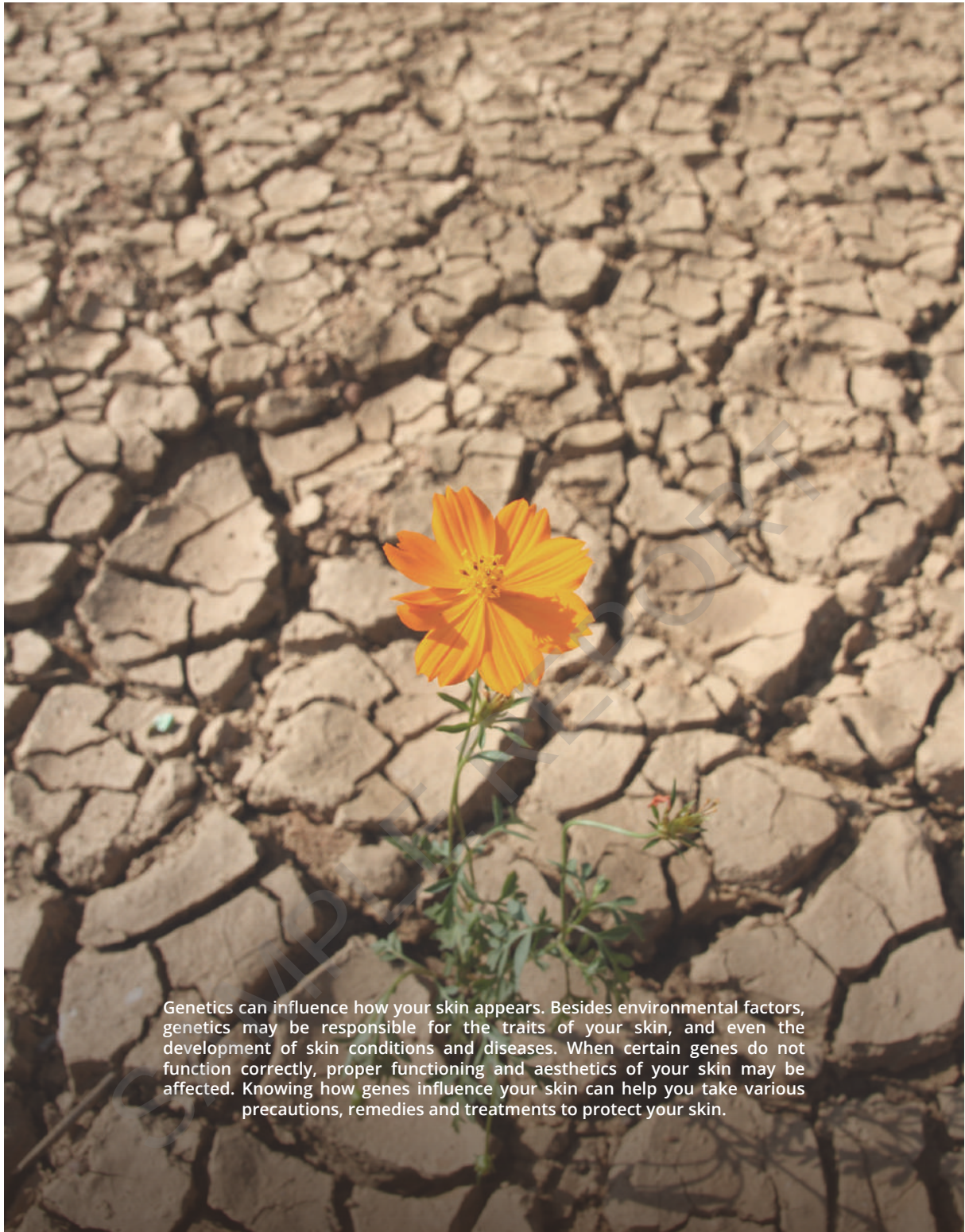
Category	Diseases/Trait	Findings	Recommendations
Derma	Response To Antioxidants	Reduced Response	A diet high in antioxidants boosts the body's defense system. Include leafy greens, cruciferous veggies, tomatoes, fatty fish, and green tea.
Wellness	LDL Cholesterol	Increased Level	Limit saturated fats and eat foods rich in omega-3 fatty acids.
Wellness	Carbohydrate Metabolism	Low Metabolism	Focus on nutrient-dense, lower-carbohydrate foods like lean meats, fish, eggs, leafy greens, whole grains, and dairy.
Wellness	Salt-Sensitivity High Blood Pressure	Increased Risk	Maintain a diet low in sodium and high in potassium, calcium, and magnesium. Stay active and maintain a healthy weight for better heart health.
Wellness	Lactose Intolerance	Higher Tendency	Lactose intolerance can be managed by switching to alternatives like soy or almond milk. Lactase supplements or probiotics may also help reduce symptoms.
Wellness	Carotene Level	Moderate Level	Natural sources of β -carotene include carrots, sweet potatoes, squash, spinach, and leafy lettuce.
Sports and Fitness	Oxidative Stress Metabolism: Antioxidants	Reduced Response	Including a variety of colorful vegetables in your diet can also help boost your overall antioxidant levels.
Sports and Fitness	Response To Sports Drinks : Lactose Metabolism	Low Metabolism	Individuals who are lactose intolerant can opt for whey protein isolate or non-whey proteins, such as pea, rice, or soy protein, as alternatives to traditional sports drinks.
Sports and Fitness	Macronutrient Metabolism: Carbohydrate Utilization	Low Utilization	To replenish your total energy reserve, eat quick carbs (such as rice) after your workout. To achieve extended satiety, eat slow carbs (such as sweet potatoes) during other meals.
Behavioral and Physical Traits	Alcohol Flush	High Response	Drinking water before and during alcohol consumption to help lessen symptoms of alcohol flush by diluting the alcohol in the body.

Your Genetic Risk Summary

Category	Diseases/Trait	Findings	Recommendations
Behavioral and Physical Traits	Alcohol Dependence	Moderate Tendency	Reducing alcohol dependence can be achieved by taking regular breaks from the drinking cycle.
Onco	Colorectal Cancer	High Risk	The risk can be reduced by maintaining a high-fiber diet, exercising regularly, and avoiding excessive alcohol.
Onco	Meningioma	Moderate Risk	Avoiding high-voltage areas and minimizing radiation exposure may reduce the likelihood of tumor growth and related long-term complications.
Onco	Breast Cancer (Female)	Moderate Risk	Regular self-exams and consulting a professional if any changes are noticed in the breast are essential for early detection.
Onco	Endometrial Cancer (Female)	Moderate Risk	Regular exercise, a balanced diet, and staying healthy help minimize this risk. Paying attention to irregularities in menstruation can also aid in early detection.
Onco	Nasopharyngeal Cancer	Moderate Risk	Prevention involves avoiding cigarette smoking, reducing diets high in salt, cutting back on alcohol consumption, and avoiding workplace pollutants like wood dust and formaldehyde.
Onco	Laryngeal Cancer	Moderate Risk	The risk can be reduced by avoiding smoking and excessive alcohol consumption, maintaining a vegetable-rich diet, and treating infections promptly.
Onco	Breast Cancer (Male)	Moderate Risk	Men with a genetic predisposition to breast cancer should encourage their female descendants to screen for BRCA1 and BRCA2 gene mutations to assess breast cancer risk.
Onco	Ovarian Cancer (Male)	Moderate Risk	Female descendants of males with genetic predisposition for ovarian cancer should go for more regular screenings for early detection.
Onco	Oral And Throat Cancers	Moderate Risk	Maintaining a healthy diet with proper nutrition supports overall oral health and lowers cancer risk.

Your Genetic Risk Summary

Category	Diseases/Trait	Findings	Recommendations
Onco	Esophageal Cancer	Moderate Risk	The risk can be reduced by quitting smoking and limiting alcohol consumption. Addressing gastric reflux (GERD) early on also helps prevent the development of esophageal cancer.
Onco	Endometrial Cancer (Male)	Moderate Risk	As endometrial cancer affects only women, if there is a strong family history, it is advisable for your daughters to have regular gynecological check-ups when they come of age.
Onco	Ovarian Cancer (Female)	Moderate Risk	The risk can be reduced by maintaining a healthy body weight, following a balanced diet, and limiting smoking and alcohol consumption.
Childhood-Onset Diseases	Teething	Delayed Development	Parents can soothe their baby's sore gums by using a teething ring or rubbing the gums with wet gauze. Avoid over-the-counter teething medications for safety.
Childhood-Onset Diseases	Juvenile Idiopathic Arthritis	Moderate Risk	This arthritis can be managed by recognizing symptoms like joint swelling and stiffness early on. A balanced diet and early treatment of infections can help reduce the severity of symptoms.



Genetics can influence how your skin appears. Besides environmental factors, genetics may be responsible for the traits of your skin, and even the development of skin conditions and diseases. When certain genes do not function correctly, proper functioning and aesthetics of your skin may be affected. Knowing how genes influence your skin can help you take various precautions, remedies and treatments to protect your skin.

Derma

ACNE RISK

0

RESULT
Risk Markers
Detected

Reduced Risk

Description

Acne is a common skin condition that happens when the hair follicles become clogged and oil gets stuck when the dead skin cells plug these pores. This leads to an acne breakout, which is commonly known as pimples or zits. Research shows that genes affect the ability of follicles to become acne resistant in adulthood.

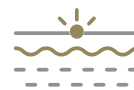
Risk Factors

- Hormonal changes
- Being a teenager
- Exposure to bacteria and germs

TYPICAL SYMPTOMS



Whiteheads
(closed plugged pores)



Blackheads
(open plugged pores)



Pimples



Painful, pus-filled lumps

RECOMMENDED INTERVENTIONS

To reduce acne risk, maintain a consistent skincare routine that includes gentle cleansing, moisturizing, and limit makeup. Avoid touching your face frequently and keep your hands clean. Use non-comedogenic products and consider over-the-counter treatments like salicylic acid or benzoyl peroxide. For persistent or severe acne, consult a dermatologist for personalized treatments such as topical retinoids, oral medications, or professional procedures.

BODY ODOR (BROMHIDROSIS)

1

RESULT Risk Markers Detected

Lower Tendency

Gene: ABCC11

Genotype: TC

Description

Body odor is the unpleasant smell produced by bacteria on the skin, which are breaking down the acids in sweat. Anyone who has reached puberty can have it. Body odor has a strong genetic basis, but can also be strongly influenced by various diseases and physiological conditions.

Risk Factors

- Hyperhidrosis
- Diabetes
- Thyroid conditions
- Kidney failure
- Liver dysfunction
- Metabolic disorders
- Hormones

TYPICAL SYMPTOMS



Excessive sweating



Skin infections on body areas prone to sweating



Smell from the body



Sudden change in body odor

RECOMMENDED INTERVENTIONS

Good hygiene with regular washing with the usage of soap, deodorant, antiperspirant to help in reducing body odor for the time being. Hair removal is also suggested to minimise odor.

CELLULITE

1

RESULT

Risk Markers
Detected

Normal Risk

Gene: HIF1A

Genotype: CC

Description

Cellulite occurs when fat deposits push through the connective tissue beneath the skin, creating a dimpled or lumpy appearance. It is more common in women than men and is influenced by genetics, aging, poor lymphatic drainage, collagen loss, weight gain, and circulation issues.

Risk Factors

- More common in women than men
- Aged over 21 years
- Obesity
- Skin conditions

TYPICAL SYMPTOMS



**Dimpled or
bumpy skin**



**Mild lumps and
depression in skin**



**Orange peel
skin texture**



**Sagged
skin appearance**

RECOMMENDED INTERVENTIONS

Cellulite can be reduced or prevented with an active lifestyle. Regular exercise and a balanced diet prevent sudden weight gain and maintain body composition. Staying hydrated also helps by keeping good circulation and lymphatic flow.

COLLAGEN INTEGRITY

1

RESULT

Risk Markers
Detected

Normal Integrity

Gene: COL1A1

Genotype: CC

Description

Collagen is a key structural protein that maintains skin strength, elasticity, and hydration. As the body ages, collagen production declines, leading to dryness, wrinkles, and reduced skin firmness. Genetic factors play a crucial role in determining collagen synthesis and degradation rates.

Risk Factors

- Older age
- Post-menopausal
- Stress
- Poor sleep

TYPICAL SYMPTOMS



Joints get stiffer



Ligaments
lose elasticity



Nails
become brittle



Skin becomes
thinner and sags

RECOMMENDED INTERVENTIONS

Maintaining collagen integrity is essential for connective tissue health. Consume foods rich in zinc, copper, vitamin C, proline, and glycine, which are readily absorbed from daily meals. Supplementation may be recommended if additional support is needed.

DRY SKIN

Description

Dry skin can result when we are exposed to dry, arid environments. It can also occur when the skin's natural hydration mechanisms and protections are damaged or defective. Dry skin can appear as flaky, rough and, at times, scaly. There are many reasons for developing dry skin, some of which can be inherited. Certain diseases and conditions like hypothyroidism can also cause dry skin.

Risk Factors

- Older age
- Diabetic
- Skin conditions

TYPICAL SYMPTOMS



Itchiness



Slight to severe scaling or peeling



Fine lines or cracks



Deep cracks that may bleed

1

RESULT

Risk Markers Detected



Normal Risk

Gene: FLG

Genotype: II

RECOMMENDED INTERVENTIONS

Skin dryness can be prevented by using humidifiers in dry weather. It is recommended that you reduce the time spent bathing, particularly in hot water. Using hypoallergenic soaps and detergents will also help alleviate dryness. Most importantly, using emollients and moisturizers will help the skin stay hydrated.

ECZEMA (ATOPIC DERMATITIS)

Description

Eczema, also known as atopic dermatitis, is a chronic inflammatory skin condition characterized by red, itchy, and scaly patches. It is common in infants and young children but can persist into adulthood. Genetic predisposition plays a significant role in its development.

Risk Factors

- From childhood
- Family history of atopy

TYPICAL SYMPTOMS



Dry and itchy skin



Red rashes and bumps on skin



Scaly, leathery patches of skin



Crusting skin

2

RESULT Risk Markers Detected

Reduced Risk

Gene: FLG

Genotype: II

Gene: FLG

Genotype: GG

RECOMMENDED INTERVENTIONS

Eczema can flare up due to sweating or stress. Avoid extreme temperatures and manage stress to reduce severity. Quality sleep is crucial for preventing flare-ups. Medications such as antihistamines, topical creams, and ointments may help manage symptoms.

FOLATE-FOLIC ACID DEFICIENCY

1

RESULT

Risk Markers
Detected

Reduced Risk

Gene: MTHFR

Genotype: TT

Description

Folic acid, also known as folate (Vitamin B9), is essential for DNA synthesis and red blood cell production. It is naturally found in foods such as leafy greens, meat, and legumes. A deficiency can lead to anemia, fatigue, and developmental issues.

Risk Factors

- Older population
- More common in women than men
- Pregnancy
- Vegetarian

TYPICAL SYMPTOMS



Extreme tiredness



Pins and
needles sensation



Problems with
memory, understanding
and judgement



Sore and
red tongue

RECOMMENDED INTERVENTIONS

Folate is typically obtained through a healthy diet, including meats, beans, and lentils. However, if deficient, folic acid supplements may be recommended.

FRECKLES

1

RESULT

Risk Markers Detected

Reduced Risk

Gene: TYR

Genotype: CC

Description

Freckles are small, pigmented skin spots that appear due to increased melanin production, often triggered by sun exposure. They are more common in individuals with fair skin and are associated with genetic variations linked to UV sensitivity. In some cases, freckles may indicate a higher risk of skin conditions like melanoma.

Risk Factors

- First visible at ages 2 to 3 years
- Light-skinned
- Red or blonde hair

TYPICAL SYMPTOMS



Round or irregularly shaped macules of brown color



Usually less than 3 mm and generally multiple



Smooth and flat texture without bumps or scales



May get darker in the sun

RECOMMENDED INTERVENTIONS

Freckles are primarily a cosmetic concern. They can be treated with dermatological procedures such as laser therapy, resurfacing, or topical treatments.

GENERALIZED PSORIASIS

5

RESULT Risk Markers Detected

Reduced Risk

Gene: IL13

Genotype: GG

Gene: IL23R

Genotype: GG

Gene: Intergenic

Genotype: AG

Gene: PSORS1C3

Genotype: GG

Gene: TNFAIP3

Genotype: TG

Description

Psoriasis is a condition affecting the skin. It is thought to be an autoimmune condition that is caused by the immune system attacking healthy skin cells by mistake. Psoriasis is characterized by the appearance of inflamed lesions on the skin. These lesions can appear scaly and dry. Patients often feel itchy, sore or burning sensations on these lesions. It can also affect the joints, making them swollen.

Risk Factors

- Appears for the first time from ages 15 to 25 years
- 5 times more frequent in Europeans than Asians

TYPICAL SYMPTOMS



Red patches of skin covered with thick and silvery scales



Dry and cracked skin that may bleed or itch



Itching, burning or soreness



Thickened, pitted or ridged nails

RECOMMENDED INTERVENTIONS

Psoriasis tends to occur in cycles, often triggered by stress or skin trauma. Doctors may prescribe corticosteroids for sudden flare-ups, complemented by topical ointments or shortwave ultraviolet B (UVB) phototherapy.

GLYCATION PROTECTION

3

RESULT

Risk Markers Detected



Normal Protection

Gene: AGER

Genotype: AA

Gene: AGER

Genotype: TC

Gene: GLO1

Genotype: AA

Description

Glycation occurs when sugars bind to proteins, leading to oxidative damage that accelerates skin aging. This process can result in wrinkles, loss of skin elasticity, and collagen degradation. Genetics influence an individual's ability to combat glycation-related damage.

Risk Factors

- Middle-age and older population
- Diabetic
- Exposure to ultraviolet rays
- Smoking

TYPICAL SYMPTOMS



Lower collagen integrity



Premature aging



Reduced ability for skin to quickly heal



Wrinkling and sagging skin

RECOMMENDED INTERVENTIONS

Glycation can be minimized by reducing sugar intake and following a diet rich in antioxidants. Avoiding UV exposure and ionizing radiation further helps prevent glycation.

KELOID RISK

1

RESULT

Risk Markers Detected

Reduced Risk

Gene: Intergenic

Genotype: TC

Description

Keloids are raised, thickened scars caused by excessive collagen production during wound healing. They can grow larger than the original injury and are more common in individuals with certain genetic backgrounds. While harmless, keloids can be aesthetically unappealing and cause discomfort.

Risk Factors

- Aged below 30 years
- Asian and African ethnicity
- Burn injury
- Previous keloid formation

TYPICAL SYMPTOMS



Thick and irregular scarring, depending on injury



Shiny, lumpy and hairless skin



Varied texture, from soft to firm and rubbery



Reddish, brown or purplish, depending on skin color

RECOMMENDED INTERVENTIONS

Keloids are difficult to treat once formed, and surgical removal has a high recurrence risk. Individuals at high risk should avoid ear piercings, surgery, and unnecessary skin trauma.

MELANIN AND PIGMENTATION

Description

Melanin is a natural skin pigment which is produced naturally in the body. It protects the skin from harmful ultraviolet (UV) rays, especially those from the sun. The pigmentation is also responsible for the color of the skin, irises and hair. Genetic factors influence how much melanin is in the skin, and this amount varies from one individual to another.

Risk Factors

- Affects individuals of all ages
- Light-skinned
- Excessive ultraviolet light exposure

TYPICAL SYMPTOMS



Skin gets darker if body produce too much melanin



Skin gets lighter if body produces too little melanin

3

RESULT

Risk Markers Detected



Normal Response

Gene: OCA2

Genotype: TC

Gene: OCA2

Genotype: CC

Gene: SLC45A2

Genotype: TC

RECOMMENDED INTERVENTIONS

Optimizing melanin production is important to prevent cell damage caused by ultraviolet (UV) exposure. Eating food rich in antioxidants, Vitamin A, Vitamin B, and Vitamin C (such as citrus, berries, and leafy green vegetables) aids in providing extra protection to the skin layer.

OMEGA-3 AND OMEGA-6 DEFICIENCY

3

RESULT Risk Markers Detected

Normal Risk

Gene: FADS1

Genotype: CC

Gene: FADS2

Genotype: GG

Gene: MYRF

Genotype: TT

Description

Omega-3 and omega-6 are polyunsaturated fatty acids that help to regulate blood pressure and prevent inflammation within the skin. In addition, these fatty acids give a healthy look to an individual, as it can prevent loss of skin moisture. Dietary intake of the right amount of omega-6 and omega-3 fatty acids is essential for optimal health. A healthy ratio of omega-6 to omega-3 fatty acids is from 1:1 to 4:1.

Risk Factors

- Affects individuals of all ages
- Low-fat diets
- Vegetarians
- Malabsorption issues

TYPICAL SYMPTOMS



Fatigue



Mood swings
or depression



Poor circulation



Poor memory

RECOMMENDED INTERVENTIONS

Increase intake of omega-3 and omega-6 fatty acids by consuming foods such as spinach, chia seeds, mangoes, lettuce, walnuts, and kidney beans.

PREMATURE WRINKLES

2 RESULT

Risk Markers Detected

Reduced Risk

Gene: LOC105374

Genotype: CC

Gene: MMP1

Genotype: DI

Description

Premature wrinkles are caused by loss of skin elasticity - with aging and ultraviolet (UV) radiation being the main causes. Signs include sagging, dullness and spotty discoloration of the skin. Genetic factors can affect an individual's skin aging, susceptibility to UV radiation and development of skin cancer. Research has shown a correlation between these factors and the appearance of wrinkles and increased pigmentation.

Risk Factors

- Middle-aged to older population
- High blood sugar
- Poor hydration
- Stress
- Smoking

TYPICAL SYMPTOMS



Fine lines and wrinkles develop on skin at a younger age



Skin becomes drier and thinner



Skin loses its flexibility and elasticity



Skin loosens and starts to sag

RECOMMENDED INTERVENTIONS

Development of premature wrinkles is strongly affected by ultraviolet (UV) radiation exposure. This exposure can be minimized by using protective sunscreen and moisturizing the skin regularly. Other lifestyle modifications include washing the face twice a day, reducing alcohol intake and quitting smoking.

RESPONSE TO ANTIOXIDANTS

5

RESULT Risk Markers Detected

Reduced Response

Gene: ATM Genotype: AG	Gene: CAT Genotype: TT	Gene: GPX4 Genotype: TT
Gene: SOD2 Genotype: AA	Gene: SOD3 Genotype: CC	

Description

Antioxidants can protect the body's cells against free radicals, which are waste products that may contribute to the development of cancer, heart disease, and other diseases. Free radicals are produced by cells as the body processes food and reacts to the environment. Oxidative stress can result if the body cannot process and remove free radicals properly. Long-term oxidative stress may damage the DNA, as well as other important molecules in the body.

Risk Factors

- Affects individuals of all ages
- Dietary insufficiency

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for low antioxidant levels. Detection is usually done via blood screening.

RECOMMENDED INTERVENTIONS

Having a diet which is high in anti-oxidant rich foods can help to support the body's anti-oxidation response. Such diet includes green leafy vegetables (e.g. cabbage, spinach), cruciferous vegetables (e.g. cauliflower and broccoli), tomatoes, fishes (e.g. salmon, tuna and sardines) and green tea.

SKIN BARRIER

0

RESULT

Risk Markers
Detected

Normal Protection

Description

The skin is made up of layers, each performs important functions to protect the body. The outermost layer of the skin, known as the skin barrier, defends the body against environmental threats while simultaneously protecting the skin from dehydrating. Without an effective skin barrier, various harmful environmental toxins and pathogens could penetrate the skin and cause adverse effects inside the body.

Risk Factors

- Humid or dry environment
- Allergens, irritants, and pollutants
- Sun exposure
- Alkaline soaps
- Exposure to harsh chemicals
- Over-exfoliation
- Psychological distress
- Genetic factors

TYPICAL SYMPTOMS



Dry, scaly skin



Itchiness and inflamed areas



Rough or discolored patches



Bacterial, viral or fungal skin infections

RECOMMENDED INTERVENTIONS

To protect your skin barrier, use a gentle cleanser and avoid over-washing. Keep your skin hydrated with a moisturizer containing ceramides or hyaluronic acid, and always apply sunscreen to shield against UV damage. Avoid harsh products with alcohol, strong acids, or fragrances, and simplify your skincare routine to prevent overloading your skin. For severe skin issues, consult a dermatologist for professional advice.

SKIN ELASTICITY

1

RESULT

Risk Markers
Detected

Normal Elasticity

Gene: MMP1

Genotype: DI

Description

Skin elasticity is the skin's ability to stretch and bounce back to its original shape. It highly depends on the quantity of collagen in one's diet. Genetics play an important factor in the degradation of collagen fibers, while environmental factors may induce or inhibit skin inflammation. Loss of skin elasticity is part of aging but poor nutrition or smoking may accelerate the process.

Risk Factors

- General population
- Affects individuals of all ages
- Exposure to sun
- Pollution
- Smoking

TYPICAL SYMPTOMS



Skin appears saggy, wrinkly or leathery



Skin is more easily damaged



Skin is less firm, plump or youthful in appearance



Skin takes time to get back to original shape when stretched

RECOMMENDED INTERVENTIONS

Although skin elasticity naturally declines with age, it can be preserved by limiting sun exposure, consuming antioxidant-rich foods, and taking collagen supplements.

SKIN LIGHTENING ABILITY

1

RESULT

Risk Markers
Detected



Normal Ability

Gene: SLC24A5

Genotype: GG

Description

Differences in skin color among individuals depend on various factors, where genes controlling the quantity and quality of melanin and sun exposure play an important role. Differences in skin and hair color are principally genetically determined and are due to variations in the amount, type, and packaging of melanin polymers. The type and amount of melanin produced is controlled by various of genes that are inherited.

Risk Factors

- Changes in production of melanin
- Drugs
- Chemical bleaching

TYPICAL SYMPTOMS



Allergic reaction



Losing hair pigmentation



Uneven skin texture



Vitiligo

RECOMMENDED INTERVENTIONS

To prevent the skin from getting darker, reduce your exposure to the sun. On the other hand, to make your skin darker, you can consider going for a natural or artificial tan (using sun protection lotion).

SUNSPOTS

Description

Sunspots are skin hyperpigmentation where areas of the skin form brown spots. It can indicate skin damage and aging. Although sunspots do not pose a health risk to the individual, they can be prevented by limiting skin exposure to ultraviolet rays (UVA and UVB). Research suggests that genetic variants increase the risk of sunspots becoming malignant if exposure to ultraviolet rays is prolonged.

Risk Factors

- Middle aged and older population
- Fair-skinned
- Exposure to ultraviolet light

TYPICAL SYMPTOMS



Flat and oval areas of pigmentation



Tan to dark brown in color



Size range from freckle size to 0.5 inch diameter



Can merge, causing them to be more noticeable

2

RESULT

Risk Markers Detected



Normal Risk

Gene: IRF4

Genotype: CC

Gene: MC1R

Genotype: AG

RECOMMENDED INTERVENTIONS

There are various home remedies that help to fade or remove sunspots, such as applying aloe vera, consuming Vitamin C, green tea and lemon juice. The exposed skin can be protected by applying SPF sunscreen as it acts as a barrier against ultraviolet (UV) radiation.

TANNING RESPONSE

Description

Tanning is a natural response to UV exposure, where melanin production increases to protect the skin. Genetic factors influence an individual's ability to tan, with those who tan easily being at a lower risk of sunburn but potentially more susceptible to vitamin D deficiency.

Risk Factors

- Affects individuals of all ages
- Fair-skinned
- Exposure to ultraviolet light

TYPICAL SYMPTOMS



Tanning response can result in skin darkening shortly after UV exposure, or delayed tanning which can take days to become visible.

4

RESULT

Risk Markers Detected



Normal Response

Gene: SLC24A5

Genotype: GG

Gene: SLC45A2

Genotype: CC

Gene: SLC45A2

Genotype: TC

Gene: TYR

Genotype: CC

RECOMMENDED INTERVENTIONS

Although tanning is a natural defense mechanism, excessive tanning can be prevented by using UV-blocking lotion and covering the skin. Oral supplementation of vitamins C, E, and A, along with consuming polyunsaturated fatty acids, can help manage tanning effects.

VARICOSE VEINS

Description

Varicose veins occur when weakened vein valves cause blood to pool, leading to enlarged, visible veins, commonly in the legs. Genetics play a significant role in susceptibility, alongside factors like obesity and pregnancy. In severe cases, varicose veins may increase the risk of thrombosis and cardiovascular complications.

Risk Factors

- Older population
- More common in women than men
- Obesity
- Pregnancy
- Standing or sitting for long periods of time

TYPICAL SYMPTOMS



Dark purple or blue veins



Veins appear twisted and bulging



Throbbing, swelling or muscle cramping in lower legs



Worsened pain after sitting or standing for a long time

1

RESULT

Risk Markers Detected



Normal Risk

Gene: MTHFR

Genotype: TT

RECOMMENDED INTERVENTIONS

Varicose veins can often be managed through self-care measures, including regular exercise, maintaining a healthy weight, and avoiding prolonged standing. In severe cases, medical treatments such as sclerotherapy (injection of a solution into the veins) may be recommended to reduce their appearance.



In your quest for better health, you will need a deeper understanding of your body. Genetic screening can identify wellness and fitness traits that may affect your quality of life. Armed with this information, you can work closely with a health or wellness practitioner to develop suitable wellness management programs to maximize your long-term health and wellbeing, because an ounce of prevention is always better than a pound of cure.

Wellness

ACHILLES TENDINOPATHY

5

RESULT Risk Markers Detected

Slightly Lower Risk

Gene: CASP8
Genotype: GG

Gene: COL5A1
Genotype: AC

Gene: COL5A1
Genotype: TC

Gene: GDF5
Genotype: AG

Gene: TNC
Genotype: AA

Description

Achilles tendinopathy occurs when there is swelling and pain in the Achilles tendon. This can result from prolonged stress on the calf muscles, or sudden movements. It can also be caused by changes which occur in the genes which are responsible for the development of connective tissues. Common injuries from Achilles tendinopathy include inflammation, tendon tears and even ruptures of the tendon.

Risk Factors

- Obesity
- High blood pressure
- Type 2 diabetes
- Prolonged steroid use
- Family history of tendinopathy

TYPICAL SYMPTOMS



Pain in heel



Stiffness in tendon



Swelling at back of ankle



Grating noise
when moving ankle

RECOMMENDED INTERVENTIONS

Achilles tendinopathy can be prevented primarily by giving your body time to recuperate. If you are active in sports or fitness, take 1 to 2 days of rest per week, and 1 to 2 months away from your sport a year. Ensure that you also include functional routines that can help to build strength and balance during your workouts. Warming up and cooling down before and after exercising also helps. Complementing your workouts with foods that are rich in protein, greens, calcium, Vitamin D and magnesium, and investing in good footwear will also help to reduce the risks of Achilles tendinopathy.

ADIPONECTIN LEVEL

Description

Adiponectin is a hormone that plays a crucial role in the body's inflammatory response and protection against oxidative stress. It also regulates glucose levels and fatty acid metabolism. Low levels of adiponectin have been associated with metabolic disorders, including obesity, hypertension, and type 2 diabetes.

Risk Factors

- Increasing age
- Family history of diabetes
- Obesity
- Sedentary lifestyle

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for low adiponectin levels.

3

RESULT

Risk Markers
Detected



Normal Level

Gene: CDH13

Genotype: TG

Gene: CDH13

Genotype: TG

Gene: Intergenic

Genotype: GG

RECOMMENDED INTERVENTIONS

Research has shown that higher levels of adiponectin have antidiabetic and antioxidant effects. It could be beneficial to have higher levels of it in the blood. High-fiber foods, fish, regular exercise, caffeine and curcumin have been linked to higher adiponectin levels in the body.

ANKYLOSING SPONDYLITIS

2

RESULT Risk Markers Detected

Low Risk

Gene: ERAP1

Genotype: TC

Gene: Intergenic

Genotype: AG

Description

Ankylosing spondylitis is a chronic inflammatory condition that can cause the small bones of the spine to fuse over time, leading to stiffness and reduced mobility. It typically begins with intermittent lower back pain and stiffness, which may worsen gradually. As the disease progresses, movement in the lower spine may become restricted. Ankylosing spondylitis is more common in men than in women.

Risk Factors

- Young adults (onset generally occurs in late adolescence or early adulthood)
- More common in men than women

TYPICAL SYMPTOMS



Lower back pain and stiffness



Hip, joint and neck pain



Difficulty breathing



Abdominal pain and diarrhea.

RECOMMENDED INTERVENTIONS

Effective intervention can be undertaken if ankylosing spondylitis is detected early. This can be done by being aware of risk factors related to lifestyle and diet. Early intervention is proven to be effective in preventing or delaying the onset of physical deformities.

ANTERIOR CRUCIATE LIGAMENT (ACL) RUPTURE

2 **RESULT**
Risk Markers
Detected

 **Slightly Lower Risk**

Gene: ACAN

Genotype: TT

Gene: MMP12

Genotype: TT

Description

The anterior cruciate ligament (ACL) is one of two key ligaments that stabilize the knee. A rupture in the ACL can cause knee instability, making standing and walking difficult. Such injuries often occur during high-impact sports involving sudden stops, pivots, or jumps. Over time, knee instability increases the risk of developing osteoarthritis.

Risk Factors

- Young age
- More common in women than men
- Participation in contact sports (football, rugby, basketball, gymnastics)
- Poor conditioning

TYPICAL SYMPTOMS



Popping sensation in knee



Loss of full range of motion



Severe pain and rapid swelling



Tenderness along the joint line

RECOMMENDED INTERVENTIONS

Anterior cruciate ligament (ACL) injuries can be prevented by sufficiently warming up and stretching before undertaking any exercise. While strengthening the muscles around the knee can help preserve the ACL, it can still be injured when we push the body too far when fatigued. A healthy diet also helps with the preservation of joints and its related structures. It would be good if wear appropriate footwear and protective gear for our sports.

CAFFEINE METABOLISM

1

RESULT

Risk Markers
Detected



Normal Metabolism

Gene: CYP1A2

Genotype: AA

Description

Caffeine is the most widely consumed stimulant in the world, found in coffee, tea, colas, and even chocolate. Caffeine works by blocking adenosine receptors in the central nervous system, which promotes alertness. However, individuals with caffeine sensitivity may experience side effects such as insomnia, headaches, anxiety, restlessness, and heart palpitations.

Risk Factors

- Liver disease
- Smoking
- Excessive alcohol consumption
- Consuming certain medications
- Diet

TYPICAL SYMPTOMS



Headache



Insomnia



Nervousness/anxiousness



Palpitations

RECOMMENDED INTERVENTIONS

While symptoms of caffeine sensitivity are mild and innocuous, prolonged symptoms can be detrimental to health. Reducing caffeine intake is advised for those experiencing sensitivity.

CALCIUM METABOLISM

3

RESULT Risk Markers Detected

Normal Metabolism

Gene: CASR

Genotype: AG

Gene: CYP24A1

Genotype: GG

Gene: Intergenic

Genotype: CC

Description

Calcium is an essential nutrient crucial for bone development and cellular signaling. Low calcium levels can lead to stunted growth, muscle cramps, brittle bones and even depression. Hypocalcemia, or poor calcium absorption, can affect people of all ages and may reduce bone mass, slowing overall growth.

Risk Factors

- Women are at higher risk than men
- Intolerance to calcium-rich foods
- Certain medications
- Hormonal changes

TYPICAL SYMPTOMS



Easy bone fractures



Muscle spasms



Numbness/tingling in hands, feet and face



Weak and brittle nails

RECOMMENDED INTERVENTIONS

Treating and dealing with a condition in which there are lower-than-average levels of calcium in your body involves increasing your consumption of calcium-rich foods or supplementation to help increase the levels of calcium available in your body.

CARBOHYDRATE METABOLISM

4

RESULT

Risk Markers Detected

Low Metabolism

Gene: Intergenic

Genotype: CC

Gene: PLIN1

Genotype: AT

Gene: PLIN1

Genotype: TC

Gene: TAS1R2

Genotype: CC

Description

Carbohydrates are the most abundant nutrients in our diet, found in grains, bread, legumes, and root vegetables. Genetics can influence how carbohydrates are metabolized, and some individuals may struggle to digest or absorb them properly. This can increase the risk of conditions such as carbohydrate malabsorption, obesity, and type 2 diabetes.

Risk Factors

- Being overweight or obese
- Diabetes
- Metabolic syndrome

TYPICAL SYMPTOMS



Failure to gain weight properly



Hypoglycemia



Kidney defects



Liver dysfunction

RECOMMENDED INTERVENTIONS

Unless an underlying condition is present, individuals with poor carbohydrate metabolism should focus on nutrient-dense, lower-carbohydrate foods like lean meats, fish, eggs, leafy greens, whole grains, and dairy. Monitoring carbohydrate intake to avoid blood sugar fluctuations and engaging in regular physical activity to support metabolic health are recommended too.

CAROTENE LEVEL

Description

Carotenes are photosynthetic pigments that give vegetables and fungi their red, orange, and yellow colors. β -carotene is the most common form consumed in the human diet and serves as precursor for Vitamin A. It also acts as an antioxidant. Genetic variations affecting β -carotene metabolism can lead to carotenemia, a condition characterized by yellowing of the skin.

Risk Factors

- Vegetarian
- Pregnant or lactating women

TYPICAL SYMPTOMS



Abdominal pain



Fatigue



Itchy skin



Unintentional weight loss

3

RESULT

Risk Markers
Detected



Moderate Level

Gene: Intergenic

Genotype: TG

Gene: SCARB1

Genotype: TT

Gene: SCARB1

Genotype: TT

RECOMMENDED INTERVENTIONS

Getting enough β -carotene is important for Vitamin A production, eye, lung, skin health, and cognitive functions. It is also known to help reduce the risk of developing certain cancers. Good sources of β -carotene are carrots, sweet potatoes, squash, spinach and leafy lettuce.

COENZYME Q10

Description

Coenzyme Q10 (CoQ10) is a naturally occurring nutrient essential for energy production in the form of adenosine triphosphate (ATP). It also supports metabolism and functions as an antioxidant, protecting cells from damage. CoQ10 is naturally found in foods such as meat, fish and nuts.

Risk Factors

- Older age
- Nutritional deficiencies, e.g. vitamin B6
- Increased demands by tissues due to disease

TYPICAL SYMPTOMS



Decreased ability to handle stress



Extreme physical fatigue



Memory lapses



Mood swings

1

RESULT

Risk Markers Detected



Normal Level

Gene: NQO1

Genotype: AG

RECOMMENDED INTERVENTIONS

CoEnzyme Q10 (CoQ10) supplementation provides health benefits, including improved symptoms in congestive heart failure patients and reduced LDL cholesterol levels in diabetics. Natural sources of CoQ10 include spinach, cauliflower, broccoli, pork, beef, chicken, legumes, nuts, and fish.

COLLAGEN FORMATION

2

RESULT Risk Markers Detected

Normal Formation

Gene: COL1A1

Genotype: CC

Gene: COL3A1

Genotype: GG

Description

Collagen is the most abundant protein in the body, making up about one-third of total proteins. It is found in connective tissues such as skin, bones, muscles and tendons. It functions as a scaffold which holds connective structures in the body together. Genetics plays a crucial role in regulating collagen formation and its degradation.

Risk Factors

- Aged 25 and above
- Vitamin C deficiency
- Smoking
- Diabetic

TYPICAL SYMPTOMS



Brittle bones



Joint pain



Loss of certain tendon functions



Loss of skin firmness and elasticity

RECOMMENDED INTERVENTIONS

Collagen production can be increased by consuming nutrients such as Vitamin C, proline, glycine and copper. These nutrients can readily be absorbed from foods that we consume regularly. Protein-rich foods such as meats and legumes also help with collagen production.

CONCUSSION

3

RESULT

Risk Markers Detected

Slightly Lower Risk

Gene: APOE

Genotype: CC

Gene: APOE

Genotype: TG

Gene: APOE

Genotype: TT

Description

A concussion occurs when the brain experiences a sudden, violent force, causing it to shake rapidly within the skull. It is classified as a mild traumatic brain injury (TBI). While not usually life-threatening, concussions can lead to serious symptoms requiring immediate medical attention. They are common in athletes participating in high-contact sports like American football, rugby, and boxing.

Risk Factors

- Falling down
- Getting involved in an accident
- Having a previous concussion
- Participating in high-risk sports

TYPICAL SYMPTOMS



Amnesia regarding traumatic event



Blurry vision



Dizziness



Ringing in the ears

RECOMMENDED INTERVENTIONS

Preventing sudden, violent movement to the head is crucial in preventing concussions. Using safety equipment such as helmets and car seats for babies can help mitigate concussions. Minimizing participation in high contact sports, such as rugby and boxing, helps reduce the risks. However, if you are actively involved, taking proper precautions is recommended.

DETOXIFICATION CAPACITY

6

RESULT
Risk Markers
Detected



Gene: COMT Genotype: AA	Gene: CYP17A1 Genotype: AG	Gene: CYP2C19 Genotype: AG
Gene: CYP2D6 Genotype: AG	Gene: GSTO1 Genotype: AC	Gene: GSTO2 Genotype: AG

Description

Xenobiotic metabolism, commonly known as detoxification, is the body's natural process of eliminating foreign substances such as antibiotics, pollutants, and food additives. The liver plays a primary role in breaking down these substances, which are then excreted through urine, feces, and sweat. If these substances accumulate, they can damage cells and DNA.

Risk Factors

- Pollution
- Liver disease
- Excessive alcohol consumption

TYPICAL SYMPTOMS



Body weight oscillation



Constant fatigue



Excessive constipation or diarrhea



Impaired digestion

RECOMMENDED INTERVENTIONS

Detoxification is a natural bodily process, but healthy habits such as adequate sleep, hydration, exercise, and reducing sugar, salt, and alcohol intake can enhance its efficiency.

ENDURANCE SPORTS PERFORMANCE

Description

Endurance is the ability of the body to sustain a given level of activity for a prolonged period of time. In the body, this is defined as the muscles' ability to resist fatigue. Endurance is generally a combination of muscular, cardiovascular and respiratory health. This can be inherited and identified as 'talent' by trainers. It can also be trained with proper and suitable exercise regimes.

Risk Factors

- Obesity
- Lack of exercise
- Overconsumption of unhealthy foods
- Have diabetes, cardiovascular diseases or respiratory diseases

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for power sports performance.

5

RESULT Risk Markers Detected

Enhanced Performance

Gene: TFAM

Genotype: GG

Gene: KDR

Genotype: TA

Gene: PPARGC1A

Genotype: TC

Gene: COL6A1

Genotype: CC

Gene: ACSL1

Genotype: AG

RECOMMENDED INTERVENTIONS

Increasing endurance performance requires fortifying the muscles, lungs and cardiovascular system. A strong, healthy diet that is rich in proteins and nutrients is recommended for increasing muscle volume. Aerobic exercises such as running, skipping rope and swimming can help with lung strength and cardiovascular health.

ENERGY UTILIZATION

2

RESULT Risk Markers Detected

Normal Utilization

Gene: IGF1

Genotype: CC

Gene: PPARG

Genotype: CC

Description

The body functions properly by utilizing energy. In the body's processes, reserves of energy are transferred and stored in the form of adenosine triphosphate (ATP). These molecules are found primarily in the mitochondria. The ability to use and replenish energy is determined by the cardiovascular, respiratory and digestive health of the individual. There have been multiple genes identified with energy capacity and utilization that can influence athletic performance.

Risk Factors

- Disease affecting energy utilization, e.g. diabetes and respiratory diseases
- Obesity
- Poor diet and lack of necessary nutrients
- Sedentary lifestyle
- Stress

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for energy utilization.

RECOMMENDED INTERVENTIONS

Optimizing proper energy utilization in the body can occur with gradual training. To get started, it is recommended to undertake moderate-intensity physical activity for at least 20 to 30 minutes for three to five times a week. This recommendation from the American College of Sports Medicine excludes time spent doing warm-ups and cool-downs.

GLAUCOMA

1

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: LOXL1

Genotype: GG

Description

Glaucoma is a common eye condition where the optic nerve, which connects the eye to the brain, becomes damaged. It is usually caused by fluid building up in the front part of the eye, which increases pressure inside the eye. Glaucoma can lead to loss of vision if it is not diagnosed and treated early. Genetic studies have suggested that more than 50 percent of glaucoma is strongly hereditary, especially among siblings.

Risk Factors

- Damage to the optic nerves
- Eye pressure
- Faulty eye drainage system
- Family history

TYPICAL SYMPTOMS



Gradual loss of side vision



Halos around lights



Having a red eye



Intense eye pain

RECOMMENDED INTERVENTIONS

Do consult a medical practitioner for treatment options, which can include prescription eye drops, oral medications, laser treatment, surgery or a combination of approaches. Laser therapy, filtering surgery, drainage tubes and minimally invasive glaucoma surgery are some examples of surgeries which are usually performed to treat glaucoma.

GLUTEN SENSITIVITY

1

RESULT Risk Markers Detected

Reduced Sensitivity

Gene: Intergenic

Genotype: TT

Description

Gluten intolerance occurs when an individual experiences adverse reactions after consuming gluten, a protein found in wheat, rye, and barley. While symptoms may resemble those of celiac disease, individuals with gluten sensitivity do not test positive for it. This condition is also referred to as non-celiac gluten sensitivity (NCGS).

Risk Factors

- Celiac disease
- Non-celiac gluten sensitivity
- Wheat allergy
- Gluten ataxia

TYPICAL SYMPTOMS



Abdominal pain



Anemia



Bloating or gas



Diarrhea or constipation

RECOMMENDED INTERVENTIONS

There is no known cure for gluten intolerance. Most people find relief from symptoms by following a gluten-free diet. You should work with your healthcare provider and a dietitian to plan your diet. You can also ask your healthcare provider about adding probiotics to your diet.

HAY FEVER

Description

Hay fever is an allergic reaction triggered by allergens, substances the immune system mistakenly identifies as harmful. It is most commonly caused by pollen from plants, which enters the body through the mouth, nose, eyes, and throat. While hay fever symptoms resemble those of a cold, it is not caused by a virus.

Risk Factors

- Allergy to pollen
- Allergy to plants
- Pollution

0

RESULT
Risk Markers
Detected

Low Risk

TYPICAL SYMPTOMS



Sneezing/coughing



Runny or blocked nose



Itchy or watery eyes



Itchy throat, mouth,
nose and ears

RECOMMENDED INTERVENTIONS

Avoid contact with pollens by keeping windows shut to minimize pollen and dust from coming in, wear sunglasses to cover your eyes when outdoors, wash your hands after touching any animals and use a dehumidifier indoors to control mold growth.

HDL CHOLESTEROL

Description

High-density lipoproteins (HDLs) are often referred to as 'good cholesterol'. This is because HDLs in blood are known to be 'transport' proteins which help transport lower density lipoproteins to the liver for processing and elimination. Higher levels of HDL are linked to better cardiovascular health and lower risks of atherosclerosis. HDLs have protective properties against diseases which include cardiovascular and metabolic diseases.

Risk Factors

- Overweight or obese
- Hypertension
- Smoking
- Excessive alcohol consumption

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for high cholesterol. Blood screening tests are required to determine cholesterol levels.

5

RESULT
Risk Markers
Detected



Normal Level

Gene: ABCA1 Genotype: TC	Gene: CETP Genotype: CC	Gene: CETP Genotype: CC
Gene: LIPC Genotype: CC	Gene: LIPC Genotype: GG	

RECOMMENDED INTERVENTIONS

High HDL cholesterol levels are influenced by genetics. However, having a healthy diet can also help increase the levels of HDL in blood. Eating foods which are rich in Omega-3 fatty acids and soluble fiber has been linked to an increase in HDL levels. Limiting saturated and trans fats also help to increase HDL. Regular exercise is also helpful.

HIGH BLOOD SUGAR (HYPERGLYCEMIA)

1

RESULT Risk Markers Detected



Normal Risk

Gene: Intergenic

Genotype: AG

Description

Hyperglycemia occurs as a result of insulin resistance or deficiency. It is characterized by increased glucose levels in the blood. It is most commonly linked to diabetes mellitus but can be present in individuals without diabetes. It is a component of metabolic syndrome. If left untreated, it can result in diabetes and other complications.

Risk Factors

- Family history of type 2 diabetes
- Overweight or obese
- Endocrine conditions
- Pancreatic conditions

TYPICAL SYMPTOMS



Frequent urination



**Numbness and pain
in fingers and toes**



**Slow healing
of skin wounds**



Frequent skin infections

RECOMMENDED INTERVENTIONS

To reduce the risks of high blood sugar, it is advisable to maintain a daily diet which is low in sugar and simple carbohydrate, and eat foods which are high in fiber and protein. Exercising frequently and regularly monitoring blood sugar is also advised. Consulting with a doctor and getting proper medication is also recommended.

HIGH TRIGLYCERIDES

4

RESULT Risk Markers Detected



Normal Risk

Gene: ZPR1

Genotype: GC

Gene: GCKR

Genotype: TT

Gene: GALNT2

Genotype: GG

Gene: APOA5

Genotype: AG

Description

Hypertriglyceridemia is characterized by elevated plasma triglyceride levels. Triglycerides are a common class of fats used to store energy, and are found circulating in the body. It is commonly caused by genetically-linked conditions such as hyperlipidemia and enzyme deficiencies. The condition can also be caused by medications and obesity. High triglyceride levels can increase the risk of diseases such as heart attack, high blood pressure, pancreatitis and stroke.

Risk Factors

- Increasing age
- Obesity
- Smoking
- Sedentary lifestyle
- Diet high in saturated fat and trans fat
- Excessive alcohol consumption

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for high triglycerides. Typically can be checked through blood tests (lipid panel screening).

RECOMMENDED INTERVENTIONS

Hypertriglyceridemia can be mitigated by avoiding food high in saturated fats and refined carbohydrates. Regular exercise also helps.

HOMOCYSTEINE METABOLISM

4

RESULT Risk Markers Detected

Normal Metabolism

Gene: BHMT

Genotype: AG

Gene: CBS

Genotype: TC

Gene: MTHFD1

Genotype: AG

Gene: MTR

Genotype: AA

Description

Homocysteine is an abundant amino acid found in the body. It is absorbed readily from foods, primarily meats. It plays a crucial role in metabolic pathways related to DNA and protein synthesis. However, high levels of homocysteine can be detrimental to the body and are commonly found in early development of cardiovascular diseases, stroke and neurocognitive disorders. Common causes of elevated homocysteine are genetic defects and low vitamin B intake.

Risk Factors

- Coronary heart disease
- Hypertension
- Psoriasis
- Kidney disease

TYPICAL SYMPTOMS



Dizziness and fatigue



Sores on tongue or in mouth



Pale skin



Tingling feeling in limbs

RECOMMENDED INTERVENTIONS

Homocysteine levels can be managed through increased consumption of foods rich in Vitamin B such as cereals, fruits and vegetables. Supplementation of zinc and glutathione will also help manage homocysteine levels.

INFLAMMATORY RESPONSE

Description

Inflammation is the body's natural defense mechanism that occurs when healing and protecting against infections, injuries, or toxins. It involves an increased blood flow to the affected area, bringing antibodies and proteins to aid in recovery. However, chronic inflammation can be harmful, leading to immune system dysregulation and increasing the risk of serious conditions such as cancer, heart disease and autoimmune disorders.

Risk Factors

- Affects individuals of all ages
- Diabetic
- Immunosuppressed
- Autoimmune disorders

TYPICAL SYMPTOMS



Flushed skin at injury location



Pain or tenderness



Swelling



Redness

6

RESULT

Risk Markers Detected

Normal Response

Gene: IL8 Genotype: TC	Gene: IL6R Genotype: TC	Gene: IL2 Genotype: CC
Gene: IL1A Genotype: CC	Gene: IL15RA Genotype: AC	Gene: CRP Genotype: CC

RECOMMENDED INTERVENTIONS

Consuming the right anti-inflammatory food is essential to prevent major diseases such as arthritis and cancer. An anti-inflammatory food diet includes olive oil, high-fiber food, nuts, fatty fish, tomatoes, green leafy vegetables and fruits such as oranges and blueberries.

LACTOSE INTOLERANCE

2

RESULT Risk Markers Detected

Higher Tendency

Gene: MCM6

Genotype: CC

Gene: MCM6

Genotype: GG

Description

Lactose intolerance is characterized by an adverse reaction to the consumption of lactose, the main carbohydrate found in milk and other dairy products. It is caused by a deficiency of the lactase enzyme. As such, those with lactose intolerance are unable to digest lactose, which results in gastric discomfort or, in more severe reactions, diarrhea.

Risk Factors

- Affects individuals of all ages
- African, Asian, Hispanic and American Indian descent
- Born prematurely
- Diseases of the small intestine

TYPICAL SYMPTOMS



Diarrhea



Nausea



Stomach cramps



Flatulence

RECOMMENDED INTERVENTIONS

Lactose intolerance can be managed by replacing lactose-containing drinks and foods with alternatives such as soy or almond milk. It is also possible to take lactase supplements or probiotics to mitigate intolerance.

LDL CHOLESTEROL

Description

Low-density lipoproteins (LDL), commonly referred to as 'bad cholesterol', is a group of lipoproteins which are responsible for the transportation of cholesterol in the body. High levels of LDLs are the main source of artery-clogging plaque, and are commonly linked to increased risk of atherosclerosis (hardening of arteries), cardiovascular diseases, hypertension and metabolic disorders.

Risk Factors

- Middle to older age
- Overweight or obese
- Diabetic
- Family history
- Cigarette smoking
- Diet high in saturated fat and trans fat

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for cholesterol level. Detection is usually done via blood screening.

2

RESULT

Risk Markers
Detected

Increased Level

Gene: CR1L

Genotype: TG

Gene: Intergenic

Genotype: CC

RECOMMENDED INTERVENTIONS

Lifestyle changes are important to decrease LDL cholesterol levels. One should limit the intake of saturated fats and eat food rich in omega-3 fatty acids. Regular exercise helps in raising high-density lipoprotein (HDL), which is good for health. Smoking and drinking habits should be limited as they pose serious health problems.

LONGEVITY

4

RESULT

Risk Markers Detected

Longer Lifespan

Gene: FOXO3

Genotype: CC

Gene: FOXO3

Genotype: TT

Gene: Intergenic

Genotype: TG

Gene: TOMM40

Genotype: AA

Description

Aging is a natural, progressive process characterized by the gradual deterioration of cellular functions over time. Longevity is influenced by factors such as telomere shortening, genetics, and lifestyle choices. While aging itself is not a disease, it increases susceptibility to degenerative conditions such as cardiovascular disease, osteoporosis, and neurodegenerative disorders.

Risk Factors

- Smoking
- Obesity
- Excessive alcohol consumption
- Poor diet
- Physical inactivity

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for longevity.

RECOMMENDED INTERVENTIONS

Individuals with the genetic markers for longevity must not take this for granted but continue maintaining a healthy lifestyle and diet. Those who indulge in smoking and drinking habits are more prone to diseases despite having longevity genetic markers.

MAXIMAL OXYGEN CONSUMPTION (VO2 MAX)

Description

VO2 max refers to the maximum amount of oxygen the body can use during intense exercise. All human biological processes require oxygen to produce energy for them to carry out daily activities. Individuals that undergo regular exercise training tend to have higher VO2 max levels. Hence, VO2 max levels are directly related to one's endurance performance.

Risk Factors

- Aged 65 years and older
- More common in women than men
- Family history
- Overweight or obese
- Diabetic
- Diet high in fat or low in fiber

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for VO2 Max levels.

3

RESULT

Risk Markers Detected



Enhanced Capacity

Gene: PPARGC1A

Genotype: TC

Gene: ADIPOQ

Genotype: TG

Gene: PPARG

Genotype: CC

RECOMMENDED INTERVENTIONS

To increase VO2 max is to increase one's endurance during an intense physical exercise session. Some of the ways to improve VO2 max levels are by performing high-intensity interval training, and switch up with aerobic activities in a single workout. These workouts help strengthen the heart muscle and increase the blood volume pumped with each beat.

OSTEOARTHRITIS

3

RESULT Risk Markers Detected

Low Risk

Gene: GDF5

Genotype: AG

Gene: MCF2L

Genotype: AA

Gene: COL6A4P1

Genotype: TC

Description

Osteoarthritis (OA) is a degenerative joint disorder that occurs when the cartilage cushioning the ends of bones wears away. This leads to painful, swollen joints with reduced mobility. OA commonly affects the hands, elbows, knees, and hips, and while it can occur at any age, it is more prevalent after age 50, affecting women more than men.

Risk Factors

- Joint injuries, even those occurred many years ago
- Repeated stress on joints
- Bone deformities
- Certain metabolic diseases, including diabetes
- Genetics
- Older age
- Gender (women have higher risk)
- Obesity

TYPICAL SYMPTOMS



Affected joints hurt during or after movement



Bone spurs can form around joint, causing a grating sensation



Loss of flexibility of joint



Swelling caused by soft tissue inflammation around joint

RECOMMENDED INTERVENTIONS

Osteoarthritis cannot be reversed, but treatments are available to alleviate the symptoms. Physical therapy and medications both help relieve symptoms, primarily the pain. Individuals can avoid the risk of osteoarthritis by having an active lifestyle and maintaining a healthy weight.

OSTEOPOROSIS

Description

Osteoporosis occurs when the body loses its bone mineral density, causing bones to become weak. The bones may easily break from a fall or even minor bumps, in severe cases. Studies suggest that women above the age of 60 years tend to suffer from this disease. The metabolism of bone tissues and the lack of calcium have been identified as a link to osteoporosis.

Risk Factors

- Aged 50 years and above
- Overweight or obese
- More common in women than men
- Calcium deficiency

TYPICAL SYMPTOMS



Fragility-related fractures



Loss of height over time



Lower back pain



Stooped posture

3

RESULT

Risk Markers Detected

Low Risk

Gene: Intergenic

Genotype: CC

Gene: JAG1

Genotype: AG

Gene: SPTBN1

Genotype: GG

RECOMMENDED INTERVENTIONS

Osteoporosis risk increases as we age. However, several lifestyle habits are effective to improve bone mass, hence lowering the likelihood of developing osteoporosis. One should exercise regularly and maintain a healthy diet in order to obtain all the essential nutrients such as calcium and Vitamin D, which can improve bone health.

PAIN TOLERANCE

Description

Pain tolerance is unique for every individual, and is influenced by various genetic and environmental factors. Besides that, expectations and past experiences play a role in how individuals feel pain as well. Increased sensitivity to pain is a risk factor for developing chronic pain, a common but debilitating condition. An individual's sensitivity to pain is crucial as sport injuries may lead to different pain conditions.

Risk Factors

- Women have lower pain threshold
- Insomnia
- Depression
- Stress

TYPICAL SYMPTOMS



The feeling of pain is dependent on individuals, therefore, is difficult to measure with certainty.

1

RESULT

Risk Markers Detected



Increased Tolerance

Gene: FAAH

Genotype: AC

RECOMMENDED INTERVENTIONS

Pain tolerance and threshold varies among individuals as it depends on the nerves and brain interactions. There are several ways to boost one's pain tolerance, such as by practicing yoga, aerobic exercise and biofeedback therapy.

POLYUNSATURATED FATTY ACIDS

Description

Polyunsaturated fatty acids (PUFA) are involved in cholesterol metabolism. They are healthy fats which can help to lower LDL (bad) cholesterol in the body, so that it does not clog the arteries. PUFAs include omega-3 and omega-6 fatty acids, which are essential for cell growth and various body functions. These fatty acids can only be obtained from food, as the body does not produce them.

Risk Factors

- Affects individuals of all ages
- Dietary consumption

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for high polyunsaturated fatty acid levels. Detection is usually done via blood screening.

2

RESULT

Risk Markers Detected

Normal Level

Gene: FADS2

Genotype: GG

Gene: FADS1

Genotype: TT

RECOMMENDED INTERVENTIONS

Polyunsaturated fatty acids (PUFA) can be obtained naturally from consuming wild fishes such as salmon and sardines. However, PUFA should only be taken in moderate amount. Try to avoid consuming high levels of PUFA.

POWER SPORTS PERFORMANCE

4

RESULT
Risk Markers
Detected

Enhanced Performance

Gene: ACTN3 Genotype: TT	Gene: AGT Genotype: AG	Gene: UCP2 Genotype: TC
Gene: UCP2 Genotype: GG		

Description

Power sports allow strong muscle contractions in a short period of time. Prolonged muscle contraction may lead to fatigue and muscle weakness. The duration of muscle contraction influences the capacity of an individual to perform power exercise. With progressive power training, this results in muscle strength gains and improved physical performance. These improvements can improve one's quality of life.

Risk Factors

- Poor sleep quality
- Physiological stress
- Lack of necessary nutrition

TYPICAL SYMPTOMS



Each individual may find certain power exercises more suitable than others, depending on muscle type, endurance level, and overall fitness.

RECOMMENDED INTERVENTIONS

Power sports performance require a long period of training to build up strong muscle contractions. Some activities that require explosive strength and power, and are good for power training, include jumping, throwing, wrestling, canoeing, kayaking and gymnastics.

RESPIRATORY INFECTIONS - SEVERITY

Description

Some individuals who suffer respiratory infections can develop more serious or even life-threatening conditions, such as acute respiratory distress syndrome. Genetic differences may account for some of these different outcomes. Depending on the genetic variants that you have, you could be more at risk for an impaired immune system, which may result in severe reactions from respiratory infections.

Risk Factors

- Undernutrition
- Vitamin A deficiency
- Exposure to air pollution
- Smoking

TYPICAL SYMPTOMS



Dizziness or loss of consciousness



Frequent and severe coughing



High fever



Wheezing or difficulty in breathing

6

RESULT Risk Markers Detected

Reduced Risk

Gene: LTA Genotype: AA	Gene: IL6 Genotype: GC	Gene: TNF Genotype: GG
Gene: ADIPOQ Genotype: AA	Gene: ADIPOQ Genotype: TC	Gene: ABO Genotype: CC

RECOMMENDED INTERVENTIONS

Practice good hydration, regular exercise, balanced diet and stress management. Also, do deep breathing exercises to enhance lung function. Performing aerobic exercises (brisk walking, jogging) can improve lung capacity and immune functions.

RESPIRATORY INFECTIONS - SUSCEPTIBILITY

Description

Not everyone responds the same way to respiratory infections. While some individuals may experience mild or no symptoms, others may develop severe complications. Susceptibility to respiratory infections depends on various factors, including genetics, immune system strength, pre-existing health conditions, and lifestyle habits.

Risk Factors

- Living in crowded areas
- Have other existing health conditions
- Low immunity
- Smoking

TYPICAL SYMPTOMS



Easily gets infected from others



Suffers frequent respiratory infections



Recurrent infections



Having different types of infections

6

RESULT

Risk Markers Detected

Reduced Susceptibility

Gene: OAS1

Genotype: AG

Gene: MUC5B

Genotype: GG

Gene: IFNG

Genotype: TA

Gene: CD209

Genotype: AA

Gene: CD14

Genotype: AG

Gene: FCGR2A

Genotype: AG

RECOMMENDED INTERVENTIONS

Do not come in close contact with anyone who is sick and avoid crowded places. Do wash your hands often or use hand sanitizers. Try to avoid touching your eyes, nose and mouth. Improve your immunity by following good health practices and habits, proper nutrition, exercise regime, etc.

RHEUMATOID ARTHRITIS

4

RESULT
Risk Markers
Detected



Gene: STAT4 Genotype: TG	Gene: CD244 Genotype: TC	Gene: CD244 Genotype: AG
Gene: PADI4 Genotype: TC		

Description

Rheumatoid arthritis is an autoimmune disease that targets the joint membranes; where, the lining of joints becomes inflamed which causes damage to the joint tissues. The main causes of this disease include genetics and infections. Symptoms can occur in phases known as flares during active periods, and remissions when symptoms lessen. It usually begins after the age of 40 years, and women are more prone to it than men.

Risk Factors

- Middle-aged
- More common in women than men
- Family history
- Smoking
- Obesity

TYPICAL SYMPTOMS



Pain or aching in joints



Same symptoms
on both sides of the body



Stiffness in joints



Tenderness and swelling
in joints

RECOMMENDED INTERVENTIONS

The onset of rheumatoid arthritis can be delayed through regular exercise and balanced nutritional intake. Supplements like fish oil, evening oil, primrose oil and blackcurrant oil help prevent the risk of rheumatoid arthritis.

SALT-SENSITIVITY HIGH BLOOD PRESSURE

Description

High blood pressure, or hypertension, is a condition which may lead to a stroke or heart attack. It happens when blood pressure against the arterial wall is more than 140mmHg (systolic) and 80mmHg (diastolic). This condition can be triggered by several factors, including excessive salt consumption, which leads to water retention, increased blood volume, and changes in artery function.

Risk Factors

- Middle to older age
- Family history of hypertension and/or cardiovascular diseases
- Overweight or obese

TYPICAL SYMPTOMS



Increases in blood pressure with higher salt intake



Pounding in ears, chest or neck



Irregular heartbeat or difficulty in breathing



Severe headaches or nosebleed

6

RESULT
Risk Markers
Detected



Increased Risk

Gene: ADD1 Genotype: TT	Gene: AGT Genotype: AG	Gene: AGTR1 Genotype: AG
Gene: CLCNKA Genotype: AA	Gene: LTBR Genotype: TT	Gene: SCNN1G Genotype: CC

RECOMMENDED INTERVENTIONS

The risk of an elevated blood pressure can be lowered by maintaining a diet low in sodium and high in potassium. One should also have adequate intakes of calcium, potassium and magnesium, while having an active lifestyle and keeping to a healthy weight.

SELENIUM

1

RESULT

Risk Markers
Detected



Gene: SEPP1

Genotype: CC

Description

Selenium is a trace mineral that makes proteins called antioxidant enzymes - which can prevent oxidative stress. It also plays a role in the regulation of thyroid hormones that control our muscle and digestive functions. A deficiency in selenium can lead to DNA damage and negatively impact the heart and nervous system. Long-term deficiency will lead to conditions such as depression, anxiety and dementia.

Risk Factors

- General population
- Crohn's disease
- Kidney disease

TYPICAL SYMPTOMS



Infertility



Muscle weakness



Mental fog



Weakened immune system

RECOMMENDED INTERVENTIONS

The first-line treatment for selenium deficiency is to eat more foods that are high in selenium content. Selenium can only be obtained from diet. Foods that are high in selenium include oysters, organ meats, tunas, Brazil nuts and eggs.

SOFT TISSUE INJURY

Description

Soft tissue injuries affect muscles, ligaments, and tendons, often resulting from strenuous exercise, sudden movements, or impact. These injuries cause pain, swelling, and bruising. Both genetic and environmental factors play a role in injury susceptibility. To reduce the risk, light exercise and warm-ups should be performed before training.

Risk Factors

- Affects individuals of all ages
- Being physically active

TYPICAL SYMPTOMS



Lump at site of injury



Muscle cramping or weakness



Pain and swelling



Visible bruising

4

RESULT

Risk Markers Detected

Low Risk

Gene: CCL2

Genotype: CC

Gene: COL5A1

Genotype: TC

Gene: IGF2

Genotype: CG

Gene: IGF2

Genotype: TG

RECOMMENDED INTERVENTIONS

Keeping the body in good physical condition helps prevent various soft-tissue injuries. Stretching the muscles before exercising can help to prepare the muscles for activities and lessen the risk of injuries. Other ways to prevent injuries include maintaining good posture and wearing proper footwear.

SPORTS INDUCED SUDDEN CARDIAC ARREST

3

RESULT Risk Markers Detected

Low Risk

Gene: BAZ2B

Genotype: TC

Gene: CDKN2B-AS1

Genotype: AG

Gene: Intergenic

Genotype: TC

Description

Sudden cardiac arrest is a condition where the heart suddenly stops beating, accompanied by difficulty in breathing and loss of consciousness. Prolonged cardiac arrest can be fatal due to the lack of oxygen and nutrient supply to vital organs. Conditions like cardiomyopathy, inborn heart defects, low potassium level, diabetes and genetic diseases are linked to sudden cardiac arrest.

Risk Factors

- More common in men than women
- Family history of cardiovascular diseases
- Smoking
- High blood pressure
- High blood cholesterol
- Obesity
- Diabetes
- Inactive lifestyle

TYPICAL SYMPTOMS



Sudden collapse



Lack of pulse



No breathing



Loss of consciousness

RECOMMENDED INTERVENTIONS

Those with genetic predispositions for sudden cardiac arrest should go for more regular heart checkups. Also, if there is a higher genetic risk for this deadly condition, competitive sports and strenuous exercise should be avoided. Living a healthy lifestyle and eating a balanced diet are recommended.

STEROID HORMONES METABOLISM

Description

Steroid hormones are steroids that act as hormones. They aid in sex maturation, metabolism and body development. The production of steroid hormones is in the adrenal gland, which helps to regulate carbohydrate metabolism, sodium and potassium level. Major steroid hormones include estrogen, androgen and progesterone. Steroid can also be artificially prescribed to withstand illness or injury.

Risk Factors

- Younger population
- Family history
- Adrenal gland dysfunction

TYPICAL SYMPTOMS



Abdominal pain



Extreme weakness



Mood changes



Nausea or dizziness

4

RESULT
Risk Markers
Detected

Normal Metabolism

Gene: CYP1B1 Genotype: TT	Gene: CYP19A1 Genotype: AG	Gene: CYP17A1 Genotype: AG
Gene: CYP17A1 Genotype: AG		

RECOMMENDED INTERVENTIONS

Steroid hormone metabolism can be improved by consuming Vitamin D supplements, zinc, and also food rich in androgens, such as cabbage and tuna.

STRESS SENSITIVITY

2

RESULT Risk Markers Detected

Reduced Sensitivity

Gene: OXTR

Genotype: AA

Gene: TPH2

Genotype: TT

Description

High levels of stress are often linked to depression, or major lifestyle diseases like heart attack and obesity. An efficient stress management allows individuals to improve their overall lifestyle as depression can make the body susceptible to various infections. Although short-term stress can help to boost one's productivity, chronic stress will result in serious illnesses.

Risk Factors

- Affects individuals of all ages
- Major life events
- Chronic illnesses or injuries
- Emotional problems

TYPICAL SYMPTOMS



Chest pain or heart racing sensation



Panic attacks, anxiety or irritability



Exhausted or have trouble sleeping



High blood pressure or muscle tension

RECOMMENDED INTERVENTIONS

Stress is an inevitable process in life, and it can be reduced with the right techniques. Individuals can practice meditation and relax themselves with aromatherapy on a regular basis. A balanced diet and adequate sleep help keep the stress level to a minimum.

TOTAL CHOLESTEROL

Description

Total cholesterol refers to the overall amount of cholesterol in the blood, including LDL and HDL cholesterol. It includes low-density lipoprotein (LDL) and high-density lipoprotein (HDL) cholesterol, among others. High cholesterol level is linked to multiple diseases worldwide, such as heart disease and atherosclerosis. Various lifestyle factors influence cholesterol levels, such as smoking, lack of exercise and poor diet.

Risk Factors

- Middle to older age
- Overweight or obese
- Hypertension
- Metabolic syndrome

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for high total cholesterol. It is typically detected through a blood test.

0

RESULT

Risk Markers
Detected



Reduced Level

RECOMMENDED INTERVENTIONS

Several lifestyle changes are required to improve one's cholesterol level. It is important to limit saturated fats, which are usually found in meat and full-fat dairy products. In addition, include foods rich in omega-3 fatty acid in daily diet. Besides that, regular exercise and reduced alcohol intake can also improve cholesterol levels. It is recommended to reduce or even stop from smoking because smoking lowers HDL cholesterol and damages blood vessels, increasing the risk of heart disease.

VISION DEGENERATION

1

RESULT Risk Markers Detected

Low Risk

Gene: TYR

Genotype: AA

Description

Age-related macular degeneration is an eye disease that can cause blurriness to the central vision. It occurs when aging results in damage to the macula, a part of the retina. This is a common condition and a leading cause of vision loss in older adults. Although it does not cause complete blindness, losing the central vision can cause difficulties in performing daily activities.

Risk Factors

- Aged 50 and above
- Family history
- Smoking
- Hypertension

TYPICAL SYMPTOMS



**Distorted images in
central line of vision**



**Difficulty seeing in
low light**



**Haziness and
blurred vision**



**Blind spot in center
of field of vision**

RECOMMENDED INTERVENTIONS

There is no cure for age-related macular degeneration (AMD), but early detection through regular eye exams is essential. Early treatment may slow progression or help prevent further vision loss. Treatments for this condition will depend on whether it is in the early or later stages. Nutritional supplements containing antioxidants may reduce the risk of this condition progressing.

VITAMIN A

1

RESULT

Risk Markers Detected



Reduced Risk

Gene: FFAR4

Genotype: TT

Description

Vitamin A is an essential nutrient for vision, cell division, growth, reproduction, and immunity. It also has antioxidant properties that protect cells from damage caused by free radicals, which are linked to diseases such as heart disease and cancer. A deficiency of Vitamin A may lead to night blindness, cornea degeneration and impaired immunity. It is not produced by the body and must be obtained from diet.

Risk Factors

- Infants and children
- Pregnant women and breastfeeding mothers are at higher risk

TYPICAL SYMPTOMS



Dry skin and eyes



Night blindness



Delayed growth



Poor wound healing

RECOMMENDED INTERVENTIONS

The amount of Vitamin A needed by the body varies as it depends on an individual's age and gender. A diet rich in Vitamin A include foods such as carrots, sweet potato, tuna, rock melon and grapefruit.

VITAMIN B2

Description

Vitamin B2, or riboflavin, is essential for various metabolic processes that maintain overall health, as well as proper growth and development. It plays a vital role in maintaining the body's energy supply. Some individuals require a higher riboflavin intake due to genetic reasons. Vitamin B2 deficiency can lead to cracked lips, anemia, mouth ulcers and dry skin. Sources of Vitamin B2 include eggs and dairy products.

Risk Factors

- Vegetarian
- Pregnancy
- Excessive alcohol consumption
- Irritable bowel syndrome
- Lactose intolerance
- Malabsorptive states

TYPICAL SYMPTOMS



Anemia



Skin cracking or dermatitis



Soreness of throat



Weakness or fatigue

0

RESULT

Risk Markers Detected

Reduced Risk

RECOMMENDED INTERVENTIONS

A sufficient intake of Vitamin B2 is essential for healthy growth and development. It can be obtained from a balanced diet, with rich sources including salmon, mushrooms, low-fat milk, almonds, spinach, avocados, and eggs.

VITAMIN B6

1

RESULT

Risk Markers Detected

Reduced Risk

Gene: NBPf3

Genotype: TC

Description

Vitamin B6, or pyridoxine, is significant to the metabolism of amino acids and lipids, and neurotransmitter and hemoglobin production. Adequate levels of pyridoxine support healthy skin and may help prevent chronic diseases. Vitamin B6 deficiency causes depression, high blood pressure and weakened immune function. Individuals with autoimmune disorders, smokers, and alcoholics are at a higher risk of vitamin B6 deficiency.

Risk Factors

- Affects individuals of all ages
- Chronic diseases
- Autoimmune diseases
- Obese
- Smoking

TYPICAL SYMPTOMS



Cracked and sore lips



Skin rashes



Tingling and pain in hands and feet



Weakened immune system

RECOMMENDED INTERVENTIONS

Most people obtain adequate Vitamin B6 from their daily diet. However, deficiencies can still occur, particularly in certain populations in less developed countries. Foods rich in Vitamin B6 include salmon, lean chicken breast, beef, sweet potatoes, bananas, and avocados.

VITAMIN B12

Description

Vitamin B12, or cobalamin, is needed for brain function, production of red blood cells and nerve tissue health. A deficiency in this vitamin can lead to depression, fatigue, anemia, cardiovascular diseases, and an increased risk of permanent nerve and brain damage. It can be obtained from beef, poultry, fish and dairy products. Vegans face a higher risk of deficiency due to the lack of animal-sourced food intake.

Risk Factors

- Since childhood
- Vegetarian
- Autoimmune diseases
- Crohn's disease
- Certain medications

TYPICAL SYMPTOMS



Pale yellow tinge to skin



Sore and red tongue or mouth ulcers



Pins and needles sensation



Decline in mental abilities

4

RESULT

Risk Markers
Detected



Normal Risk

Gene: CUBN

Genotype: AA

Gene: FUT2

Genotype: AA

Gene: FUT2

Genotype: GG

Gene: TCN1

Genotype: AA

RECOMMENDED INTERVENTIONS

Vitamin B12 is primarily obtained from animal-based foods such as clams, liver, beef, poultry, Swiss cheese, eggs, mackerel, and crab.

VITAMIN C

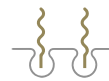
Description

Vitamin C, or L-ascorbic acid, helps to maintain bones, skin, blood vessels, and acts as an antioxidant. It plays a crucial role in controlling infections, healing wounds, and supporting the immune system by stimulating white blood cell activity. Deficiency in this vitamin can lead to oxidative stress that can cause cell damage.

Risk Factors

- Affects individuals of all ages
- Poor diet
- Smoking
- Excessive alcohol consumption

TYPICAL SYMPTOMS



Corkscrew-shaped body hair



Dry and damaged skin, or being easily bruised



Rough and bumpy skin



Spoon-shaped fingernails with red spots/lines

2

RESULT

Risk Markers Detected



Normal Risk

Gene: SLC23A1

Genotype: CG

Gene: SLC23A2

Genotype: GG

RECOMMENDED INTERVENTIONS

Vitamin C plays a vital role in maintaining overall health and is commonly found in various foods. Good sources include broccoli, oranges, papaya, strawberries, and tomatoes. Additional supplementation may be taken if necessary.

VITAMIN D

Description

Vitamin D, or calciferol, is crucial for healthy bone formation, as it regulates calcium and phosphate metabolism. It also aids in wound healing and proper immune function. Vitamin D can be obtained from sun exposure, foods and supplements. Studies suggest that Vitamin D may play a role in the prevention and treatment of various conditions, including type 1 and type 2 diabetes, hypertension, osteoporosis, and multiple sclerosis.

Risk Factors

- Darker skin
- Elderly
- Overweight or obese
- Living far from the equator
- Always using sunscreen when going out
- Staying indoors

TYPICAL SYMPTOMS



Bone pain



Fatigue



Mood changes



Muscle weakness,
aches or cramps

6

RESULT

Risk Markers Detected

Reduced Risk

Gene: CYP2R1 Genotype: TG	Gene: CYP2R1 Genotype: AG	Gene: CYP2R1 Genotype: AG
Gene: CYP2R1 Genotype: AG	Gene: GC Genotype: TG	Gene: NADSYN1 Genotype: GG

RECOMMENDED INTERVENTIONS

In addition to being synthesized through sunlight exposure, Vitamin D can also be obtained from dietary sources such as salmon, eggs, fortified foods, and coconut milk to help prevent deficiency.

VITAMIN E

3

RESULT

Risk Markers Detected



Normal Risk

Gene: APOA5

Genotype: AG

Gene: CYP4F2

Genotype: TC

Gene: SCARB1

Genotype: AA

Description

Vitamin E acts as an antioxidant and is essential for the proper functioning of organs. It helps reduce cellular damage and may play a role in preventing heart disease, diabetes, and cancer. Vitamin E deficiency is caused by irregularities in metabolism and may lead to abnormal liver functions. It can be obtained naturally in various foods.

Risk Factors

- Affects individuals of all ages
- Chronic pancreatitis
- Cystic fibrosis
- Crohn's disease

TYPICAL SYMPTOMS



Immune system problems



Muscle weakness



Numbness and tingling



Vision deterioration

RECOMMENDED INTERVENTIONS

Since Vitamin E must be obtained from the diet, it is important to consume foods rich in this nutrient, such as sunflower seeds, almonds, spinach, olive oil, shrimp, kiwifruit, and trout, to reduce the risk of deficiency.



Whether you are a professional athlete, a leisure gym-goer, or an aspiring Olympian, you can train more efficiently by understanding your genetic strengths and weaknesses. With genetic screening, you can discover which forms of exercise will produce the best results for you, and how your genetics may affect your performance in sports or fitness training. Harness your genetic strengths today to improve your sports and fitness performance, and reach your potential.

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BONE AND JOINT HEALTH: ACHILLES TENDINOPATHY

5

RESULT Risk Markers Detected

Slightly Lower Risk

Gene: CASP8
Genotype: GG

Gene: COL5A1
Genotype: AC

Gene: COL5A1
Genotype: TC

Gene: GDF5
Genotype: AG

Gene: TNC
Genotype: AA

Description

Achilles tendinopathy is a medical condition that causes pain and swelling in the Achilles tendon. Excessive tension on the calf muscles as a result of extended usage or rapid, violent movements tends to increase the occurrence of the condition. Tendon injuries are rather frequent and can vary from simple irritation to rips and ruptures. Achilles tendinopathy is more common in older people and sportsmen. Changes in the genes that regulate the development, composition, and structure of connective tissues can predispose a person to Achilles tendon injury.

TYPICAL SYMPTOMS



Pain in heel



Stiffness in tendon



Swelling at back of ankle



Grating noise when moving ankle

RECOMMENDED INTERVENTIONS

A systematic workout routine aimed at increasing the strength of your Achilles tendon is essential. Incorporate single-leg movement, stability, and balance training into your everyday workout routine. Regular visits to a physiotherapist for muscle relaxation around the ankle and calves are essential for intense athletes. Foods rich in Vitamin C, such as lemon, broccoli, and tomatoes, aid in collagen formation, while adequate calcium and Vitamin D consumption are essential for tendon health.

BONE AND JOINT HEALTH: ANTERIOR CRUCIATE LIGAMENT (ACL) RUPTURE

2

RESULT Risk Markers Detected

Slightly Lower Risk

Gene: ACAN

Genotype: TT

Gene: MMP12

Genotype: TT

Description

The anterior cruciate ligament (ACL) is important in knee joint stabilization, as it keeps the bones around the knee in proper alignment during movement and prevents them from sliding. When a knee is subjected to a sudden and strong impact, particularly during a sudden stop with a change of direction, these ligaments can be damaged, resulting in pain, misalignment, and inflammation, which, if not treated promptly, can lead to structural changes in the cartilages and osteoarthritis. An ACL tear can be a partial or full tear.

TYPICAL SYMPTOMS



Popping sensation in knee



Loss of full range of motion



Severe pain and rapid swelling



Tenderness along the joint line

RECOMMENDED INTERVENTIONS

Medical intervention, including surgery, may be a possibility depending on the severity of the damage. Consider consulting with your doctor about your ACL damage. Otherwise, single leg stability, balance, and mobility for strength training are critical for preventing ACL injuries. Strong hamstrings and glutes have been shown to reduce the occurrence of ACL injuries. Intensive physiotherapy can assist with ACL injury mediation and prevention.

BONE AND JOINT HEALTH: CALCIUM METABOLISM

Description

Calcium helps in regulating blood pressure, promoting blood clotting, maintaining brain functions, increasing insulin sensitivity, and engaging in muscle contraction and relaxation, along with promoting healthy bones and teeth. According to one study, a calcium-rich diet for extended, intense endurance cycling may aid in the prevention of bone resorption. Inadequate weight-bearing activities combined with malnutrition, or over-training with high volume and intensity, may result in reduced blood calcium levels, further reducing bone mineral density and increasing the prevalence of osteoporosis and osteoarthritis.

3

RESULT

Risk Markers Detected



Normal Metabolism

Gene: CYP24A1

Genotype: GG

Gene: Intergenic

Genotype: CC

Gene: CASR

Genotype: AG

TYPICAL SYMPTOMS



Easy fracturing of bones



Muscle spasms



Numbness/tingling in hands, feet and face



Weak and brittle nails

RECOMMENDED INTERVENTIONS

Supplementation with calcium is generally safe, however, do consult your doctor or sports nutritionist first. Consuming food high in calcium, such as broccoli, kale, citrus fruits, dairy products, fish, nuts, eggs can help in boosting your overall calcium level.

BONE AND JOINT HEALTH: OSTEOARTHRITIS

Description

Obesity, type of employment, muscular weakness, past injury, diet, hormonal variables, and hereditary factors are all associated with osteoarthritis (OA), the most frequent disease of the synovial joints in adults. In comparison to osteoporosis, which is the loss of bone mass that increases the risk of fractures, osteoarthritis is a complicated illness that causes joint pain while also reducing joint mobility and function. Tenderness and stiffness around the joints of the fingers, hips, knees, and ankles are common symptoms of OA.

3

RESULT Risk Markers Detected

Low Risk

Gene: MCF2L

Genotype: AA

Gene: GDF5

Genotype: AG

Gene: COL6A4P1

Genotype: TC

TYPICAL SYMPTOMS



Affected joints hurt during or after movement



Bone spurs can form around joint, causing a grating sensation



Loss of flexibility of joint



Swelling caused by soft tissue inflammation around joint

RECOMMENDED INTERVENTIONS

Weight-bearing exercise is essential for preventing OA or relieving its symptoms. Stronger muscles and bones can assist in increasing bone mineral density, which can help postpone the onset of OA. Structural activities that provide stress on the spine can aid in growing total bone mass. Multi-joint exercises are essential for OA treatment. Calcium, magnesium, Vitamin C, D, and K-rich foods are all important in the management of OA.

BONE AND JOINT HEALTH: OSTEOPOROTIC FRACTURES

Description

The loss of bone mineral density is referred to as osteoporosis. Low bone mineral density, often known as porous bones, increases the risk of bone fracture. Until a bone fracture occurs, there are no evident symptoms of osteoporosis. The pathophysiology of osteoporosis is connected to bone tissue metabolism, which includes bone production and resorption. When activity and diet are out of balance, bone health suffers, which can lead to bone stress injuries and early osteoporosis. Both of these can result in morbidity, lost training and competition time.

6

RESULT Risk Markers Detected

Low Risk

Gene: COLEC10

Genotype: AA

Gene: Intergenic

Genotype: AA

Gene: Intergenic

Genotype: CC

Gene: JAG1

Genotype: AG

Gene: RANK

Genotype: AA

Gene: SPTBN1

Genotype: GG

TYPICAL SYMPTOMS



Fragility-related fractures



Loss of height over time



Lower back pain



Stooped posture

RECOMMENDED INTERVENTIONS

If you already have osteoporosis, check with your doctor before beginning any exercise routine. Otherwise, exercise is a tried-and-true method for treating osteoporotic fractures. Concentrate on multi-joint and structural exercises that target your primary muscle groups. Stronger muscles and bones can assist in alleviating osteoporosis discomfort. Calcium, magnesium, Vitamin D, and Vitamin K are essential for osteoporosis.

BONE AND JOINT HEALTH: VITAMIN D

Description

Vitamin D regulates immunological functions, protein synthesis, muscle functions, cardiovascular functions, inflammatory responses, cell proliferation, and musculoskeletal functions. According to studies, most athletes do not get enough Vitamin D, which can lead to an increase in the occurrence of osteoporosis and osteoarthritis. Furthermore, greater Vitamin D serum levels are linked to increased muscular strength, lower injury rates, and improved athletic performance. It is critical to know that natural foods high in Vitamin D are found in animal liver, salmon, tuna, and red meat. Thus, Vitamin D fortification is a common practice in the food sector.

6

RESULT Risk Markers Detected

Optimal Utilization

Gene: GC Genotype: TG	Gene: NADSYN1 Genotype: GG	Gene: CYP2R1 Genotype: TG
Gene: CYP2R1 Genotype: AG	Gene: CYP2R1 Genotype: AG	Gene: CYP2R1 Genotype: AG

TYPICAL SYMPTOMS



Bone pain



Fatigue



Mood changes



Muscle weakness, aches or cramps

RECOMMENDED INTERVENTIONS

Sunlight exposure for at least 15 to 30 minutes per day, numerous times per week, can provide a natural supply of Vitamin D. If your work environment does not allow for direct sunshine exposure, Vitamin D supplementation may be beneficial. To increase Vitamin D intake, consume more red meat, as well as fatty seafood such as tuna and salmon. For vegetarians or vegans, consume extra fortified tofu, cereals, and soy milk to increase total Vitamin D level.

ENERGY METABOLISM AND UTILIZATION: DIETARY CHOLINE TENDENCY

1

RESULT
Risk Markers
Detected



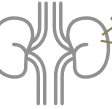
Normal Tendency

Gene: PEMT

Genotype: CC

Description
Choline is a crucial nutrient for physically active people who participate in long periods of intense activity. Choline transfers lipids out of the liver for energy conversion, and it is required for cholesterol clearance from the liver. Choline, together with Vitamin Bs, provides fuel and generates energy for the body. Choline is also required for the formation of acetylcholine, a neurotransmitter in our bodies, and hence plays an important role in improving accuracy, attention, and memory.

TYPICAL SYMPTOMS

-  **Anxiety and restlessness**
-  **Fatty liver**
-  **Muscle damage**
-  **Kidney damage**

RECOMMENDED INTERVENTIONS

Meat, fish, dairy, and eggs are the best sources of choline in everyday diet. Vegetarians/vegans may consider eating more broccoli, cauliflower, soy, and potatoes, as these plant-based foods contain more choline. Beans, nuts, seeds, and whole grains are also advised as staple foods for improving choline consumption in active people.

ENERGY METABOLISM AND UTILIZATION: PHOSPHOROUS METABOLISM

1

RESULT
Risk Markers
Detected



Gene: ALPL

Genotype: TC

Description

Phosphorus is needed to make energy-storing adenosine triphosphate (ATP). It benefits the heart, kidneys, muscles, and nerves. It also aids in bone and joint health by producing and maintaining bone and teeth. Phosphorus deficiency or excess can occur depending on diet and health conditions like diabetes, heavy alcohol consumption, or eating disorders. Research shows that a person's genotype may affect serum phosphorus levels. Unfavorable genotype reduce alkaline phosphatase activity, a specific enzyme that aids in phosphorus metabolism, causing phosphorus deficiency.

TYPICAL SYMPTOMS



Anxiety



Bone pain



Loss of appetite



Stiff joints

RECOMMENDED INTERVENTIONS

A general, well-balanced diet with adequate protein intake can help mitigate phosphorus deficiency. Meat, fish, and organ meat are all phosphorus-rich foods. Legumes, peas, and almonds can supplement vegetarian/vegan diets. Unless one is on a highly restricted diet with medication that interferes with calcium and phosphorus metabolism, phosphorus levels should be more than sufficient for daily activities.

ENERGY METABOLISM AND UTILIZATION: VITAMIN B12

4

RESULT Risk Markers Detected



Normal Utilization

Gene: CUBN

Genotype: AA

Gene: FUT2

Genotype: AA

Gene: FUT2

Genotype: GG

Gene: TCN1

Genotype: AA

Description

Vitamin B12 (cobalamin) is the generic term for a group of cobalt-containing corrinoid compounds that are required for optimal metabolism and function in all animal organs and systems. They also play a role in DNA methylation, a process that causes variation in gene expression. Its role in the creation of red blood cells, maintenance of healthy immunological function, better brain signal transmission, and creatine synthesis are all important for athletic performance.

TYPICAL SYMPTOMS



Pale yellow tinge to skin



Sore and red tongue or mouth ulcers



Pins and needles sensation



Decline in mental abilities

RECOMMENDED INTERVENTIONS

Vegans and vegetarians may be at risk of low Vitamin B12 levels since foods rich in this vitamin are mostly found in animal products. If you are a vegetarian or vegan, you may want to take B-complex vitamins. Eggs, livers, tuna, beef, yogurt, and fortified cereals can be a staple meal for increasing Vitamin B12 levels.

ENERGY METABOLISM AND UTILIZATION: VITAMIN B6

1

RESULT

Risk Markers
Detected

Optimal Utilization

Gene: NBPf3

Genotype: TC

Description

Vitamin B (containing B6, B12, and folate) aids in energy use, amino acid metabolism, red blood cell maintenance, and tissue regeneration. They aid in the breakdown of homocysteine. Plasma homocysteine concentrations rise when these vitamins are in short supply, increasing the risk of cardiovascular diseases. B6 is usually associated with energy generation and stress tolerance. B6 aids in the metabolism of iron from food, resulting in increased iron bioavailability, hemoglobin, and oxygen for the working muscles.

TYPICAL SYMPTOMS



Cracked and sore lips



Skin rashes



Tingling and pain in hands
and feet



Weakened immune function

RECOMMENDED INTERVENTIONS

When dietary intake is inadequate or endurance is challenged, Vitamin B6 supplementation may be beneficial. If you or someone in your family has a history of anemia, consult your doctor. Female athletes may require Vitamin B complex, iron, and folate supplements. Foods rich in B6 include chickpeas, fortified cereals, tuna, and salmon.

ENERGY METABOLISM AND UTILIZATION: VITAMIN B9 - FOLATE TENDENCY

Description
Folate has a range of functions in the body, including acting as a coenzyme in DNA synthesis and energy metabolism. Folate is also involved in the metabolic pathways that regulate the amino acid homocysteine metabolism. The combination of low folate and excessive homocysteine can increase the risk of cardiovascular diseases. According to studies, those who carry an unfavorable risk allele may have low folate levels due to poor Vitamin B9 metabolism.



TYPICAL SYMPTOMS

-  **Anemia**
-  **Fatigue**
-  **Muscle weakness**
-  **Reduced eyesight**

RECOMMENDED INTERVENTIONS

Consultation with your doctor for a serum folate blood test is recommended in order to check if you need a Vitamin B9 or folate supplement. Foods high in folate, such as legumes, greens, fruits, and organ meat, may aid with folate supplementation. Enriched food, such as fortified cereals, can help you obtain more of folate.

MACRONUTRIENT METABOLISM: CARBOHYDRATE UTILIZATION

4

RESULT

Risk Markers
Detected

Low Utilization

Gene: Intergenic

Genotype: CC

Gene: PLIN1

Genotype: TC

Gene: PLIN1

Genotype: AT

Gene: TAS1R2

Genotype: CC

Description

Carbohydrates are essential for good health. They offer the most direct source of energy for our internal processes and physical activities, in the form of glucose. Our brain relies primarily on carbohydrates as its main source of energy. Those who follow a zero-carb or low-carb diet are effectively forcing their bodies to convert glycogen, a form of stored glucose, back into glucose in order to meet their energy needs. However, these diets are not suited for everyone; as some of us utilize carbohydrates more efficiently than others, which may lead to health problems.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

It is critical to understand your overall carbohydrate consumption, depending on your physical activities and lifestyle, in order to perform at your best. Speak to your sports nutritionist for advice. To replenish your total energy reserve, eat quick carbs (such as rice) after your workout. To achieve extended satiety, eat slow carbs (such as sweet potatoes) during other meals.

MACRONUTRIENT METABOLISM: CHOLESTEROL RESPONSE TO DIETARY FAT

Description

Saturated, monounsaturated, polyunsaturated, and trans fats are all essential dietary fats that regulate your body's daily functions and energy production. It is a contentious topic in health and fitness since some research shows that eating more dietary fat raises total cholesterol levels. However, a new study reveals that some people may have a genetic predisposition to the harmful consequences of excessive dietary fat consumption, while others may benefit from it. When a person of this variant type consumes a lower fat diet, their total cholesterol and low-density lipoproteins (LDL) levels decrease.

O

RESULT

Risk Markers
Detected



Favorable Response

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

In general, saturated fats should not account for more than 10% of your total daily calories from fat. Increase your consumption of plant-based fats and fatty fish, which are high in monounsaturated and polyunsaturated fats, as they help in regulating cholesterol levels. Avoid trans fat since it has been related to a rise in bad cholesterol levels, which can contribute to heart problems.

MACRONUTRIENT METABOLISM: FAT UTILIZATION

3

RESULT

Risk Markers Detected



Normal Utilization

Gene: APOA5

Genotype: AG

Gene: PPARG

Genotype: CC

Gene: PPM1K

Genotype: CC

Description

Fat is needed for energy, cell and tissue protection, hormone coordination, and nutrition metabolism. Most people think that fat is harmful and will adopt a low-fat or fat-restricted diet. However, this can cause hormone imbalance, reduced immunity, and nutrient malabsorption. Some athletes may also adopt a high-fat and high-protein diet, like the ketogenic or paleo diet, to optimize their body composition. Yet, not all genotypes use fat equally and some people may even gain weight on a high-fat diet, according to research.

TYPICAL SYMPTOMS



Dry skin



Itchy eyes



Constant chilliness



Dry hair and hair loss

RECOMMENDED INTERVENTIONS

Consult a doctor or sports nutritionist to determine your daily fat needs. A diet strong in healthy fats, like the Mediterranean diet, may support weight loss, heart health, and blood pressure regulation. Good sources of healthy fats include fatty food such as salmon and tuna, nuts, olives and oils, dark chocolate, eggs, and seeds.

MACRONUTRIENT METABOLISM: INSULIN RESPONSE TO DIETARY FAT

0

RESULT
Risk Markers
Detected


Favorable Response

Description

Insulin is commonly associated with sugar and carbohydrates. On the other hand, dietary fat also induces insulin response and can lead to insulin resistance and fat accumulation; which can lead to health problems. A range of variables can influence insulin response, including the types and quantities of fat, carbohydrates and proteins ingested, the quantity and intensity of physical activity, stress, and resting levels. Low-fat diets have been demonstrated to aid in maintaining optimal insulin sensitivity in certain people, but they may not be suitable for everyone, as a person's genotype may indicate how insulin sensitivity responds to dietary fat consumption. Insulin sensitivity is a measure of a person's ability to avoid or improve insulin resistance. The more responsive your insulin sensitivity is, the better your body can utilize fat as energy, losing weight.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Insulin sensitivity can be addressed by eating a balanced diet rich in omega-3 and 6 unsaturated fats.

MACRONUTRIENT METABOLISM: POLYUNSATURATED FAT METABOLISM

3

RESULT

Risk Markers
Detected



Normal Metabolism

Gene: FADS1

Genotype: TT

Gene: FADS1

Genotype: GG

Gene: FADS2

Genotype: GG

Description

Polyunsaturated fatty acids (PUFA) are fatty acids that have two or more double bonds in their structure. PUFAs are phospholipids found in cell membranes that have a role in cholesterol metabolism. Omega-3 polyunsaturated fatty acids (PUFAs) including alpha-linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA) have been shown to lessen overall inflammation in our bodies. According to research, PUFA supplementation may aid in the recovery of eccentric movement after exercise. We do not process PUFA in the same way. Some people may have stronger enzyme activity, allowing for faster PUFA metabolism, whilst others may not digest PUFA as quickly.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Fish-rich diets are high in EPA and DHA, while vegetarians and vegans can choose plant-based oils such as flaxseed or soybean oils. If you think you might require a fish oil supplement, consider consulting a doctor.

MACRONUTRIENT METABOLISM: PROTEIN UTILIZATION

0

RESULT
Risk Markers
Detected

Optimal Utilization

Description

Protein is an essential macronutrient for our bodies' crucial processes. Proteins, in the form of enzymes, hormones, or nutrients, assist in forming new muscles, repairing and rebuilding tissues, and coordinating physiological activities. Different people may require different amounts of protein, depending on their everyday life. As protein is necessary for muscle development and repair after training and workouts, athletes require more than an ordinary person. A high-protein diet is also used by many active athletes to increase their lean body tissues and reduce fat. However, the effectiveness of the method may be substantially determined by a person's DNA profile. Some people with the appropriate genetic combination may lose more weight by eating more protein, as their bodies use protein more efficiently.

TYPICAL SYMPTOMS



Increased risk of bone fractures



Loss of muscle mass or muscle weakness



Low blood pressure and low heart rate



Weight loss

RECOMMENDED INTERVENTIONS

Consulting a sport nutrition specialist or dietitian may be beneficial in figuring out your daily protein needs.

MACRONUTRIENT METABOLISM: VITAMIN E

3

RESULT Risk Markers Detected



Normal Metabolism

Gene: APOA5

Genotype: AG

Gene: CYP4F2

Genotype: TC

Gene: SCARB1

Genotype: AA

Description

Vitamin E is a fat-soluble vitamin that is required for good health. Vitamin E acts as an antioxidant, inhibiting lipid peroxidation in cell membranes, preventing oxidative DNA damage, lowering carcinogen formation, and reducing inflammation after intense exercise. Vitamin E levels in the blood have been linked to central obesity, and it is also linked to maintaining adequate immunological function and blood viscosity. Vitamin E supplementation has also been demonstrated in a study to help remove lactate quicker during long-distance running at high altitudes.

TYPICAL SYMPTOMS



Immune system problems



Muscle weakness



Numbness and tingling



Vision deterioration

RECOMMENDED INTERVENTIONS

It is suggested that high-performance athletes get a blood test and consult with their physician and sports nutritionist about supplements. Vitamin E is abundant in plant-based oils, such as seeds, nuts, and colorful vegetables including pumpkin, carrots, and bell peppers.

MENTAL ABILITY AND PERFORMANCE: IMPULSE CONTROL AND TASTE PREFERENCE

0

RESULT
Risk Markers
Detected

High Tendency

Description

It takes self-control to eat a healthy, portion-controlled diet. While some people are conscious of their impulsivity, the vast majority of people tend to eat without consideration. According to studies, fatty foods excite the brain's pleasure centers, making it more difficult to resist the impulse to consume more. Those with a strong willpower to resist temptation and craving may have genes that help them do so. According to studies, some may have higher genetic inclination to a greater desire for excitement and a preference for stronger tasting foods. These individuals may have a preference for high-calorie, fatty meals and are prone to overeating, placing them at risk for obesity.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Mindful eating and alternative eating strategies can be powerful tools to circumvent the craving to eat more. Consider talking to an exercise professional, nutritionist or dietitian for improving your eating strategies.

MENTAL ABILITY AND PERFORMANCE: INTRINSIC MOTIVATION TO EXERCISE

0

RESULT
Risk Markers
Detected

High Motivation

Description

Personal drive or intrinsic motivation is essential for enjoying sports or exercises without relying on outside motivation. Intrinsic motivation occurs when an individual finds an activity rewarding or pleasant on an intrinsic level. Intrinsic drive may be inherent in the genetic makeup of a person. They push themselves because they want to see how far they can go. They appreciate sports and fitness activities because of how they feel afterward, in addition to all the other measurable benefits. According to research, people with certain genetic composition are more likely to push themselves to achieve better, higher, or longer goals.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Those with a low level of intrinsic motivation might break down their fitness or sports goals into smaller, more manageable milestones. Participating in sporting events such as marathons and football matches might assist in boosting the motivation to achieve greater fitness objectives.

MENTAL ABILITY AND PERFORMANCE: SLEEP DURATION

1

RESULT

Risk Markers Detected



Normal Duration

Gene: ABCC9

Genotype: GG

Description

Sleep is essential for both physical and mental health. Sleep is necessary for immunity, metabolism, learning, memory, and other vital functions. Although most people associate sleep with characteristics like age, gender, lifestyle, diet, caffeine, alcohol use, overall health, or stress level, studies suggest that a person's genetic composition may also influence their total sleep length. Having a genetic combination that favors extended sleep duration means that the person can recover more quickly from daily oxidative stress and metabolic demands.

TYPICAL SYMPTOMS



Memory loss



Irritability



Mood swings



Difficulty staying focused

RECOMMENDED INTERVENTIONS

Aside from visiting a sleep specialist about an underlying sleep problem, you can try to enhance your sleep by lowering general caffeine and alcohol intake, setting a dim, cool, and pleasant sleeping environment, or limiting the use of devices before bed.

MENTAL ABILITY AND PERFORMANCE: STRESS SENSITIVITY

2

RESULT Risk Markers Detected

Low Sensitivity

Gene: TPH2

Genotype: TT

Gene: OXTR

Genotype: AA

Description

Anxiety and stress are prevalent emotional states among athletes at all levels of competition. Response to stressful sports-related circumstance, that the individual sees as potentially stressful, which results in a variety of cognitive evaluations, behavioral responses, and/or physiological arousals is classified as sport-related anxiety. A certain amount of stress is necessary to improve performance in sports and training, but too much stress might cause cognitive incoherence and performance decline. Sports-related stress can cause a variety of reactions, such as anxiousness before a game, stumbling through coordination during a game, or a lack of communication and collaboration in team sports.

TYPICAL SYMPTOMS



Chest pain or heart racing sensation



Panic attacks, anxiety or irritability



Exhausted or have trouble sleeping



High blood pressure or muscle tension

RECOMMENDED INTERVENTIONS

Self-awareness of your emotional states is critical for stress management. If you require systematic and appropriate structure in the management of stress in any form, seek expert assistance. Magnesium-rich foods, such as dark chocolate, bananas, and pumpkin seeds, may help with stress management. Yoga or meditation may also be beneficial.

MUSCLE BUILDING CAPACITY: TESTOSTERONE

1

RESULT Risk Markers Detected



Normal Level

Gene: Intergenic

Genotype: TT

Description

The testes and adrenal glands produce testosterone, a steroid hormone that is important for assessing muscle size, strength, and function, as well as maintaining reduced body fat levels. Low testosterone levels can impair your health and have been linked to heart disease, metabolic syndrome, type 2 diabetes, osteoporosis, muscle loss, and increased mortality risk. Although lifestyle factors like excessive alcohol intake, smoking, and being overweight reduce testosterone, research also suggests that testosterone levels are primarily genetic.

TYPICAL SYMPTOMS



Low sex drive



Bad mood and irritability



Fatigue and inability to focus



Muscle loss and decrease in strength

RECOMMENDED INTERVENTIONS

Keeping a healthy weight and BMI, engaging in weighted resistance training, or being more physically active are all proven methods for maintaining testosterone levels. Shellfish, chia seeds, pumpkin seeds and dairy foods (e.g. cheese and yogurt) may naturally improve your testosterone levels.

MUSCLE BUILDING CAPACITY: ZINC METABOLISM

1

RESULT

Risk Markers
Detected



Normal Metabolism

Gene: LINC01420

Genotype: CC

Description

Zinc is an essential trace element that plays a key role in immune function, protein synthesis, wound healing, reproduction, and growth. Zinc is also essential for testosterone production. Zinc deficiency hinders immune function and has been associated with cardiovascular disease and diabetes. Vegetarians may have a higher risk of zinc deficiency since red meat and poultry are readily available sources of zinc, while zinc from plant sources is less likely to be absorbed. Genetic variants may influence your natural serum zinc levels.

TYPICAL SYMPTOMS



High zinc levels may cause toxicity - resulting in nausea, vomiting, loss of appetite and headaches



Low zinc serum levels may cause low libido, hair loss, loss of appetite and irritability

RECOMMENDED INTERVENTIONS

Oysters, carbs, shrimps, meats, beans, nuts, legumes, and fortified grains are all great sources of zinc. Consider zinc supplementation if you are a vegetarian/vegan or elderly.

MUSCLE PERFORMANCE: COPPER METABOLISM

1

RESULT Risk Markers Detected


Normal Metabolism

Gene: SELENBP1

Genotype: TT

Description

Iron absorption, red blood cell synthesis, immunity, energy generation, bone, connective tissue, neuron, and blood vessel functions all require copper. Low copper levels can cause heart disease, bone and joint difficulties, as well as weakened immune system. Although rare, copper poisoning can be caused by the genetically heritable Wilson's disease; copper insufficiency on the other hand, is much more common and can be caused by too much zinc supplementation, dietary inadequacies, or genetic variations.

TYPICAL SYMPTOMS



Anemia



Irregular heartbeat



Osteoporosis and bone fractures



Thyroid problems

RECOMMENDED INTERVENTIONS

Taking multivitamins with a trace amount of copper can help with low copper levels. Otherwise, a person's supplement usage should be discussed with their doctor, since zinc, iron, and Vitamin C supplements may decrease copper absorption. Copper-rich foods include shellfish, tree nuts, mushrooms, legumes, and dark green leafy vegetables, which can be consumed daily.

MUSCLE PERFORMANCE: IRON METABOLISM

2

RESULT Risk Markers Detected

Normal Metabolism

Gene: HFE

Genotype: CC

Gene: Tmprss6

Genotype: AG

Description

Iron is an energy-producing nutrient that is essential for making hemoglobin, a protein in red blood cells that helps carry oxygen for energy metabolism. Iron overload is linked to insulin resistance, diabetes, heart disease, Alzheimer's disease, and cancer, and it can also induce hemochromatosis. Diet, gender, age, and daily activities may affect iron levels; while genetic variances can affect how much iron the body absorbs and recycles.

TYPICAL SYMPTOMS



Iron overload may cause abdominal pain, weight loss and hyperpigmentation



Low iron can cause iron deficiency anemia, which reduces energy output

RECOMMENDED INTERVENTIONS

Plant-based protein may have less heme iron, which helps reduce high iron levels. Organ meat is a useful source of iron for those with low iron levels. Female athletes may need iron supplements due to menstrual blood loss.

MUSCLE PERFORMANCE: MAGNESIUM METABOLISM

1

RESULT Risk Markers Detected



Normal Metabolism

Gene: MUC1

Genotype: TC

Description

Magnesium is important for blood sugar regulation, muscular contraction and relaxation, and heart rhythm. Its insufficiency has been linked to type 2 diabetes and high blood pressure. Though low magnesium is usually caused by a lack of magnesium in the diet, large intakes of alcohol, soda, and caffeine, as well as taking drugs that interfere with its absorption, can cause magnesium levels to drop.

TYPICAL SYMPTOMS



Fatigue



Mood swings



Muscular cramps



Nausea

RECOMMENDED INTERVENTIONS

Dark chocolate, dark leafy greens, nuts and seeds, fatty fish, avocado, legumes, whole grains, yogurt, soy foods, and bananas are all good sources of magnesium minerals. Be mindful of consumption of alcoholic drinks and carbonated beverages as they may cause low magnesium levels.

OXIDATIVE STRESS METABOLISM: ANTIOXIDANTS

Description

Reactive oxygen species (ROS), which are produced as a result of cell metabolism, affect cell structure and function, contributing to muscle injury, immunological dysfunction, and weariness. Exercise-induced ROS may be helpful because they stimulate glycogen resynthesis, reduce infection susceptibility, and may even improve athletic performance by initiating and promoting adaptive responses to training. The amount to which reactive species are harmful or beneficial is determined by the individual's activity duration, intensity, fitness qualities, and nutritional state. Antioxidant supplements have been demonstrated to be an effective countermeasure to exercise-induced oxidative stress and muscle injury in studies.

5

RESULT

Risk Markers Detected

Reduced Response

Gene: SOD3

Genotype: CC

Gene: GPX4

Genotype: TT

Gene: SOD2

Genotype: AA

Gene: ATM

Genotype: AG

Gene: CAT

Genotype: TT

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Consult your doctor about antioxidant supplements, since there are a variety of options available, including Vitamins A, B, C, E, etc. Including a wide range of colorful vegetables in your diet can assist in enhancing your overall antioxidant levels.

OXIDATIVE STRESS METABOLISM: COENZYME Q10

1

RESULT

Risk Markers
Detected

Normal Level

Gene: NQO1

Genotype: AG

Description

Coenzyme Q10 (CoQ10) can affect cellular bioenergetics and protect cells from oxidative stress induced by free radicals. CoQ10 is a fat-soluble, vitamin-like molecule found in all cells. It plays a key function in the mitochondrial oxidative pathway, ATP synthesis, antioxidant regeneration, regulating the stability, fluidity, and permeability of cell membranes, encouraging cell growth, and suppressing cell death. Supplementing with CoQ10 may aid in improving aerobic power, anaerobic capacity, and overall recovery. Aged athletes or individuals may also benefit from CoQ10 supplements for overall health and performance.

TYPICAL SYMPTOMS



Decreased ability to handle stress



Extreme physical fatigue



Memory lapses



Mood swings

RECOMMENDED INTERVENTIONS

Foods rich in CoQ10 include organ meat, cattle, chicken, mackerel, and sardines. To increase their CoQ10 levels, vegetarians and vegans can eat more spinach, broccoli, cauliflower, legumes, nuts, and seeds. Adding oils to your meals (olive oils, cooking oils) might help enhance total CoQ10 absorption.

OXIDATIVE STRESS METABOLISM: VITAMIN A

1

RESULT

Risk Markers
Detected

Reduced Risk

Gene: FFAR4

Genotype: TT

Description

Vitamin A is well-known for its function in eye health, but it is also a powerful antioxidant that fights free radicals created by oxidative stress, and has been linked to bone health. Vitamin A is present in the body as retinol, retinal, and retinoic acid. Vitamin A is predominantly stored in the liver, kidney, and adipose tissue as long-chain fatty esters and provitamin carotenoids. Because Vitamin A may autoxidize as oxygen tension rises, it is most effective as an antioxidant at low oxygen tensions, which are typical of physiological tissue values. However, taking too much Vitamin A might be harmful.

TYPICAL SYMPTOMS



Dry skin and eyes



Night blindness



Delayed growth



Poor wound healing

RECOMMENDED INTERVENTIONS

Vitamin A supplements should be discussed with your doctor or sports nutritionist. Vitamin A supplementation should not be self-prescribed because it might have dangerous negative effects. Increase your intake of meat, fatty fish, eggs, red bell peppers, sweet potatoes, and carrots for a natural supply of Vitamin A.

OXIDATIVE STRESS METABOLISM: VITAMIN C

2

RESULT Risk Markers Detected

Normal Risk

Gene: SLC23A1

Genotype: CG

Gene: SLC23A2

Genotype: GG

Description

Vitamin C is necessary for immunological function, collagen synthesis, and cortisol production, as well as the removal of free radical species that cause cell damage. Vitamin C supplementation may improve athletic performance by lowering the harmful effects of free radical species. It has been hypothesized that athletes undergoing intense training and competition are unable to maintain appropriate vitamin tissue levels, even if they ingest the recommended daily doses. Vitamin C supplementation has been demonstrated to improve aerobic threshold in studies.

TYPICAL SYMPTOMS



Corkscrew-shaped body hair



Dry and damaged skin, or being easily bruised



Rough and bumpy skin



Spoon-shaped fingernails with red spots/lines

RECOMMENDED INTERVENTIONS

Increase your intake of citrus fruits including oranges, kiwis, lemons, grapefruits, and other vegetables such as bell peppers, strawberries, tomatoes, broccoli, cabbage, and cauliflower, which are all natural sources of Vitamin C. Vitamin C supplementation can help, but bear in mind that too much Vitamin C might cause nausea and diarrhea.

RESPONSE TO CARDIOVASCULAR TRAINING: EXERCISE HEART RATE RESPONSE

1

RESULT

Risk Markers
Detected



Normal Response

Gene: CREB1

Genotype: AG

Description

The fundamental function of the heart is to transport oxygenated and deoxygenated blood for vital functions throughout the body. Your heart beats quicker while you exercise to keep your working muscles supplied with oxygen. Your body becomes more effective at using oxygen as you get fitter and thus heart rate shall not suddenly increase. A healthier person may have a lower resting heart rate, as the heart can pump more oxygenated blood with fewer beats per minute. This indicates that heart health has improved, as well as endurance and exercise capacity.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

How much your workout heart rate responds to endurance training is mostly determined by your genetics. Before starting targeted heart rate training, talk to your doctor to make sure you do not have any underlying cardiovascular issues. An organized moderate and high-intensity interval training with a targeted heart rate training can systematically increase your heart rate response to cardio exercise.

RESPONSE TO CARDIOVASCULAR TRAINING: FAT LOSS RESPONSE TO CARDIO

1

RESULT

Risk Markers
Detected

Optimal Response

Gene: LPL

Genotype: TT

Description

Fat loss occurs when our body uses stored fat as the primary energy source. To use stored fat as an energy source, our bodies must first deplete the readily available glucose in our blood and muscles. More rigorous endurance exercise or high-intensity interval training can help tap into your fat reserves more quickly. During cardio exercise, different genotypes may alter how rapidly one may reach the fat fuel supply. Some people with favorable genotypes may lose fat more quickly with cardio exercise, while others may need to push themselves for more or longer higher-intensity cardio sessions to see results.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

According to the American Heart Association, a weekly cardio workout of at least 150 minutes of moderate effort, or 75 minutes of strong intensity, is recommended for heart and overall health. Consult an exercise professional if you want a more structured cardio training session based on your genotype.

RESPONSE TO CARDIOVASCULAR TRAINING: FITNESS RESPONSE TO CARDIO

3

RESULT

Risk Markers
Detected

High Response

Gene: AMPD1

Genotype: GG

Gene: NRF1

Genotype: AA

Gene: PPARGC1A

Genotype: TC

Description

The fitness response to cardio exercise assesses the likelihood of improving fitness as a result of engaging in cardiovascular activity. Fitness implies the body is more effective at utilizing oxygen, allowing people to push harder in their sports without becoming exhausted. Being able to withstand more activity implies burning more calories, which helps in losing weight. The more fit a person is, the more oxygen they can take in, and the harder and longer they can exercise without being exhausted. The VO2 max is used to assess a person's fitness. The more fit a person is, the more oxygen they can take in, and the harder and longer they can exercise without being exhausted, resulting in a higher VO2 max.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Improvement of VO2 max is determined by characteristics such as a person's size, gender, and DNA. A mix of moderate to high-intensity endurance exercise and the appropriate amount of resistance training can considerably improve your VO2 max. Consult an exercise specialist for advice on how to effectively train and enhance your VO2 Max.

RESPONSE TO CARDIOVASCULAR TRAINING: GLUCOSE RESPONSE TO CARDIO

1

RESULT

Risk Markers
Detected



Normal Response

Gene: PPARG

Genotype: CC

Description

Our major source of energy is glucose, which is produced by the breakdown of carbohydrates in the food. Insulin resistance and diabetes may occur if blood glucose levels increase and remain high. Muscles utilise stored glucose as energy, and the blood glucose supply is then accessed to replace the muscular reserves, which helps to control blood glucose levels. According to research, those with more advantageous genotype have quicker blood glucose clearance, whereas those with less favorable genotype have a slower response, or a lower reduction in glucose levels.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Incorporate resistance and strength training into your routine to boost your glucose clearance as more muscle cells equals quicker glucose consumption. Structure your workout routine to push yourself harder anytime you reach a comfort zone, since increased intensity burns glucose more quickly. Consume more fiber rich and low glycemic index food to help boost satiety and insulin sensitivity.

RESPONSE TO CARDIOVASCULAR TRAINING: HDL RESPONSE TO CARDIO

Description

High-density lipoprotein (HDL), or 'good cholesterol' is necessary for transporting cholesterol to the liver and assisting in its removal from the bloodstream. Excess cholesterol in the blood can contribute to plaque formation, which can lead to heart disease. As a result, having higher HDL levels is advantageous. Studies show that regular exercisers have higher HDL serum levels. Although training regimes, structures of training, length, intensity, and number of sessions all affect HDL serum levels post-training, individuals with the more 'favorable' genotype observed higher-than-average increases in HDL levels, whereas those with 'unfavorable' genotype saw a lesser response in HDL levels increment.

5

RESULT Risk Markers Detected



Normal Response

Gene: ABCA1

Genotype: TC

Gene: CETP

Genotype: CC

Gene: CETP

Genotype: CC

Gene: LIPC

Genotype: CC

Gene: LIPC

Genotype: GG

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Pushing yourself above your comfort zone or remaining longer in the burning zone during a cardio workout will help improve your HDL serum levels in general. A fiber-rich, or plant-based diet high in natural healthy fats, can aid in the management of HDL blood levels and total cholesterol.

RESPONSE TO CARDIOVASCULAR TRAINING: INSULIN SENSITIVITY RESPONSE TO CARDIO

2

RESULT Risk Markers Detected

Normal Response

Gene: PPARG

Genotype: CC

Gene: LIPC

Genotype: CC

Description

Insulin is a hormone that transports glucose, a major source of energy, throughout our bodies through the circulatory system. Cells in healthy individuals are sensitive to insulin activity, and blood glucose levels are regulated within a healthy range. Insulin resistance develops as insulin sensitivity diminishes. This raises blood glucose levels, which may lead to diabetes. Exercise improves glucose absorption in our muscle cells, which helps to increase insulin sensitivity. By boosting glucose consumption, exercise may help prevent diabetes and increase fat reduction. According to research, individuals with the more advantageous genotype have greater gains in insulin sensitivity from the same amount of exercise, whereas those with the less favorable genotype saw less change.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Increased workout sessions will undoubtedly enhance insulin sensitivity while combining strength and endurance training is found beneficial. Exercising in the correct heart rate zone and for an optimal duration can further enhance insulin sensitivity.

RESPONSE TO CARDIOVASCULAR TRAINING: TRIGLYCERIDES RESPONSE TO CARDIO

4

RESULT

Risk Markers Detected



Normal Response

Gene: APOA5

Genotype: AG

Gene: GALNT2

Genotype: GG

Gene: GCKR

Genotype: TT

Gene: ZPR1

Genotype: GC

Description

Triglycerides are a form of fat which circulate in our body and are used as energy fuel. When a person consumes more triglycerides than they burn, the excess may cause arterial blockage and hardening, leading to heart disease. Excess triglycerides may also be stored as fat, resulting in weight gain. One of the most effective strategies to decrease triglycerides is to engage in regular aerobic activity. According to studies, a person's genetic profile determines how their triglycerides react to cardio exercise i.e. those with more favorable genetic variants have larger triglyceride reductions, while those with less favorable variants may experience smaller reductions or even increased triglyceride levels after exercise.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Increasing fiber intake and consuming a diet rich in healthy fats such as olives and avocados can help lower triglyceride levels. Regardless of genetic variations, increased aerobic exercise can help lower bad cholesterol levels and should remain a priority in daily lifestyle.

RESPONSE TO RESISTANCE TRAINING: BODY COMPOSITION

Description

Body composition refers to bones, water, fat mass, and lean tissues that comprise the human body. The amount of body fat a person possesses in proportion to muscles is often the focus of training and sports performance. Maintaining a healthy body composition is beneficial, since it can help reduce the risk of cardiovascular disease, diabetes, and cancers. Resistance exercise aids in the development and maintenance of lean muscle tissue, lowering total body fat percentage. In most sports, a person's overall body composition improves their strength and speed. Many elements, such as major muscle fibers, hormones, and the type of strength training performed, as well as genetic combinations, will impact how various body compositions adapt to a resistance training program. Those with stubborn genetic variants may benefit less from resistance exercise, while others get instant results.

6

RESULT Risk Markers Detected

Optimal Response

Gene: ETV5 Genotype: TT	Gene: GNPDA2 Genotype: AG	Gene: LINGO2 Genotype: AA
Gene: MAP2K5 Genotype: TC	Gene: NEGR1 Genotype: TC	Gene: PVRL2 Genotype: GG

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

For best body composition results, consult a fitness expert regarding suitable resistance training amount and intensity.

RESPONSE TO SPORTS DRINKS: CAFFEINE METABOLISM

1

RESULT Risk Markers Detected

 **Normal Metabolism**

Gene: CYP1A2

Genotype: AA

Description

Caffeine is acclaimed as the finest energy stimulant for athletic performance and training, whether it comes from coffee, tea, or is added as an ergogenic aid in sports beverages. To obtain these benefits, current guidelines indicate consuming low-to-moderate dosages of caffeine, ranging from 3 to 6 mg/kg, 60 minutes before exercise. Caffeine is not digested in the same way by everyone; some of us metabolize it faster, resulting in an increase in energy production, while others, owing to genetic variations, may have heart palpitations and sleeplessness.

TYPICAL SYMPTOMS



Headache



Insomnia



Nervousness/anxiousness



Palpitations

RECOMMENDED INTERVENTIONS

It is suggested that caffeine to be avoided on a regular basis or that decaffeinated sports drinks or beverages to be consumed by lower caffeine tolerance individuals. Caffeine intake should not exceed 400 mg per day, which is around 4 cups of 250 mL black coffee for those with moderate or higher metabolism levels.

RESPONSE TO SPORTS DRINKS: LACTOSE METABOLISM

2

RESULT Risk Markers Detected

Low Metabolism

Gene: MCM6

Genotype: CC

Gene: MCM6

Genotype: GG

Description

Dairy products, along with the use of whey protein, have been marketed as a beverage to aid in muscle synthesis, recovery, and regular consumption by people from all walks of life, including athletes, busy working adults, and children who want to grow taller and bigger. Lactose intolerance, on the other hand, can make dairy products and whey protein drinks harmful to certain people. Technology has aided in the separation of whey from the general lactose-containing makeup, resulting in 'whey isolate' for sports nutrition.

TYPICAL SYMPTOMS



Diarrhea



Nausea



Stomach cramps



Flatulence

RECOMMENDED INTERVENTIONS

Lactose-intolerant individuals can use whey protein isolate or non-whey proteins such as pea, rice, or soy protein as alternatives to traditional sports drinks. Dairy products should be avoided or consumed in moderation in your daily diet.

SPORTS-INDUCED INJURY RISK: CONCUSSION

Description

Concussions are brain injuries produced by physical trauma to the head as a result of a severe activity or accident. Concussions are frequent among athletes who participate in contact sports, including rugby, football, boxing, and ice hockey. They are vulnerable to recurrent head injuries, which, if not given the time to recover, can have cumulative effects on the brain, resulting in brain damage and difficulties. According to research, genetic variations may affect how quickly individuals recover from concussions. Higher risk allele expression may imply a delayed recovery from concussion damage.

3

RESULT

Risk Markers
Detected



Normal Risk

Gene: APOE	Gene: APOE	Gene: APOE
Genotype: TG	Genotype: CC	Genotype: TT

TYPICAL SYMPTOMS



Amnesia regarding
traumatic event



Blurry vision



Dizziness



Ringling in the ears

RECOMMENDED INTERVENTIONS

It is recommended to combine agility and change of direction training. Multi-directional movement training can help individuals respond to sudden changes in direction during sporting events. Weight-bearing and strength training are also essential in improving injury prevention. Foods rich in Vitamin C, antioxidants, calcium, and magnesium can aid in the repair of brain injuries.

SPORTS-INDUCED INJURY RISK: SOFT TISSUE INJURY

4

RESULT Risk Markers Detected

Reduced Risk

Gene: CCL2 Genotype: CC	Gene: COL5A1 Genotype: TC	Gene: IGF2 Genotype: CG
Gene: IGF2 Genotype: TG		

Description

Musculoskeletal soft tissue injuries are frequent in athletes, particularly those who participate in vigorous or contact sports. However, these injuries impact the general population as they are caused by a mix of genetic and environmental factors that may lead to long-term health problems. The majority of soft tissue injuries occur during eccentric muscular contractions, which are distinguished by greater muscle lengthening and tension. This can damage muscle fibers, resulting in weakness, soreness, reduced range of motion, and an increase in muscle proteins in the blood.

TYPICAL SYMPTOMS



Lump at site of injury



Muscle cramping or weakness



Pain and swelling



Visible bruising

RECOMMENDED INTERVENTIONS

Weight bearing and strength training are essential for strengthening muscles and tissues. Regular exercise, whether aerobic or strength-based, can assist in improving muscular resilience to injury. Consuming foods rich in calcium, magnesium, polyunsaturated fatty acids, and vitamins C, D, E, and K may help reduce soft tissue inflammation.

TRAINING MODES AND ENERGY: ENDURANCE CAPACITY

Description

Endurance training is a 20-minute or longer session of exercise or physical activity that relies only on the production of aerobic energy. Individual with higher endurance capacity can engage in physical activity for longer periods of time without experiencing muscle weakness or tiredness as their bodies can withstand a higher lactic acid build-up. According to research, some people are genetically predisposed to excel at endurance training, while others may need to work harder or put in more effort to achieve the same results.

5

RESULT Risk Markers Detected

High Capacity

Gene: ACSL1

Genotype: AG

Gene: COL6A1

Genotype: CC

Gene: KDR

Genotype: TA

Gene: PPARGC1A

Genotype: TC

Gene: TFAM

Genotype: GG

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

A systematic training regime can help to improve your endurance potential, similar to losing weight and gaining muscles. To improve your maximum endurance potential, gradually push your aerobic capacity by increasing the length of your sessions or engage in high-intensity interval training to improve your overall recovery time.

TRAINING MODES AND ENERGY: ENERGY UTILIZATION

Description

Adenosine triphosphate (ATP) is required for biological functions in the human body. Physical exertion is dependent on ATP reserves, which are constantly changing as a result of metabolic turnover. When energy demands increase, such as during exercise, free ATP breaks down to provide the needed energy. According to research, not everyone uses energy in the same manner. Some people may be able to use energy more effectively, allowing them to perform better in sports and training. Those who use energy inefficiently, on the other hand, may become weary more easily during periods of high energy usage.

2 RESULT Risk Markers Detected

Normal Efficiency

Gene: IGF1

Genotype: CC

Gene: PPARG

Genotype: CC

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Although our bodies use energy in different ways, depending on our genetics, regular exercise has been shown to enhance our total energy usage. Your overall energy uptake will be improved by a structured program targeted to boost your aerobic and anaerobic capability. Exercising in a semi-fasted state may be beneficial, depending on whether your body favors carbohydrate or fat metabolism.

TRAINING MODES AND ENERGY: EXERCISE INTENSITY- POWER CAPACITY

Description

Power training primarily utilizes anaerobic energy production that employs creatine monophosphate (CP) and adenosine triphosphate (ATP) as energy fuel during an intense bout of exercise/physical activity that lasts between 10 and 60 seconds. Research shows that your muscle fiber type, fast and slow muscle fiber proportions, and genetic variations may influence your exercise intensity for power training. Individuals with higher power capacity have more capacity to perform powerful muscle contractions for a longer amount of time, while individuals with lower power capacity may fatigue faster during power movement.

4

RESULT

Risk Markers Detected

High Capacity

Gene: UCP2

Genotype: TC

Gene: UCP2

Genotype: GG

Gene: ACTN3

Genotype: TT

Gene: AGT

Genotype: AG

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

While natural improvement in power capacity may be limited, a structured training program focused on exercise intensity can help condition muscles for faster recovery and increased performance. A structured high intensity interval training (HIIT) is also recommended to gradually improve your exercise intensity.

WEIGHT MANAGEMENT: WEIGHT AND SATIETY

0

RESULT
Risk Markers
Detected


Normal Tendency

Description

Satiety is induced by the expansion of your stomach, and the interplay of the 'satiety hormones' leptin and ghrelin. These hormones regulate hunger and appetite during digestion, with leptin suppressing hunger and ghrelin enhancing appetite. All of these signals are combined in the brain and processed to indicate fullness and satisfaction. Your genes also influence how effectively your body creates and reacts to satiety signals. Your genotype may interact to cause greater or lower satiety after a meal, causing hunger or appetite. All of these factors will have an impact on whether you lose or gain weight.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Even though genetics may play a role in satiety and hunger, managing portion sizes can help reduce sensitivity to food triggers. Making wiser and healthier meal choices or substitutions may also help you stay on track for the duration of your exercise and fitness journey.

WEIGHT MANAGEMENT: WEIGHT LOSS METABOLISM

3

RESULT

Risk Markers
Detected



Normal Metabolism

Gene: LEPR

Genotype: CG

Gene: LEPR

Genotype: AG

Gene: LEPR

Genotype: AA

Description

Metabolism is the total sum of all biological events that take place in our bodies. We evaluate metabolism in terms of calories burnt because energy is used during metabolism. Resting Metabolic Rate (RMR) is the amount of calories burned just by laying down or sitting while doing nothing. In inactive persons, RMR accounts for approximately 70% of daily calorie burn. A favorable genotype may assist in boosting your RMR, leading to greater energy consumption and quicker weight reduction with influence from food, activity, and lifestyle. Given equal circumstances, individuals with less favorable genotypes may find it more difficult to reduce weight.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

In order to lose weight, you must first understand your total daily energy expenditure (TDEE). Consult a fitness specialist to design a training program aligned with your TDEE requirements. A good night's sleep and a reduced stress level have both been demonstrated to increase the amount of energy expended in metabolism.

WEIGHT MANAGEMENT: WEIGHT LOSS TENDENCY

1

RESULT

Risk Markers
Detected



Normal Tendency

Gene: PPARG

Genotype: CC

Description

Anyone can lose weight from a regular, structured exercise program together with a healthy diet. However, studies have shown that not everyone loses weight at the same rate. Those with less favorable genetic variants still lose weight over time, albeit at a slower rate than those with favorable genotypes. Following an organized diet and fitness regimen has been shown to help people lose weight.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Consider hiring an exercise professional and strictly stick to an organized training regimen. Portion control aids in calorie reduction and prevention of overeating. High-fiber and whole-food foods may promote satiety and help you avoid overeating. Breaking down key objectives into smaller, manageable steps can improve adherence to your weight loss plan.

WEIGHT MANAGEMENT: WEIGHT REGAIN TENDENCY

4

RESULT

Risk Markers
Detected

Low Tendency

Gene: NEGR1

Genotype: TC

Gene: PPARG

Genotype: CC

Gene: GNPDA2

Genotype: AG

Gene: TMEM18

Genotype: GG

Description

Weight management is about managing the equation of weight loss and weight regain. Most people find losing extra weight challenging, but maintaining the weight loss can be even harder. Following a large weight reduction, your metabolism may decrease, making it more difficult to keep the pounds off. As your body attempts to achieve equilibrium, your hunger may rise. All of these factors may be impacted by your genetics, with some individuals being more prone to increased hunger and a slower metabolism, increasing the risk of weight gain.

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for this disease/trait.

RECOMMENDED INTERVENTIONS

Maintain focus on your fitness objectives. Understand that fitness extends beyond the numbers on the scale. Aim to build lean muscle over time, as having more lean muscle is a defensive mechanism that helps you burn more fat and calories.



DNA contains genetic information for each person, including physical features associated with how you look and also behaviors that you exhibit. A trait is a specific characteristic of a person, and children inherit a range of traits from their parents when copies of genes are passed down. Knowing your genetic traits will help you to understand what makes you, 'YOU'. Besides genetic factors, traits can also be influenced by your environment.

Behavioral & Physical Traits

ABILITY TO MEMORIZE

1

RESULT

Risk Markers
Detected



Normal Ability

Gene: COMT

Genotype: GG

Description

Memory is a vital cognitive function that involves storing, retrieving, reasoning, comprehending, and learning information. An individual's ability to memorize is influenced by genetic factors, which play a significant role in cognitive performance, emotional memory, and educational achievement. Certain genetic variations may contribute to enhance memory capabilities, such as superior recall, improved visual memory retention, and faster acquisition of new skills.

Risk Factors

- Injury to head
- Aging
- Side effects of medications
- Psychological trauma

TYPICAL SYMPTOMS



Difficulty concentrating



Struggling to learn new things



Repetitive conversations



Frequently misplacing items

RECOMMENDED INTERVENTIONS

While genetics may influence memory capacity, there are effective strategies to enhance memory regardless of genetic factors. Regular exercise has been shown to support cognitive function and improve memory. A healthy diet can also play a crucial role, as studies indicate that consuming fatty fish like salmon, which is rich in omega-3 fatty acids, can boost memory. Adequate sleep is another essential factor for memory consolidation. Additionally, engaging in mentally stimulating activities and challenging the brain can help enhance memory retention and cognitive abilities.

ADDICTIVE PERSONALITY

1

RESULT

Risk Markers
Detected

Lower Tendency

Gene: CHRNA3

Genotype: GG

Description

Individuals who have addictive personality disorder tend to develop addictions to either substances or behaviors. These addictions can manifest as an unhealthy fixation or obsession towards something. Genetics have been identified as being connected to certain types of addictions. However, environmental factors and social influence can also contribute to the development of addictions.

Risk Factors

- Affects individuals of all ages
- Environmental factors
- Influence from family or friends

TYPICAL SYMPTOMS



Adventurous and risk-taking



Difficulty in self-regulation



Obsessive compulsive



Related to others with addictive personality

RECOMMENDED INTERVENTIONS

Those who have signs of addiction, toward any behavior or substance, should seek professional help to curb the addiction. However, treatment or rehabilitation for the addiction will depend on how severe the condition is.

ALCOHOL DEPENDENCE

4

RESULT Risk Markers Detected



Moderate Tendency

Gene: OPRM1

Genotype: AG

Gene: DRD2

Genotype: AC

Gene: ALDH2

Genotype: AA

Gene: ANKK1

Genotype: AG

Description

Alcohol dependence is a chronic disease that can progressively get worse. Those who suffer from alcohol dependency are likely to experience withdrawal symptoms when they do not consume alcohol. Excessive consumption of alcohol can eventually lead to chronic liver disease and liver cancer. It also affects regular social functioning, such as driving and interaction with other people.

Risk Factors

- Family history of alcohol dependence
- Have mental health problems, e.g. depression, anxiety

TYPICAL SYMPTOMS



**Strong urge to
drink alcohol**



**Unable to limit
alcohol consumption**



**Difficulty in fulfilling
major tasks due to
alcohol abuse**



**Spend a lot of time
drinking**

RECOMMENDED INTERVENTIONS

Alcohol dependence risk can be lowered by taking a regular break from the normal drinking cycle. One may face withdrawal symptoms like anxiety and jitteriness during treatment and rehabilitation. However, by breaking the drinking cycle, the body will be less accustomed to alcohol, hence reducing the dependency.

ALCOHOL FLUSH

Description

Some individuals who drink alcohol experience an unpleasant phenomenon called the alcohol flush reaction. The primary feature of the alcohol flush reaction is a red face, or flush. However, this can also be accompanied by hives, nausea, low blood pressure, worsening of asthma, or an episode of migraine.

Risk Factors

- Genetics
- Aldehyde dehydrogenase 2 deficiency

2 **RESULT**
Risk Markers
Detected
High Response

Gene: ALDH2

Genotype: AA

Gene: CCDC63

Genotype: CC

TYPICAL SYMPTOMS



Asthma Attack



Hives



Migraine



Nausea

RECOMMENDED INTERVENTIONS

Tips to reduce alcohol flush include drinking water before and during alcohol consumption to help lessen symptoms of alcohol flush by diluting the alcohol in the body; eating a heavy meal before drinking to slow the absorption of alcohol into the bloodstream; avoiding drinks that contain toxic substances found in certain alcoholic beverages that can intensify alcohol flush symptoms; and staying away from consuming dark liquors, red wine, and brandy; taking an antihistamine to help reduce symptoms of alcohol flush; and reducing alcohol consumption.

APPETITE REGULATION

0

RESULT
Risk Markers
Detected

Normal Regulation

Description

Hunger is a natural process that signals an individual's need to eat. However, some people have genetic disruptions affecting structures like the melanocortin-4 receptor (MC4R), causing dysregulation in appetite signaling, which is linked to a greater risk of obesity. Certain genetic variations are associated with reduced satiety after eating, leading to overeating. These disruptions in appetite regulation have also been linked to various non-communicable diseases.

Risk Factors

- Affects individuals of all ages
- Family history of eating disorder
- Mental health conditions
- Diabetes

TYPICAL SYMPTOMS



Persistent overeating



Binge eating



Severe obesity



**Craving high calorie
or sugary food**

RECOMMENDED INTERVENTIONS

To prevent appetite dysregulation from leading to obesity, individuals should adopt mindful eating practices and consume protein- and fiber-rich foods for better satiety. Portion control strategies, such as using smaller plates, can help regulate intake. Regular exercise and adequate sleep also play key roles in appetite regulation.

BODY / VOLUME (HAIR)

Description

Hair volume refers to the fullness of the hair at the scalp, while hair body is the movement of the hair. An individual's hair thickness and texture are highly dependent on genetics. Accumulated oil produced by the scalp will cause the hair to appear flat and without volume. To support healthy hair growth, one should maintain a balanced diet rich in vitamins and minerals, such as biotin, zinc and iron.

Risk Factors

- Have male/female pattern baldness
- Malnutrition and vitamin deficiencies
- Long period of significant stress

TYPICAL SYMPTOMS



Hair appears flat and lacks lift



Hair looks thin or sparse in some areas



Hair is fragile and prone to breakage



Scalp is more visible due to reduced hair density

1

RESULT

Risk Markers
Detected



Normal Volume

Gene: C1orf127

Genotype: CC

RECOMMENDED INTERVENTIONS

There are many different ways to obtain a more volumized hair. One can opt for shorter hair and layering them appropriately to get the volume needed. There are also various shampoos and conditioners that help increase hair volume.

BODY HAIR

Description

The amount of body hair is influenced by genetics. Hair follicles have different sensitivity to the androgen group of hormones, primarily testosterone. Higher levels of the androgen hormone during puberty can cause finer hair to grow into thicker and darker hair over various parts of the body. A condition known as hirsutism occurs when women experience excessive hair growth, particularly in a male-like pattern.

Risk Factors

- Family history of conditions which cause excessive hair growth
- Hormonal imbalance
- Obesity

TYPICAL SYMPTOMS



Change in growth, thickness or distribution of hair



Sudden or progressive changes in hair patterns



Excessive facial and body hair



Stiff or dark body hair in uncommon places

0

RESULT

Risk Markers Detected

Normal Body Hair

RECOMMENDED INTERVENTIONS

Excessive body hair growth in women (hirsutism) can be a symptom of underlying health conditions. If you are experiencing it, especially recently, it is recommended to consult your doctor.

EMOTIONAL QUOTIENT (EQ)

0

RESULT
Risk Markers
Detected



Normal Level

Description

Emotional Quotient (EQ) refers to one's ability to perceive, use, evaluate, and handle emotions. Higher EQ indicates the ability to recognize one's own emotions as well as those of others, and then use the emotional information to guide their behavior. This includes effectively distinguishing different emotions and adapting one's responses to different situations. While emotional intelligence is shaped by life experiences, genetics also play a role. Research suggests that approximately 10% of our empathy may be attributed to genetic factors.

Risk Factors

- Lack of emotional awareness
- Family environment
- Reduced social exposure
- Trauma

TYPICAL SYMPTOMS



Difficulty recognizing emotions



Difficulty handling stress and conflict



Lack of self-awareness



Poor communication skills

RECOMMENDED INTERVENTIONS

Improving EQ can be done through several ways, including understanding and recognizing your own emotions, thoughts and behaviors; trying to understand and share the feelings of others; developing effective communication, conflict resolution and relationship-building skills; managing your emotions in challenging situations; building the ability to bounce back from emotional setbacks and stress; practicing being non-judgmental in the moment; asking others for feedback and insights on how you can improve your emotional intelligence; and reflecting on past experiences and learn from them to grow emotionally.

EXTROVERSION

0

RESULT

Risk Markers
Detected

Lower Tendency

Description

Extroversion is a personality trait that is associated with behaviors that include sociability, high levels of talkativeness, and being comfortable around people. It is the opposite of introversion, where an individual person prefers being alone. Research suggests that extroversion and introversion are roughly 50% genetically based. The studies also believe that apart from genetics, environmental factors play an important role in determining the level of extroversion.

Risk Factors

- Environmental factors
- Higher dopamine responsiveness
- Lower level of cortical arousal

TYPICAL SYMPTOMS



Energetic around people



Anxious/bored when alone



Wide social circle



Outgoing

RECOMMENDED INTERVENTIONS

Some ways to motivate extroverts include providing opportunities for social interactions as extroverts often find motivation through such interactions; encouraging public recognition as extroverts usually thrive on attention and recognition; setting group goals as extroverts often work well in groups and enjoy the sense of camaraderie that comes with working towards a common goal; offering a variety of new experiences as extroverts often enjoy trying new things; fostering a positive and energetic work environment as extroverts are often energized by such work environment.

FOOD ADDICTION

1

RESULT Risk Markers Detected



Normal Tendency

Gene: ANKK1

Genotype: AG

Description

Food addiction is characterized primarily by the overconsumption of palatable foods, which are foods rich in sugars, salts and fats. The consumption of these foods stimulates the reward center of the brain, which leads to a release of feel-good hormones such as dopamine. This release can override the body's regular signals for satiety and satisfaction, leading to overeating and obesity.

Risk Factors

- Affects individuals of all ages
- Have a close relative with an eating disorder
- Have mental health issues (depression, anxiety, obsessive-compulsive disorder)

TYPICAL SYMPTOMS



Continuing eating even when not hungry



Decreased energy and has chronic fatigue



Digestive disorders



Eating until feeling ill

RECOMMENDED INTERVENTIONS

Overcoming food addiction is a long and challenging process. Some have gained success in 'therapeutic fasting' and following a strict diet that prioritizes whole grains, fruits and vegetables. However, the treatment approaches still depend on the severity of the individual's condition.

GENERAL COGNITIVE ABILITY

4

RESULT Risk Markers Detected

Normal Ability

Gene: Intergenic

Genotype: CC

Gene: Intergenic

Genotype: AA

Gene: Intergenic

Genotype: TG

Gene: LOC101927

Genotype: AA

Description

General cognitive ability is often referred to as 'general intelligence' and it can predict outcomes like educational and occupational levels, far better than any other behavioral traits. This is one of the most inheritable behavioral traits; therefore, genes usually play an important role in the heritability of general intelligence. General cognitive ability indicates an individual's ability to reason, plan, solve, and comprehend complex ideas. These qualities are quite essential for human survival and adaption.

Risk Factors

- Alzheimer's disease
- Aging
- Mood disorders (e.g. depression)
- Substance abuse (e.g. drugs and alcohol)
- Traumatic brain injuries

TYPICAL SYMPTOMS



Difficulty concentrating



Poor memory



Slower processing speed



Language difficulties

RECOMMENDED INTERVENTIONS

You can enhance your cognitive function by engaging in activities that challenge your brain, such as reading, solving puzzles, learning a new skill, and getting regular exercise. Additionally, maintaining a healthy lifestyle, including a balanced diet and getting enough sleep, can help support overall brain function.

HAIR DRYNESS / BRITTLENESS

1

RESULT

Risk Markers
Detected



Normal Risk

Gene: TCHHL1

Genotype: GG

Description

Dry hair can occur when the scalp is unable to retain moisture, produce sebum or is exposed to harsh and dry conditions. Most commonly, it is caused by a lack of sebum that coats the hair, causing it to become dull and frizzy. Dry hair can be caused by genetics but usually the condition is aggravated by lack of vitamins to the hair or using harsh hair products.

Risk Factors

- Environmental factors, e.g. prolonged sun exposure, pollution or regular swimming in chlorinated/salty water
- Hair care practices, e.g. excessive heat styling and chemical treatments
- Poor nutrition
- Hormonal changes

TYPICAL SYMPTOMS



Dull hair



Frizzy hair



Split ends



Brittle texture

RECOMMENDED INTERVENTIONS

Dry hair can be remedied by reducing the frequency of hair washing. Supplementation with Vitamin A, Vitamin C and biotin can also help. Trimming unhealthy hair can also help the growth of healthy hair.

HAIR GRAYING

Description

Hair graying is the result of reduced melanin pigment in the roots of the hair. It is expected to occur naturally as one gets older; however, premature graying can occur when the hair follicles lose pigment cells, which can happen even before the age of 20 years. The causes of hair graying can be due to genetics or other external factors that affect melanin production in the hair.

Risk Factors

- Vitamin deficiencies
- Oxidative stress
- Smoking
- Certain medical conditions, e.g. autoimmune diseases

0

RESULT
Risk Markers
Detected

Normal Tendency

TYPICAL SYMPTOMS



Loss of natural hair colour



Rapid shedding of thicker dark colored hairs



Growth of thinner white hairs



Premature graying

RECOMMENDED INTERVENTIONS

With regards to hair graying, once a follicle stops producing melanin, it is irreversible. However, steps can be taken to prevent it from worsening. Having a healthy diet rich in Vitamin B12 and Vitamin D will help maintain melanin production. Fruits and vegetables which are rich in antioxidants can also help maintain melanin production.

HAIR THICKNESS

1

RESULT

Risk Markers
Detected



Normal Hair

Gene: C1orf127

Genotype: CC

Description

Hair thickness is the diameter of each individual hair strand, which can be categorized as coarse, medium or fine. Thicker hair shafts lend itself to the appearance of fuller, more luscious hair. Besides genetics, there are other factors that can influence the appearance of a person's hair, such as hormones, medications, chemicals, gender and age. As a person ages, hair strands normally get thinner.

Risk Factors

- Hormonal imbalance
- Aging
- Gender differences (Men tend to have thicker hair strands than women)

TYPICAL SYMPTOMS



Coarse hair strands



Hair strand is
clearly visible



Diameter of hair is
120 microns



Ponytail circumference
is 4 inches or more

RECOMMENDED INTERVENTIONS

Hair thickness is influenced primarily by genetics. However, diet can play a role too. Eating foods rich in iron and Vitamins A, B, C, D and E help maintain hair growth and health. Using the appropriate hair tools and products can also help with maintaining hair health.

HAIR THINNING

1

RESULT

Risk Markers Detected

Reduced Risk

Gene: Intergenic

Genotype: TC

Description

Hair thinning refers to a gradual reduction in hair density, which may manifest as reduced volume, finer hair strands, or visible thinning patches. While it does not always lead to baldness, it can contribute to hair becoming fragile and brittle over time. This condition can be caused by both genetic and outside factors that can affect an individual's hair health, such as hormones, medication and chemicals in hair products.

Risk Factors

- Family history
- Aging
- Certain medical conditions
- Stress
- Poor nutrition

TYPICAL SYMPTOMS



Hair is fragile and brittle



Reduced hair volume and thickness



Scalp may become more visible



Hair appears weaker or less shiny

RECOMMENDED INTERVENTIONS

Although hair thinning is a natural process, it can be managed by adopting healthy lifestyle habits. Avoiding excessive heat, harsh chemical treatments, and tight hairstyles can help reduce breakage. A balanced diet rich in biotin, iron, zinc, protein, and vitamins (A, B, C, D, and E) supports healthy hair growth.

HAIR TYPE

Description

Hair types are mainly genetically determined. There are generally four types of hair, which are straight, wavy, curly or coiled hair. Studies have shown that the type of hair is influenced by different genes that determine hair texture and thickness in individuals of different ethnic backgrounds. Hair type can also be influenced by external factors.

Risk Factors

- Genetics
- Hormonal changes
- Environmental factors
- Heat styling & chemical treatments

TYPICAL CHARACTERISTICS



Straight hair type



Wavy hair type



Curly hair type



Coily hair type

1

RESULT

Risk Markers
Detected

Straight Hair

Gene: TCHHL1

Genotype: GG

RECOMMENDED INTERVENTIONS

An individual's hair type is based on the curl pattern of the hair, which is determined by the hair follicle. Although hair type is determined by genetics, curl patterns can be easily altered using heat or chemicals. One should always consult a professional to know which hairstyle suits best.

HARM AVOIDANCE BEHAVIOR

2 **RESULT**
Risk Markers
Detected

Lower Tendency

Gene: SLA

Genotype: TT

Gene: APBA2

Genotype: AG

Description

Harm avoidance is characterized by feelings such as excessive worrying, pessimism, shyness, doubt, fearfulness and fatigue. This condition is correlated to low levels of serotonin. Individuals with harm avoidance tendencies are known to exhibit behaviors that actively avoid negative outcomes. They also tend to avoid social interactions for fear of rejection. Harm avoidance is a risk factor for anxiety disorders and depression.

Risk Factors

- Affects individuals of all ages
- Have a close relative with mental health issues (depression, anxiety, etc.)
- History of eating disorder

TYPICAL SYMPTOMS



Anticipatory worry



Fear of uncertainty



Shyness



Susceptibility to fatigue

RECOMMENDED INTERVENTIONS

Harm avoidance is a personality trait that correlates with anxiety and depression. More often than not, individuals with harm avoidance behavior tend to have high social phobia. This personality trait requires help from medical professionals to prevent the progression of other related diseases.

INFORMATION PROCESSING POWER

2

RESULT Risk Markers Detected

Normal Ability

Gene: Intergenic

Genotype: TC

Gene: Intergenic

Genotype: TG

Description

Information processing power refers to the brain's ability to take in, interpret, store, and respond to information efficiently. It is a fundamental cognitive function that affects learning, decision-making, problem-solving, and overall mental performance. Faster and more effective information processing allows individuals to react quickly to new information, while slower processing can lead to challenges in academics, work, and daily life.

Risk Factors

- Neurological disorders
- Sensory processing disorders
- Older age
- Brain injuries
- Chronic stress and lack of mental stimulation

TYPICAL SYMPTOMS



Difficulty understanding



Frequent forgetfulness



Difficulty following instructions



Slow reading and comprehension

RECOMMENDED INTERVENTIONS

Improving information processing involves maintaining a healthy lifestyle, including regular physical exercise, proper sleep, and a balanced diet rich in brain-supporting nutrients. Engaging in cognitive activities such as reading, puzzles, and learning new skills helps stimulate neural pathways. Social interaction and stress management techniques, such as meditation, also support cognitive function. In cases of medical conditions affecting processing speed, professional intervention, such as cognitive therapy and medication, may be needed.

INSOMNIA

Description

Insomnia is a common problem faced by many, where the individual has regular sleeping problems. The affectee may find it difficult to fall asleep at night, or may wake up multiple times when sleeping. Researchers have estimated that heritability accounts for 31% to 58% of a person's likelihood of experiencing insomnia. Genes that may cause insomnia are related to the same processes that regulate the sleep-wake.

Risk Factors

- Caffeine consumption
- Drug use
- Jet lag
- Stress and anxiety
- Uncomfortable sleeping environment

TYPICAL SYMPTOMS



Waking up during the night



Difficulty in falling asleep



Waking up too early



Not feeling well-rested after sleeping

2 **RESULT**
Risk Markers
Detected

Normal Risk

Gene: Intergenic

Genotype: AA

Gene: SLC2A13

Genotype: GG

RECOMMENDED INTERVENTIONS

Some tips to help prevent insomnia include sticking to a schedule, create a sleep-conducive environment, exercising regularly during the daytime, limiting caffeine and alcohol consumption, relaxing before going to bed, avoiding naps during the late afternoon of early evening, and getting exposure to natural light during the day. It is important to note that everyone is different, and what works for one person may not work for another. It may take some trial and error to find what works best for you.

INSTANT GRATIFICATION

1

RESULT

Risk Markers
Detected



Normal Tendency

Gene: GPM6B

Genotype: CC

Description

Instant gratification is the desire to experience a rewarding and immediate benefit while foregoing a future benefit. This could lead to harmful outcomes for the individual, which can affect their health. Individuals prone to instant gratification are more likely to be obese or will abuse substances, such as excessive alcohol drinking and usage of drugs.

Risk Factors

- Affects individuals of all ages
- Environmental factors

TYPICAL SYMPTOMS



Seize any opportunity
for instant pleasure



Tendency to be
more impulsive



Putting off a work task
to do something else
more enjoyable



Inability to control
emotional urges

RECOMMENDED INTERVENTIONS

Instant gratification may bring both benefit and harm to the individual, depending on the situation. A simple guide to overcome instant gratification is to try and make a conscious decision regarding whether to do something, instead of just following impulse decisions. By learning from past mistakes, one can prevent impulse decisions and learn to enjoy the moment without following sudden urges.

INTELLIGENCE QUOTIENT (IQ)

1

RESULT Risk Markers Detected



Normal Level

Gene: SNAP25

Genotype: AG

Description

Intelligent quotient measures an individual's ability to reason and find solutions to problems. It is a comparison between an individual and the performance of others of the same age group in the same test. The results for IQ are drawn from specifically-designed tests. Twin studies suggest that there is a link between the IQ of a person and their genetic makeup. Heritability of IQ is estimated to be between 57% to 73%.

Risk Factors

- Fetal alcohol syndrome
- Genetic conditions (Down syndrome)
- Injuries to the head
- Problems during childbirth

TYPICAL SYMPTOMS



Struggles with problem-solving and critical thinking



Slow learning pace



Poor academic/occupational performance



Limited adaptability to new situations

RECOMMENDED INTERVENTIONS

It is important to note that IQ is not a fixed trait, and that individuals have the ability to improve their cognitive abilities to a certain extent. Some of the methods which you can utilize to enhance your cognitive abilities and potentially improve your IQ include engaging in lifelong learning, exercising regularly, getting enough sleep each night, eating a healthy and balanced diet, challenging your brain and reducing your stress level.

LUMBAR DISC DISEASE(LOWER BACK PAIN)

0

RESULT
Risk Markers
Detected


Normal Risk

Description

Lumbar disc disease is characterized by pain in the lower back, pain or numbness in the leg, and localized tenderness at the affected intervertebral discs. It is a medical condition that usually affects the lower back or lumbar region. Due to aging, the tissues in the intervertebral discs degenerate over time and lose their elasticity, which causes the discs to harden and become out of shape, resulting in this condition.

Risk Factors

- Aging
- Sedentary lifestyle
- Heavy lifting

TYPICAL SYMPTOMS



Intermittent or continuous back pain



Lower back pain while standing, bending or sitting



Spasm of the back muscles



Loss of muscle strength

RECOMMENDED INTERVENTIONS

Back pain which is caused by lumbar disc disease can be hereditary. Knowing your family history can be helpful in assessing the risk. Obesity is a common factor that can accelerate lumbar disc degeneration. Therefore, it is recommended to maintain a healthy weight. Do not put your back under undue and sudden stress. If you need to lift heavy objects in your vocation, it is recommended to invest in a back brace. When exercising, make sure you warm up and prevent undue strain, particularly when lifting weights. Use proper techniques and make sure to rest between sets.

MALE PATTERN BALDNESS

2

RESULT Risk Markers Detected

Normal Risk

Gene: LOC100270

Genotype: AG

Gene: Intergenic

Genotype: TC

Description

Male pattern baldness is the hair loss pattern most commonly seen in men as they age. It is related to a change in the male sex hormones (androgens). It has a strong genetic factor and does not cause harm. With male pattern baldness, hairs begin to weaken and the hair follicle shrinks, resulting in shorter and finer strands of hair.

Risk Factors

- Genetics or family history
- Hormonal influences such as androgens, including dihydrotestosterone (DHT)

TYPICAL SYMPTOMS



Receding hairline



Circular area on back of head thins and expands over time



Thinner strands around crown of head and temples



Horseshoe pattern of hair around sides of head

RECOMMENDED INTERVENTIONS

Hair loss may be a part of the aging process, but it can be psychologically distressing. Male pattern baldness occurs due to changing hormone levels. Treatments may include hair transplantation, hair growth medications (such as minoxidil), and laser and light therapies.

MEMORY SKILLS

1

RESULT Risk Markers Detected



Normal Ability

Gene: SORCS1

Genotype: GG

Description

Memory is associated with how an individual acquires, stores, and retrieves information when required. It is an essential cognitive function that helps individuals to learn and understand their environment. Memory is understood as an informational processing system and comprises sensory processor, short-term memory, and long-term memory. It is essential for learning because without memories, learning and understanding will be affected greatly. Genetic variations are known to affect learning and memory.

Risk Factors

- Injury to head
- Old age
- Reaction to medicine
- Trauma

TYPICAL SYMPTOMS



Trouble following instructions



Struggles with learning new information



Reduced ability to concentrate



Frequent forgetfulness

RECOMMENDED INTERVENTIONS

To improve memory skills and make it easier to retain and recall information, it is recommended to practice active listening, use visualization techniques, digest information in smaller chunks, repeat information out loud, obtain sufficient sleep and exercise, utilize mnemonic devices, play memory games, focus on current tasks, and associate new information to things that you already know.

MONOBROW

0

RESULT
Risk Markers
Detected



Normal Risk

Description

A monobrow or unibrow is the fusion of the eyebrows at the bridge of the nose. It is an uneventful condition in which the hairs that grow above the bridge of the nose are around the same thickness as the eyebrows. It has been proven to be genetically linked and can, in some cases, be linked to other medical conditions.

Risk Factors

- Genetics
- Medical conditions, e.g. Cornelia de Lange syndrome

TYPICAL SYMPTOMS



Two eyebrows meet in the middle, above bridge of nose



Hair above bridge of nose is the same color and thickness as eyebrows

RECOMMENDED INTERVENTIONS

A monobrow may be aesthetically displeasing for some people, and they might choose to remove or reshape it. Some treatments for these procedures include waxing, threading, shaving and laser hair removal.

MOTION SICKNESS

1

RESULT Risk Markers Detected

Normal Risk

Gene: Intergenic

Genotype: AG

Description

Motion sickness occurs when the brain is unable to process the signals from the sensory organs while traveling in a moving vehicle, due to a mismatch between actual and expected motion. This can result in nausea, dizziness and lightheadedness. Anyone can be susceptible to motion sickness and it can occur in varying degrees. It is most common in pregnant women and children.

Risk Factors

- Affects individuals of all ages
- Women are affected more than men
- Alcohol or drug use
- Pregnancy
- History of migraine, anxiety, frequent vomiting or nausea

TYPICAL SYMPTOMS



Cold sweats



Dizziness



Headache



Increased saliva,
nausea and vomiting

RECOMMENDED INTERVENTIONS

Motion sickness symptoms can be reduced by looking at the horizon and nibbling on some light snacks when traveling in a vehicle. Additionally, listening to music can help distract from the discomfort, and avoid reading or looking at devices while in a moving vehicle.

NICOTINE DEPENDANCE

0

RESULT
Risk Markers
Detected

Normal Risk

Description

Nicotine is the primary chemical in tobacco that makes smoking addictive and difficult to quit. It produces a temporary pleasing effect in the brain by stimulating dopamine release. However, this effect is short-lived, leading to cravings for another cigarette. Over time, the body develops tolerance, requiring more nicotine to achieve the same effects. Nicotine dependence occurs when an individual becomes reliant on nicotine and struggles to quit. A complex mix of environmental, behavioral and genetic factors appears to raise the risk of nicotine dependence.

Risk Factors

- Tobacco usage
- Family history
- Mental illness
- Drug abuse
- Social factors

TYPICAL SYMPTOMS



Anxiety and panic attacks



Trouble sleeping



Irritability and frustration



Difficulty concentrating

RECOMMENDED INTERVENTIONS

To reduce nicotine dependence, you can seek professional help, consider nicotine replacement therapy, practice relaxation methods (such as deep breathing or meditation), avoid situations or habits that trigger the urge to smoke, and engage in regular exercise.

OPTIMISM AND EMPATHY

0

RESULT Risk Markers Detected

Normal Tendency

Description

Optimism is a thought that the best outcome will happen, even if it is not likely. It may lead to better health and success for an optimistic individual. Empathy is the ability to understand and share the feelings of another individual's situation. Studies have shown that genetics and environmental factors may determine one's optimism level. However, it can be improved through personal effort.

Risk Factors

- Poor mental health
- Neurological conditions, e.g. anosognosia
- Fear and stress
- Socioeconomic status
- Low self-esteem

TYPICAL SYMPTOMS



Optimistic:
Confident and hopeful
about the future



Pessimistic:
Believe that the
worst will happen



Empathetic:
Has the ability to
understand feelings
of others



Apathetic:
Lack motivation to do
anything or just don't
care

RECOMMENDED INTERVENTIONS

Optimism is crucial for a healthier lifestyle and stress-free environment. An individual should analyze their thoughts and think about their capability to help others. By minimizing negative thoughts, they will be able to start afresh every day without high stress.

POSITIVE AFFECTIVITY

0

RESULT
Risk Markers
Detected

Normal Tendency

Description

Positive affectivity is a trait that reflects individual differences in positive emotional experience, such as enthusiasm, sociability, and happiness. People high in positive affectivity experience better moods. They are generally more cheerful, enthusiastic, and confident. Higher levels of positive affect are usually associated with positive outcomes. Genome-wide association studies estimate that positive affectivity has a heritability component of almost 60 percent.

Risk Factors

- Anxiety
- Depression
- Emotional disturbances
- Trauma

TYPICAL SYMPTOMS



Lack of enthusiasm



**Reduced motivation
& energy**



**Higher susceptibility
to stress**



Limited social engagement

RECOMMENDED INTERVENTIONS

Positive thinking can be achieved through various methods that can alleviate negative feelings. These methods include replacing negative thoughts with positive ones, socializing with individuals that have a positive outlook in life, exercising to improve your mood, getting sufficient sleep, reducing stress levels, eat properly, get involved in activities that bring you joy, and try to smile as often as you can.

READING AND SPELLING

Description

Reading is the process where one applies the sound-symbol relationship and successfully mixes them together to read a word. Spelling, on the other hand, is the ability to segment words into individual sounds and then use the correct sound-symbol relationship to produce words in written forms. Results from studies have suggested that reading and spelling disorder should be regarded as a complex disorder that is strongly influenced by genetic factors.

Risk Factors

- Hearing impairment
- Lack of interest
- Visual impairment

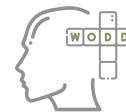
TYPICAL SYMPTOMS



Difficulty recognizing and sounding out words



Trouble understanding



Incorrect spelling of common words



Poor handwriting

2

RESULT

Risk Markers
Detected

Normal Ability

Gene: Intergenic

Genotype: TC

Gene: LOC105376

Genotype: AA

RECOMMENDED INTERVENTIONS

The treatment for reading and spelling difficulties include helping the child develop auditory and vocal skills, encouraging them to write about experiences that they enjoyed, use hands-on-activities and modern teaching methods to develop their interest in reading and spelling.

REWARD DEPENDENCE

Description

Reward dependence is the tendency to be more sensitive towards obtaining socially defined rewards or getting approval from others. Individuals with a high reward dependence level face higher peer pressure and the risk of developing personality disorders. Those who lack approval may seek other rewards by overeating or substance abuse. Reward dependence is associated with an inherited personality trait.

Risk Factors

- Genetics
- Hormones, e.g. low levels of norepinephrine

TYPICAL SYMPTOMS



**Learn from
reward signals**



**Need constant
social approval**



**Repeat actions
associated with rewards**



**Increase
in sociability**

3

RESULT Risk Markers Detected



Normal Tendency

Gene: LOC107984

Genotype: AG

Gene: MIS18A

Genotype: TC

Gene: Intergenic

Genotype: TC

RECOMMENDED INTERVENTIONS

Individuals with high reward dependence personality must avoid situations that involve seeking reward as the end goal. Individuals who seek reward excessively may end up facing adverse consequences.

SENSITIVE PERSONALITY

1

RESULT

Risk Markers Detected



Normal Tendency

Gene: DRD2

Genotype: AC

Description

Sensitive personality refers to individuals who have a heightened awareness of their surroundings and experience emotions more deeply. This personality trait has been linked to dopamine activity in the brain and can be influenced by genetic factors or past experiences. Research suggests that genetics account for nearly half of the variability in an individual's sensitivity.

Risk Factors

- Genetics
- Life experiences

TYPICAL SYMPTOMS



Constantly overthink and worry about many things



Have emotional reaction to everything



Often feel tired and overwhelmed



Sensitive to criticism

RECOMMENDED INTERVENTIONS

A sensitive personality may interfere with one's daily interaction with others. This can be managed by taking time to recharge and minimizing external stimuli. Building strong social support, maintaining a healthy lifestyle, and boosting self-confidence can enhance emotional stability. If sensitivity significantly impacts daily life, seeking professional help or cognitive behavioral therapy (CBT) can be beneficial.

SLEEP DEPTH

2

RESULT Risk Markers Detected

Normal Depth

Gene: TLN2

Genotype: AG

Gene: LOC124906

Genotype: TC

Description

During sleep, the body cycles through different sleep stages, with deep sleep being essential for brain health and function. This stage allows the brain to rest, recover and consolidate memories while also regulating hormone balance. Research suggests that certain genetic variations can influence the duration and intensity of deep sleep in individuals.

Risk Factors

- Genetics & neurobiological factors
- Stress & mental health conditions
- Substance use (e.g. caffeine)
- Poor sleep habits & environment

TYPICAL SYMPTOMS



Feeling tired even after a full night's sleep



Difficulty concentrating



Daytime sleepiness



Frequent waking during the night

RECOMMENDED INTERVENTIONS

When sleeping, your body needs the deep sleep stage the most, as it is responsible for making you feel recharged. Obtaining the optimal amount of exercise at the correct time, adjusting the time you go to bed, maintaining a proper sleep schedule, having a comfortable sleep environment, playing soft white noise in the background, and avoiding caffeine consumption may increase your deep sleep.

SLEEP DURATION

1

RESULT

Risk Markers Detected



Normal Duration

Gene: ABCC9

Genotype: GG

Description

Sleep duration is the quantity of time that an individual spends sleeping. This may be measured as a one-time sleep period or for sleep experienced over 24 hours. Sleep duration and quality are genetically inherited traits to a certain extent, as studies have indicated that a substantial amount of variability in sleep characteristics is genetically determined - where approximately 10 to 40 percent of the variation in sleep duration may be inherited.

Risk Factors

- Anxiety
- Medical problems (e.g. asthma)
- Medication use
- Physical pain
- Substance abuse
- Uncomfortable sleeping environment

TYPICAL SYMPTOMS



Persistent fatigue



Difficulty concentrating



Daytime sleepiness



Decreased cognitive performance

RECOMMENDED INTERVENTIONS

To improve sleep duration, maintain a consistent sleep schedule by going to bed and waking up at the same time every day, including weekends. Avoid sleeping when too hungry or too full, as it can cause discomfort. Ensure a comfortable sleep environment and address any physical discomfort for better rest. Limit naps to 30 minutes to prevent interference with nighttime sleep.

SLEEP LATENCY

Description

Sleep latency, or sleep onset latency, is the time it takes a person to fall asleep after turning off the lights. On average, a healthy person takes between 10 to 20 minutes to fall asleep. An individual's sleep latency can be an indicator of the overall quality of their sleep. Quickly falling asleep during the day and rapidly entering Rapid Eye Movement (REM) sleep can be signs of conditions, such as narcolepsy and idiopathic hypersomnia.

Risk Factors

- Old age
- Anxiety
- Depression
- Usage of medication
- Uncomfortable environment

TYPICAL SYMPTOMS



Idiopathic hypersomnia



Insomnia



Narcolepsy



Sleepy Apnea

0

RESULT

Risk Markers
Detected

Normal Latency

RECOMMENDED INTERVENTIONS

Treatment for unusual sleep latency includes setting a regular time for going to bed, and turning off all noise-producing gadgets and devices around bedtime. Exercising moderately for a few hours before bedtime can help to shorten sleep latency and improve the duration and depth of sleep.

SLEEP QUALITY

2

RESULT Risk Markers Detected

Normal Quality

Gene: CACNA1C

Genotype: TC

Gene: SLC2A13

Genotype: TT

Description

Sleep quality refers to an individual's overall satisfaction with their sleep experience. It is determined by factors such as sleep efficiency, sleep latency, sleep duration, and wakefulness after sleep onset. Good quality sleep improves brain performance, mood and health. Science shows that sleep is influenced heavily by the environment as well as genetic factors. These factors can also play a role in the occurrence of many sleeping disorders.

Risk Factors

- Health condition
- Medication use
- Aging
- Sleep hygiene
- Substance abuse
- Working at nights

TYPICAL SYMPTOMS



Persistent fatigue



Difficulty concentrating



Daytime sleepiness



Frequent nighttime awakenings

RECOMMENDED INTERVENTIONS

Sleep quality can be improved by optimizing the sleeping environment until the body gets used to the stable 'biological clock'. Regular exercise in the day and maintaining a consistent sleep schedule help improve the sleep cycle. Avoid consuming large meals, caffeine, or alcohol before bedtime. Minimize electronic device use in the bedroom, such as televisions, smartphones, and computers, as they can disrupt sleep.

SWEET TOOTH

Description

Genetic factors affect how individuals perceive sweetness, as it depends on the number of sweet receptors they have. Individuals with a sweet tooth have a higher tolerance for sugary food. However, high sugar consumption may lead to complications such as weight gain, diabetes and fatty liver. These health problems may then cause more serious complications like a heart attack or stroke.

Risk Factors

- Affects individuals of all ages
- Genetic predisposition

TYPICAL SYMPTOMS



Individuals taste sweetness differently, therefore, is difficult to measure with certainty.

0

RESULT
Risk Markers
Detected



Lower Tendency

RECOMMENDED INTERVENTIONS

Sweet cravings can be controlled by consuming a balanced meal or snack every few hours, and eating fruits as a substitute. Drinking water and going for a walk also helps in curbing cravings for sweet snacks.

Cardiovascular diseases are the leading cause of mortality in the world. The earlier cardiovascular diseases are detected, the easier it is to treat them. Besides cardiovascular diseases, our metabolic health is also important. Being metabolically healthy means that your body can respond properly to food, which may lower your risks of a range of non-communicable diseases. Know your genetic risks early, so that you can take necessary steps to minimize your health risks.



Cardio & Metabolic

ABDOMINAL AORTIC ANEURYSM

3

RESULT Risk Markers Detected



Slightly Lower Risk

Gene: DAB2IP

Genotype: AA

Gene: Intergenic

Genotype: AG

Gene: LRP1

Genotype: GG

Description

Abdominal aortic aneurysm occurs when a portion of the wall of the aorta, the largest artery in the body, weakens. This causes the aorta to expand abnormally at any segment. It occurs most commonly in the lower portion of the aorta. Aneurysms can go undetected for long periods of time. However, as the vessel expands, patients experience some discomfort and pain in their abdomen or back. When an aneurysm ruptures, it can cause massive internal bleeding or sudden death.

Risk Factors

- Age 65 years and older
- Five times more common in men than women
- Smoking
- Hypertension

TYPICAL SYMPTOMS



**Pulsating swelling
in center of abdomen**



Abdominal pain



Back pain



Coughing up blood

RECOMMENDED INTERVENTIONS

Risks of developing an abdominal aortic aneurysm can be reduced by treating underlying conditions which can cause it. This includes hypertension, diabetes, atrial fibrillation, coronary artery disease and high cholesterol. Lifestyle changes such as eating foods that are rich in nutrients and low in sugars and saturated fats can help to prevent and control these conditions. However, if needed, doctors may prescribe appropriate medications to manage the conditions.

ARRHYTHMOGENIC RIGHT VENTRICULAR DYSPLASIA

0

RESULT
Risk Markers
Detected

Low Risk

Description

Arrhythmogenic right ventricular dysplasia, is a genetic condition in which the muscles in the lower right ventricle are replaced by fatty or fibrous scar tissues. This causes the right ventricle to be distended and have poor contractions. Patients with this condition commonly experience arrhythmias and can pass away suddenly.

Risk Factors

- Aged less than 40 years
- Family history
- Strenuous exercise
- Competitive sports

TYPICAL SYMPTOMS



Ventricular arrhythmias



Palpitations



**Dizziness,
lightheadedness
or fainting**



**Sudden
cardiac death**

RECOMMENDED INTERVENTIONS

Arrhythmogenic right ventricular dysplasia has no known curative treatments to date. However, the disease can be managed with medical intervention. Individuals with ARVD are advised to go for regular check-ups.

ATRIAL FIBRILLATION

3

RESULT

Risk Markers Detected

Slightly Lower Risk

Gene: Intergenic

Genotype: TC

Gene: ZFH3

Genotype: TC

Gene: ZFH3

Genotype: TT

Description

Atrial fibrillation is a condition in which contractions at the atria of the heart are uncoordinated and irregular, causing problems with the synchronization of the atria and ventricles. This causes difficulty in getting blood out of the heart. This condition is characterized by a rapid heart rate. Patients with atrial fibrillation can experience palpitations, shortness of breath and a feeling of weakness.

Risk Factors

- Aged 65 years and older
- Hypertension
- Diabetes
- Metabolic syndrome
- Obesity
- Hyperthyroidism

TYPICAL SYMPTOMS



Chest pain



Confusion



Palpitations
- irregular heartbeat



Weakness,
dizziness and fainting

RECOMMENDED INTERVENTIONS

Atrial fibrillation can be genetically linked. Screening for genetic predispositions can help in managing and preventing onset of this disease. It is usually managed by treating underlying risk factors including obesity, hypertension, high cholesterol and diabetes.

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

3

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: CHRNA3

Genotype: TC

Gene: FAM13A

Genotype: TT

Gene: MMP9

Genotype: GG

Description

Chronic obstructive pulmonary disease (COPD) is a group of medical conditions where airflow to and from the lungs is compromised due to chronic inflammation of the airways that damages the lungs. Individuals tend to develop the condition after prolonged exposure to irritants and particulate matter, typically cigarette smoke. Difficulty breathing, cough and wheezing are common symptoms of COPD.

Risk Factors

- Aged 45 and above
- Smoker

TYPICAL SYMPTOMS



Fatigue



Persistent coughing



Shortness of breath



Wheezing

RECOMMENDED INTERVENTIONS

Risk of developing chronic obstructive pulmonary disease (COPD) can be reduced by minimizing exposure to irritants and pollutants. Smokers and those around them are at increased risk of developing COPD.

CORONARY HEART DISEASE

5

RESULT Risk Markers Detected

Slightly Lower Risk

Gene: CDKN2B-AS1

Genotype: AG

Gene: Intergenic

Genotype: AG

Gene: Intergenic

Genotype: GC

Gene: LPA

Genotype: TC

Gene: MIA3

Genotype: AC

Description

Coronary heart disease occurs when the major blood vessels responsible for transporting blood to and from the heart are damaged or diseased. The most common cause of this is the buildup of plaque in the blood vessels or chronic inflammation. This compromises the supply of oxygen to the heart and removal of waste products from it. This disease can lead to a heart attack.

Risk Factors

- Aged 55 years and older
- Diabetic
- Obese
- Smoker
- Sedentary lifestyle
- Excessive alcohol consumption

TYPICAL SYMPTOMS



Pain
around chest



Shortness
of breath



Fatigue



Heart attack

RECOMMENDED INTERVENTIONS

Coronary heart disease can be mitigated by consuming a diet which is low in fats and salt but rich in whole grains, fruits and vegetables. Regular exercise and managing stress effectively can help reduce the risks as well.

DEEP VEIN THROMBOSIS AND PULMONARY EMBOLISM

1

RESULT Risk Markers Detected



Slightly Lower Risk

Gene: DAB2IP

Genotype: AA

Description

Deep vein thrombosis is a condition in which one or more clots form in the veins deep in the body. This can occur in individuals who have underlying medical conditions that cause clotting, or when a person is in the same position for long periods of time, such as sitting in an airplane. It can be a life-threatening condition as clots can become dislodged and affect other organs such as the lungs, causing a pulmonary embolism.

Risk Factors

- Aged 45 years and above
- Major surgery
- Immobility
- Obese
- Frequent long-haul flights

TYPICAL SYMPTOMS



**Coughing
up blood**



**Lightheaded
or dizziness**



**Rapid
breathing**



Rapid pulse

RECOMMENDED INTERVENTIONS

Risks of developing deep vein thrombosis and pulmonary embolism can be reduced by using compression socks. It is also recommended to move during long haul flights. If not possible, simple mobility exercises help to reduce the chances of clot formation.

DILATED CARDIOMYOPATHY

3

RESULT Risk Markers Detected



Gene: BAG3	Gene: PRKCA	Gene: ZBTB17
Genotype: TT	Genotype: TC	Genotype: CC

Description

Dilated cardiomyopathy primarily affects adults aged 20 to 60 years. It occurs when the left ventricle becomes unable to pump blood efficiently. This is usually caused by dilation or thinning of the ventricle. The condition progressively affects other areas of the heart and eventually leads to heart failure. Arrhythmias, blood clots and valve problems are common complications.

Risk Factors

- More common in men than women
- Aged between 20 to 50 years
- Family history
- Immune system disorders, such as lupus
- Neuromuscular disorders
- Hypertension

TYPICAL SYMPTOMS



Fainting



Heart rhythm problems



Shortness of
breath with exertion



Swelling in the
legs and other areas

RECOMMENDED INTERVENTIONS

Risk for dilated cardiomyopathy is often inherited and unavoidable. Making good lifestyle choices is the best way to mitigate and minimize the risk of developing it. At-risk individuals are advised to not smoke, drink or consume drugs. Regular exercise is crucial; however, seek a doctor's advice to determine what exercises you can do. Getting enough sleep will allow your body to regenerate and recuperate.

DYSLIPIDEMIA

1

RESULT

Risk Markers Detected

Low Risk

Gene: Intergenic

Genotype: GG

Description

Dyslipidemia is the condition in which one or more of the lipids in the blood are not balanced. These lipids include cholesterol, low-density lipoproteins (LDL) and high-density lipoproteins (HDL). Dyslipidemia is a risk factor in developing cardiovascular disease, obesity and metabolic syndrome. It can be caused by genetic factors, lifestyle factors or a combination of both.

Risk Factors

- Aged over 50 years
- Overweight or obese
- Diabetic
- Family history
- Smoker
- Diet high in saturated fat and trans fat

TYPICAL SYMPTOMS



There are no obvious signs and symptoms for dyslipidemia. Detection is usually done via blood screening.

RECOMMENDED INTERVENTIONS

Dyslipidemia can be treated through making lifestyle changes. Elevated levels of lipids in blood plasma can be reduced through exercising and eating a healthy diet. Generally, consumption of foods rich in Omega-3, such as salmon and cod, can help in regulating lipid levels in the blood. Avoiding refined sugars or replacing them with complex carbohydrates, such as vegetables and whole grains, will also help.

GALLSTONE

0

RESULT

Risk Markers
Detected

Low Risk

Description

Gallstones are hard, crystalline deposits that can form in the gallbladder. They are caused primarily because of the overproduction of cholesterol or bilirubin in the gallbladder, which leads to crystallization. It is generally identified when the bile duct is blocked and causes pain in the abdomen. Gallstones can be in different sizes - from as small as a grain of sugar to as large as a ping-pong ball.

Risk Factors

- More common in women than men
- Family history
- Overweight or obese
- Diabetic
- Diet high in fat or low in fiber

TYPICAL SYMPTOMS



**Back pain between
shoulder blades**



**Pain in
right shoulder.**



**Sudden and intensifying
pain in center of abdomen**



**Sudden pain in upper
right portion of abdomen**

RECOMMENDED INTERVENTIONS

Gallstones can be avoided by monitoring your dietary intake. Diets rich in salt and fat can contribute to gallstone formation. Drinking enough water can also help mitigate the risk of gallstone formation.

GESTATIONAL DIABETES

0

RESULT
Risk Markers
Detected

Low Risk

Description

Gestational diabetes is a condition that affects a woman's blood glucose levels during pregnancy. It can occur anytime during pregnancy but is most common in the third trimester. It is caused by the body's inability to meet the increased requirement of insulin during pregnancy. It can occur in 2% to 8% of pregnant women who do not have any previous diabetic history.

Risk Factors

- Female only
- Pregnant
- History of diabetes
- Obesity

TYPICAL SYMPTOMS



Increased and frequent urination



Increased thirst



Weight loss even with increased appetite



Yeast infections

RECOMMENDED INTERVENTIONS

Gestational diabetes can occur in any woman. However, it can be mitigated with a healthy diet and exercise. Women who have had gestational diabetes are advised to have regular meals and monitor their blood sugar, as well as increase intake of dietary fiber.

GOUT

Description

Gout is the most common form of arthritis. This condition is caused by high levels of uric acid in the blood; where the uric acid crystallizes and forms crystal deposits in the joints and tendons. It causes intense pain, swelling and stiffness in joints - most commonly in the joint of the big toe. It occurs most commonly in men but can affect women post menopause.

Risk Factors

- More common in men than women
- Aged 40 years and above
- Excessive alcohol consumption
- Obesity
- Diet high in purines
- Diabetes
- Hypertension

TYPICAL SYMPTOMS



Intense joint pain



Joint stiffness



Joint swelling



Discomfort in affected part

3

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: GCKR

Genotype: TT

Gene: GCKR

Genotype: TT

Gene: SLC2A9

Genotype: AA

RECOMMENDED INTERVENTIONS

Risk of gout is best mitigated by limiting your intake of foods which are rich in purines, such as tomatoes, red meats and organ meats. Also reduce drinking alcoholic beverages and drinks sweetened with fructose.

GRAVES' DISEASE (HYPERTHYROIDISM)

4

RESULT Risk Markers Detected

Low Risk

Gene: LOC105378

Genotype: AG

Gene: TSHR

Genotype: TC

Gene: TSHR

Genotype: AG

Gene: Intergenic

Genotype: TT

Description

Graves' disease is an autoimmune condition affecting the thyroid. This disease develops when autoantibodies (antibodies that mistakenly target an individual's own tissue or organ) stimulate the thyroid, causing an overproduction of thyroid hormones. This condition is typically more common in women but can affect anyone below the age of 40. Having another form of autoimmune disease can also increase the risk of having Graves' disease.

Risk Factors

- More common in women than men
- Aged 60 years and above
- Family history of thyroid disease
- History of autoimmune disease
- Recent pregnancy

TYPICAL SYMPTOMS



**Fine tremor of
hands or fingers**



**Enlargement of
the thyroid gland**



Bulging eyes



**Thick and red skin on
shins or top of feet**

RECOMMENDED INTERVENTIONS

Graves' disease has a very strong genetic link; therefore, screening for genetic risks is an effective preventive measure. If you have a risk for this disease, it is recommended for your family members to undergo genetic screening as well; so that signs of the disease can be found out early and treatment can then start, if necessary.

HEART ATTACK

Description

Atherosclerosis of coronary arteries, or the deposition of fatty materials on the arterial wall, is the major underlying cause of heart attack. With time the plaque narrows the blood vessel. In extreme situations, the buildup can completely block blood flow through the artery. The blockage leads to a shortage in oxygen supply and, if prolonged, to a damage to the heart. In severe cases, death of parts of the heart muscle occurs affecting its physiological function.

Risk Factors

- Diabetes or metabolic syndrome
- High blood cholesterol or triglyceride levels
- High blood pressure
- Men aged above 45 years and women aged above 55 years
- Obesity
- Smoking

TYPICAL SYMPTOMS



Pain or squeezing sensation in chest or arms



Heartburn, nausea, indigestion or abdominal pain



Shortness of breath or sudden dizziness



Cold sweat or fatigue

6

RESULT Risk Markers Detected



Slightly Lower Risk

Gene: CDKN2B-AS

Genotype: AG

Gene: CDKN2B-AS

Genotype: AG

Gene: Intergenic

Genotype: AG

Gene: Intergenic

Genotype: GC

Gene: LPA

Genotype: TC

Gene: MIA3

Genotype: AC

RECOMMENDED INTERVENTIONS

You may mitigate the risk of heart attack by treating underlying medical conditions (e.g. hypertension, diabetes, coronary artery diseases), having a balanced diet, and maintaining a body mass index (BMI) of between 18.5 and 24.9.

HYPERTENSION

Description

Blood pressure is the pressure exerted against the walls of the blood vessels as it travels through the body, and is indicative of the pressure exerted by the heart to circulate blood. Hypertension, or commonly known as high blood pressure, is characterized by a marked, prolonged increase in blood pressure. It is usually linked to conditions such as strokes, aneurysms and heart attacks.

Risk Factors

- East or Southeast Asians
- Family history
- High body mass index (BMI)
- Stress
- Smoking
- Excessive alcohol consumption

TYPICAL SYMPTOMS



Chest pain



Dizziness



Headache



Nose bleed

3

RESULT
Risk Markers
Detected



Low Risk

Gene: UMOD	Gene: CSK	Gene: CNM2
Genotype: AA	Genotype: CC	Genotype: AG

RECOMMENDED INTERVENTIONS

Hypertension can be mitigated by maintaining a diet low in salt and fat. Foods rich in Omega-3 fatty acids are also beneficial. Exercise and regular monitoring can help mitigate and manage blood pressure.

HYPOTHYROIDISM

4

RESULT
Risk Markers
Detected

Slightly Lower Risk

Gene: Intergenic Genotype: GG	Gene: PDE8B Genotype: AA	Gene: VAV3 Genotype: TC
Gene: ZBTB10 Genotype: TC		

Description

Hypothyroidism is a condition where the thyroid is unable to produce enough thyroid hormones or precursors to support bodily function. This leads to a slowed metabolism, which can affect daily activities and essential bodily functions. As a result, individuals may experience symptoms such as fatigue, weight gain, slow heart rate, cold intolerance and dry skin.

Risk Factors

- More common in women than men
- Older than aged 60 years
- Exposure to radiation to the neck
- Prior thyroid surgery
- Family history of thyroid or autoimmune diseases
- Caucasian or Asian

TYPICAL SYMPTOMS



Increased sensitivity to cold



Puffy face



Muscle and joint aches tenderness and stiffness



Brittle fingernails and hair

RECOMMENDED INTERVENTIONS

Hypothyroidism can be caused by various factors such as autoimmune disease, radiation therapy or certain medications. It is difficult to prevent hypothyroidism, but early detection is important for effective treatment. People in the high-risk group should get tested for this disease as well.

METABOLIC SYNDROME

Description

Metabolic syndrome is a cluster of risk factors that can lead to the development of cardiovascular diseases, type 2 diabetes and stroke. Related conditions include increased blood pressure, elevated blood sugar, excess abdominal fat, and abnormal levels of cholesterol or triglyceride. The exact cause of metabolic syndrome is not known; however, this condition is closely linked to genetics, dietary habits, obesity and physical inactivity.

Risk Factors

- Risk increases with age
- Obesity
- Diabetes
- Non-alcoholic fatty liver disease
- Polycystic ovary syndrome
- Sleep apnea

TYPICAL SYMPTOMS



High blood sugar



High cholesterol



Increased blood pressure



Obesity

3

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: MTMR9

Genotype: AA

Gene: CDH13

Genotype: TG

Gene: Intergenic

Genotype: CC

RECOMMENDED INTERVENTIONS

A lifelong commitment is required to prevent conditions that can cause metabolic syndrome. Metabolic syndrome can be avoided by scheduling regular checkups, maintaining a healthy diet and adopting a physically active lifestyle.

NON-ALCOHOLIC FATTY LIVER DISEASE

1

RESULT

Risk Markers Detected

Low Risk

Gene: LOC157273

Genotype: AA

Description

Non-alcoholic fatty liver disease is characterized by the abnormal accumulation of fats in liver cells (hepatocytes). The disease is becoming increasingly more common globally, particularly in Western countries. It is the most common form of chronic liver disease affecting a quarter of the United States. Those with this disease are at risk of developing non-alcoholic steatohepatitis, which can lead to liver failure.

Risk Factors

- Aged over 50 years
- Overweight/obese
- Type 2 diabetes
- Hypertension
- Metabolic syndrome
- Smoking

TYPICAL SYMPTOMS



Enlarged liver
(early stage)



**Pain in upper
right abdomen**



**Swelling of the legs
due to fluid retention**



**Yellow discoloration of
skin and whites of eyes**

RECOMMENDED INTERVENTIONS

Lifestyle modifications play a major role in preventing liver damage and reducing liver function. One should reduce smoking and avoid high calorie intake to maintain a healthy weight. The regulation of blood lipids is important, as the liver is involved in lipid metabolism. This includes consuming polyunsaturated oils and healthy saturated oils like virgin coconut oil.

OBESITY

Description

Obesity is the condition in which an individual has a body mass index (BMI) of more than 30. It is characterized by an increase of overall body fats and possible central adiposity. It is a known contributor to a range of diseases, and is considered one of the largest epidemics of the 21st century.

Risk Factors

- Family history
- High-fat diet
- Diabetic
- Smoking
- Sedentary lifestyle
- Stress

TYPICAL SYMPTOMS



Underweight range =
BMI below 18.5



Normal range =
BMI 18.5 to 24.9
is normal



Overweight range =
BMI 25.0 to 29.9



Obesity range =
BMI 30.0 and higher

1

RESULT

Risk Markers Detected



Low Risk

Gene: BDKRB2

Genotype: TC

RECOMMENDED INTERVENTIONS

A change in diet is highly related to the occurrence of obesity. Long-term commitment is required to prevent obesity such as exercising regularly and following a healthy diet plan.

PERIPHERAL ARTERIAL DISEASE

Description

Peripheral arterial disease is a circulatory problem caused by the narrowing of arteries. As fatty deposits cause the artery wall to thicken, it becomes less elastic. This disease can lead to reduced blood flow to the limbs. When the arms and legs do not receive enough blood, it can result in pain when walking. This disease is linked to risk factors such as aging, diabetes and obesity.

Risk Factors

- Aged over 50 years
- Hypertension
- Diabetes
- High levels of homocysteine
- High cholesterol
- Smoking

TYPICAL SYMPTOMS



No pulse or a weak pulse in legs or feet



Pain in the legs when walking and exercising



Sores on toes, feet or legs that will not heal



Tingling sensation or numbness in feet

3

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: DAB2IP

Genotype: AA

Gene: Intergenic

Genotype: AG

Gene: Intergenic

Genotype: GC

RECOMMENDED INTERVENTIONS

The most effective way to prevent peripheral arterial disease is to maintain a healthy lifestyle. Besides that, it is important to control blood pressure and cholesterol with medications, to prevent further complications.

TYPE 1 DIABETES

4

RESULT Risk Markers Detected

Slightly Lower Risk

Gene: ERBB3

Genotype: TG

Gene: IL2RA

Genotype: TT

Gene: Intergenic

Genotype: TC

Gene: STAT4

Genotype: TG

Description

Type 1 diabetes increases blood sugar volume in insulin-dependent individuals. It can be indicated by a glucose blood volume of more than 126 mg/dl (over 7 mmol/l). Symptoms of type 1 diabetes include frequent thirst and urination, weight loss and blurred vision. Treatment focuses on managing blood sugar levels, diet and lifestyle, as there is no cure for type 1 diabetes.

Risk Factors

- History of autoimmune disease
- Family history: parent or sibling with type 1 diabetes

TYPICAL SYMPTOMS



Extreme hunger



Fatigue
and weakness



Increased thirst
and frequent urination



Unintended
weight loss

RECOMMENDED INTERVENTIONS

Given that type 1 diabetes is highly linked to genetics, genetic screening is crucial to detect the risk markers in a child so that further diagnosis and treatment can start for this disorder.

TYPE 2 DIABETES

6

RESULT Risk Markers Detected

Low Risk

Gene: ARAP1 Genotype: AA	Gene: CDKAL1 Genotype: TT	Gene: Intergenic Genotype: TC
Gene: Intergenic Genotype: AG	Gene: Intergenic Genotype: CC	Gene: Intergenic Genotype: AC

Description

Type 2 diabetes is an impairment in the way the body regulates and uses sugar as energy. It is caused by insulin resistance in the body. This impairment results in a high level of blood glucose, and is deemed to be the most common form of diabetes. Individuals with this disorder suffer decreased life expectancy and face a higher risk of cardiovascular diseases.

Risk Factors

- Being overweight or obese
- Aged 45 years or older
- Family history of diabetes
- Have high blood pressure
- Not physically active
- History of heart disease or stroke

TYPICAL SYMPTOMS



**Increased hunger
and unintended weight loss**



**Increased thirst
and frequent urination**



**Numbness or
tingling in limbs**



**Slow-healing sores
or frequent infections**

RECOMMENDED INTERVENTIONS

Type 2 diabetes can be minimized by maintaining a healthy diet and reducing the amount of daily sugar consumption. Individuals with a family history of type 2 diabetes should monitor their blood glucose regularly for early detection and treatment.

Cancer is caused by damage to DNA, and it manifests as uncontrollable growth of abnormal cells which spread to surrounding tissues and organs. Among the major factors that contribute to cancer are genetics, lifestyle, infection and the environment. While little can be done to change genetic factors that contribute to cancer, knowing your genetic risks can help to narrow down the types of cancers to look out for. Earlier and regular screening or other appropriate actions can then be taken to address the risks.



Onco

ACUTE LYMPHOBLASTIC LEUKEMIA

Description

Acute Lymphoblastic Leukemia occurs when white blood cells grow uncontrollably. These cells overtake the bone marrow and cause the production of other blood cells to be stifled. Eventually, these cells migrate to the bloodstream and to other organs. It is classified as a blood cancer which develops most commonly during childhood, particularly before the age of 5 years.

Risk Factors

- Children aged 2 to 5 years
- More common in boys than girls
- Radiation exposure

TYPICAL SYMPTOMS



**Pale skin
(anemia)**



**Fatigue
and weakness**



**Easy bruising
and bleeding**



**Unexplained and
prolonged fever**

3

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: ARID5B

Genotype: TC

Gene: ARID5B

Genotype: TG

Gene: GATA3

Genotype: AA

RECOMMENDED INTERVENTIONS

There are no known preventive measures for acute lymphoblastic leukemia, but reducing exposure to high-dose radiation and certain chemicals may help lower risk. While avoiding infections does not prevent acute lymphoblastic leukemia, it helps reduce complications in individuals with weakened immune systems.

BASAL CELL CARCINOMA

1

RESULT

Risk Markers Detected

Low Risk

Gene: CLPTM1L

Genotype: CC

Description

Basal cell carcinoma is one of the most common forms of skin cancer. It affects the basal cells, which are the cells responsible for producing new skin cells, as old ones are sloughed off. Most basal cell carcinomas are thought to be caused by long-term exposure to ultraviolet radiation, such as from the sun. It usually appears as a change in the skin, such as lesions on the face, scalp and neck.

Risk Factors

- Exposure to ultraviolet rays
- Aged over 50 years
- Fair-skinned
- Male
- Chronic infections and skin inflammation

TYPICAL SYMPTOMS



Growths with a slightly raised border



Open sores that bleed, ooze or crust repeatedly



Patches of red and irritated skin



Raised areas of skin that are shiny or pearly

RECOMMENDED INTERVENTIONS

Basal cell carcinoma can be avoided most easily by avoiding exposure to ultraviolet (UV) rays. This can be achieved by applying sunscreen with a sun protection factor (SPF) rating higher than 30, avoiding tanning beds and wearing protective layers such as caps and long sleeve clothing. It is also recommended to do regular skin examination on yourself and discuss any changes with your doctor.

BLADDER CANCER

Description

Bladder cancer occurs when the cells which line the bladder are defective. This causes uncontrolled growth which can lead to cancer. These cells can also be found in the ureter and kidneys. It is often detected by the presence of blood in the urine, painful urination, and back pains. It can be detected early and effectively treated. However, recurrence is common.

Risk Factors

- Aged 55 years and above
- More common in men than women
- Frequent bladder infection
- Family history of bladder cancer or Lynch syndrome
- Smoking

TYPICAL SYMPTOMS



Blood in urine



Fatigue



Frequent and painful urination



Abdominal or back pain

5

RESULT Risk Markers Detected

Low Risk

Gene: CASC11

Genotype: GG

Gene: Intergenic

Genotype: AG

Gene: Intergenic

Genotype: TT

Gene: Intergenic

Genotype: TC

Gene: PSCA

Genotype: TC

RECOMMENDED INTERVENTIONS

The primary risk factor for bladder cancer is smoking, and quitting significantly reduces risk. Avoiding occupational exposure to carcinogenic chemicals, staying hydrated, and consuming a diet rich in fruits and vegetables may also help prevent the disease.

BREAST CANCER (FEMALE)

Description
Breast cancer is the formation of cancerous cells in the breast and it is the most common cancer diagnosed among women. Cancerous growths usually occur within the milk ducts of breasts, with symptoms such as lumps, redness and nipple discharge. Breast cancer risk increases with age and family history. However, survival rates have increased due to early detection and better treatment approaches.

- Risk Factors**
- Mainly women
 - Rarer in men
 - Risk increases with age
 - Family history of breast cancer
 - Period starts at a younger age
 - Menopause at a younger age
 - Having first child at an older age
 - Never been pregnant
 - Postmenopausal hormone therapy

TYPICAL SYMPTOMS



Breast or nipple pain



Nipple discharge (other than breast milk)



Palpable mass or lump in breast



Redness or thickening of nipple or breast skin

6

RESULT

Risk Markers Detected

Moderate Risk

Gene: ADAM29 Genotype: AC	Gene: FGFR2 Genotype: AG	Gene: FGFR2 Genotype: AG
Gene: Intergenic Genotype: AA	Gene: Intergenic Genotype: AG	Gene: Intergenic Genotype: TC

RECOMMENDED INTERVENTIONS

Breast cancer is highly treatable if discovered in the early stages. Hence, screening for genetic risk is important if there is a family history of breast cancer. Individuals should also conduct regular self-examination of the breast and consult a professional if there are any changes in the breast.

BREAST CANCER (MALE)

Description
Although rare, a man's breast cells can still develop cancer despite not having any milk-producing ducts. If a man is diagnosed with breast cancer, the chances of his daughters or granddaughters inheriting it will be higher due to the genetic predisposition. Most men are diagnosed with breast cancer when they are 50 years old and above.

- Risk Factors**
- Mainly women
 - Rare in men
 - Risk increases with age
 - Family history of breast cancer

TYPICAL SYMPTOMS



Breast or nipple pain



Palpable mass or lump in breast



Redness or thickening of nipple or breast skin



Swelling of all or part of breast

6

RESULT
Risk Markers Detected

Moderate Risk

Gene: ZNF365 Genotype: TC	Gene: Intergenic Genotype: AA	Gene: Intergenic Genotype: AG
Gene: Intergenic Genotype: TC	Gene: ADAM29 Genotype: AC	Gene: FGFR2 Genotype: AG

RECOMMENDED INTERVENTIONS

Male with genetic predisposition for breast cancer should encourage their female descendants to screen the BRCA1 and BRCA2 genes for breast cancer risks, in addition to doing regular checkups.

CERVICAL CANCER (FEMALE)

4

RESULT Risk Markers Detected

Low Risk

Gene: Intergenic

Genotype: TG

Gene: IL6

Genotype: GG

Gene: IL12B

Genotype: TG

Gene: CTLA4

Genotype: AG

Description

Cervical cancer occurs when the cells in the lining of the cervix are damaged, causing them to grow uncontrollably, resulting in cancer. It can be caused by genetics, or external factors. A major cause of this cancer is due to infection by the human papillomavirus (HPV), although a large number of women infected by HPV may not develop cervical cancer.

Risk Factors

- Women only
- Aged between 20 and 50 years
- Family history of cervical cancer
- Multiple sexual partners
- Multiple pregnancies
- Smoking

TYPICAL SYMPTOMS



Abnormal vaginal bleeding



Pain during intercourse



Pain in the
lower abdomen



Unusual discharge
from vagina

RECOMMENDED INTERVENTIONS

The most effective way to prevent cervical cancer is through vaccination against human papillomavirus (HPV). This will significantly reduce the risk of developing cervical cancer. Regular pap smears are recommended for females when they come of age. If sexually active, the frequency should be increased. Practicing safe sex also helps mitigate risks. Not smoking, reducing alcohol consumption and having a healthy lifestyle can also reduce the risks.

CERVICAL CANCER (MALE)

Description

Although men will not develop cervical cancer, they risk passing down the genes to their daughters and granddaughters. Should a male individual tests positive for cervical cancer risk, it is recommended that his female descendants check for genetic predispositions to this cancer.

Risk Factors

- Does not exist

TYPICAL SYMPTOMS



Men do not develop cervical cancer, therefore, there are no symptoms.

4

RESULT

Risk Markers Detected

Low Risk

Gene: CTLA4

Genotype: AG

Gene: IL12B

Genotype: TG

Gene: IL6

Genotype: GG

Gene: Intergenic

Genotype: TG

RECOMMENDED INTERVENTIONS

While men do not develop cervical cancer, those with a strong family history should encourage their daughters to undergo regular gynecological check-ups when they reach the appropriate age. Additionally, HPV vaccination is important for both men and women, as it not only reduces the risk of cervical cancer in women but also protects men from HPV-related cancers such as penile, anal, and throat cancers.

CHRONIC LYMPHOCYTIC LEUKEMIA

1

RESULT
Risk Markers
Detected



Gene: IRF4

Genotype: AG

Description

Leukemia is the cancer of the blood. Chronic lymphocytic leukemia (CLL) occurs when the bone marrow produces too many lymphocytes. It is the most common type of leukemia in adults, typically does not have symptoms, and the cancer will usually progress slowly. However, when symptoms develop, they typically are enlarged but painless lymph nodes, splenic enlargements, frequent infections and night sweats, among others.

Risk Factors

- Family history
- More common in men
- Older age
- Exposure to herbicides, radiation and viral infections

TYPICAL SYMPTOMS



Enlarged lymph nodes



Fever



Night sweats



Pain or a sense of fullness after meals

RECOMMENDED INTERVENTIONS

There is no foolproof way of preventing chronic lymphocytic leukemia. Certain preventive measures for this cancer include minimizing exposure to radiation, herbicides and insecticides, and avoiding tobacco products.

COLORECTAL CANCER

Description

Colorectal cancer begins in the colon or rectum. Most of this cancer starts as growths, called polyps, on the inner lining of the colon or rectum. These polyps may be discovered first before they turn cancerous - which can take 10 to 15 years. Therefore, early screening is important to get a more effective treatment.

Risk Factors

- Aged over 50 years
- Family history
- Genetic conditions (hereditary non-polyposis colon cancer)
- Turcot syndrome and Peutz-Jeghers syndrome
- Low fiber diet
- Excessive alcohol consumption
- Obesity

TYPICAL SYMPTOMS



Feeling that bowel doesn't empty completely



Persistent abdominal discomfort (cramps, pain, bloating)



Persistent change in bowel habits (constipation, diarrhea, change in stool texture)



Rectal bleeding or blood in stool

6

RESULT Risk Markers Detected

High Risk

Gene: C11orf93 Genotype: AC	Gene: CDH1 Genotype: AG	Gene: Intergenic Genotype: AA
Gene: Intergenic Genotype: TC	Gene: Intergenic Genotype: TT	Gene: Intergenic Genotype: TG

RECOMMENDED INTERVENTIONS

Colorectal cancer treatment includes surgery, radiation therapy, and chemotherapy. The treatment approach varies between individuals, depending on their overall health. The risk of colorectal cancer can be reduced by maintaining a high-fiber diet, exercising regularly, and avoiding excessive alcohol consumption. Regular screening through colonoscopies, starting at age 45 or earlier for those with a family history, is crucial for early detection and prevention.

ENDOMETRIAL CANCER (FEMALE)

Description

Endometrial cancer is a hormone-related cancer which affects the inner lining of the uterus, the endometrium. It occurs when the cells of this lining grow uncontrollably. There are multiple causes for the overgrowth; however, it is most commonly related to overexposure to estrogen. Common symptoms are excessive vaginal bleeding, bleeding between periods and pelvic pain. The risk of this cancer increases with age.

Risk Factors

- Female only
- Late menopause
- Never been pregnant
- Use of oral contraceptives
- Ovarian tumors
- Polycystic ovarian syndrome
- History of breast cancer
- Family history of endometrial cancer and non-polyposis colon cancer
- Obesity

TYPICAL SYMPTOMS



Abnormal, watery or blood-tinged discharge from vagina



Pain during intercourse



Prolonged periods or bleeding between periods



Vaginal bleeding after menopause

5

RESULT

Risk Markers
Detected



Moderate Risk

Gene: NME1

Genotype: TT

Gene: CYP19A1

Genotype: AC

Gene: ESR1

Genotype: AA

Gene: HNF1B

Genotype: AG

Gene: PGR

Genotype: TT

RECOMMENDED INTERVENTIONS

Risk of endometrial cancers can be mitigated by maintaining a healthy lifestyle. One of the known causes of overproduction of estrogen is obesity. Keeping healthy, exercising and maintaining a healthy diet will help minimize this risk. Being aware of irregularities in menstruation will also help.

ENDOMETRIAL CANCER (MALE)

Description

Endometrial cancer is a hormone-related cancer which affects the inner lining of the uterus. While men are not at risk of developing the cancer, the risk can be passed on to your daughters and granddaughters. Should a male individual tests positive for endometrial cancer risk, it is recommended that his female descendants check for genetic predispositions to this cancer.

Risk Factors

- Does not exist

TYPICAL SYMPTOMS



Men do not develop endometrial cancer, therefore, there are no symptoms.

5

RESULT

Risk Markers
Detected



Moderate Risk

Gene: CYP19A1

Genotype: AC

Gene: ESR1

Genotype: AA

Gene: HNF1B

Genotype: AG

Gene: NME1

Genotype: TT

Gene: PGR

Genotype: TT

RECOMMENDED INTERVENTIONS

Although you are not at risk as endometrial cancer only affects women, however, should there be a strong family history of this cancer, your daughters should go for regular check-ups at the gynecologist when they come of age.

ESOPHAGEAL CANCER

4

RESULT Risk Markers Detected

Moderate Risk

Gene: Intergenic

Genotype: TT

Gene: HECTD4

Genotype: AA

Gene: ADH7

Genotype: CC

Gene: ADH1B

Genotype: CC

Description

Esophageal cancer occurs when the cells in the esophagus, a tube connecting the stomach to the throat, grow uncontrollably. It usually occurs because of damage to the cells lining the inner esophagus. This disease can be brought on by cigarette smoking, excessive alcohol drinking or gastric reflux. There are many subtypes of esophageal cancer; however, adenocarcinoma is one of the most common ones.

Risk Factors

- Aged over 45 years
- More common in men than women
- Excessive alcohol consumption
- Smoking
- Gastroesophageal reflux disease
- Barrett's esophagus

TYPICAL SYMPTOMS



**Trouble
swallowing food**



Chest pain



Heartburn



Vomiting blood

RECOMMENDED INTERVENTIONS

Esophageal cancer can be mitigated by reducing or stopping habits such as smoking and drinking alcohol. Early intervention of gastric reflux issues also helps to prevent esophageal cancer.

GALLBLADDER CANCER

1

RESULT

Risk Markers Detected

Low Risk

Gene: OGG1

Genotype: CG

Description

The gallbladder is a small, pear-shaped organ nestled under the liver. It is responsible for the production of bile, which is crucial in the digestion of fats. Cancer occurs in the gallbladder when the cells in the organ grow unabated. Gallbladder cancer is rare and can go undetected. It is usually detected in the later stages when prognosis can be dire.

Risk Factors

- Older age
- More common in men than women
- History of gallstone disease
- Other gallbladder diseases and conditions

TYPICAL SYMPTOMS



Abdominal bloating



Abdominal pain
(upper right
portion of abdomen)



Nausea



Yellowing of the
skin and whites
of the eyes

RECOMMENDED INTERVENTIONS

Risk for developing gallbladder cancer can be lowered by treating the underlying conditions that are thought to contribute to it, including gallstones, chronic inflammation and polyps.

GLIOMA DEVELOPMENT (BRAIN TUMOR)

3

RESULT Risk Markers Detected

Slightly Lower Risk

Gene: TERT

Genotype: CC

Gene: PHLDB1

Genotype: AG

Gene: CDKN2B-AS

Genotype: AG

Description

Gliomas are one of the most common tumors which occur in the brain. They occur when the glial cells surrounding the neuron become defective. Gliomas affect brain functions and can be life-threatening, depending on where they occur in the brain. There are various types of gliomas and they are usually categorized depending on the types of cells that cause the tumor.

Risk Factors

- Can affect individuals of all ages
- Certain types of gliomas (ependymomas and pilocytic astrocytomas) are more common among children and young adults
- Exposure to radiation
- Inherited nervous system disorders

TYPICAL SYMPTOMS



Nausea and vomiting



Personality or
behavioural changes



Drowsiness, confusion,
memory loss or coma



Changes in hearing,
taste or smell

RECOMMENDED INTERVENTIONS

There are no established prevention strategies for gliomas, but minimizing exposure to high-dose radiation and identifying genetic predisposition through family history may help assess risk.

HEPATOCELLULAR CARCINOMA (LIVER CANCER)

Description

Hepatocellular carcinoma is the most common type of cancer originating in the liver. It is caused by damage to the liver cells due to prolonged or sudden trauma. The cancer has a higher incidence rate in countries which have elevated cases of hepatitis B and C cases. It occurs most commonly as a result of cirrhosis, chronic hepatitis or toxin exposure.

Risk Factors

- Hepatitis B or C infection
- Excessive alcohol consumption
- Obesity
- Diabetes

TYPICAL SYMPTOMS



Enlarged liver



Enlarged spleen



Swelling or fluid build-up in abdomen



Yellowing of skin and eyes

6

RESULT

Risk Markers
Detected



Slightly Lower Risk

Gene: CTLA4 Genotype: AG	Gene: EGF Genotype: GG	Gene: HSPA5 Genotype: TC
Gene: KIF1B Genotype: AG	Gene: LINC01605 Genotype: TC	Gene: LOC101929 Genotype: AG

RECOMMENDED INTERVENTIONS

The risk of developing hepatocellular carcinoma increases in individuals with prolonged chronic liver diseases, such as hepatitis B and C. Liver scarring from cirrhosis and fatty liver disease further elevates this risk. Preventive measures include minimizing exposure to toxins and substances that burden the liver while maintaining a healthy diet and body weight. Vaccination against hepatitis B, antiviral treatment for hepatitis C, and limiting alcohol consumption significantly lower the risk of liver cancer.

HODGKIN'S LYMPHOMA

2

RESULT Risk Markers Detected

Slightly Lower Risk

Gene: Intergenic

Genotype: TC

Gene: TCF3

Genotype: AG

Description

Hodgkin's lymphoma is a cancer affecting the lymphatic system. It affects a specific type of lymph cells, primarily the B-lymphocytes. It is characterized by enlarged lymph nodes and proliferation of lymph cells outside the lymphatic system. It can also present with cold sweats, itching and fatigue. There are many forms of this disease, depending on the location and composition of the tumor.

Risk Factors

- Aged between 15 and 30 years, and over 55
- More common in men than women
- Family history of lymphoma
- Previous infection with infectious mononucleosis

TYPICAL SYMPTOMS



Enlarged lymph nodes



Fever which may come and go



Night sweats



Unexpected weight loss

RECOMMENDED INTERVENTIONS

Hodgkin's lymphoma is primarily caused by damage to the mechanisms involved in the creation of lymphocytes. This can occur as a result of prolonged, underlying infection or ionizing radiation. To mitigate the risk of developing the condition, it is recommended to minimize exposure to radiation and seek immediate medical attention when dealing with infections, particularly the Epstein Barr Virus (EBV) and Human Immunodeficiency Virus (HIV).

KIDNEY CANCER

Description

Kidney cancer, or renal cell carcinoma, is a type of cancer that begins in the kidneys. It can occur when the cells lining the small filtration tubules in the kidney propagate unabated due to infection or genetic damage. It can cause blood in urine, tiredness, loss of weight and persistent pain in the sides of the body.

Risk Factors

- Risk increases with age
- Smoking
- Obesity
- Hypertension
- Excessive alcohol consumption
- Family history

TYPICAL SYMPTOMS



Blood in the urine



Low red blood cell count



Lower back pain on one side



Lump on the side or lower back

3

RESULT Risk Markers Detected

Low Risk

Gene: APOE

Genotype: TG

Gene: EPAS1

Genotype: AA

Gene: LOC102724

Genotype: GG

RECOMMENDED INTERVENTIONS

Smoking, obesity, and high blood pressure are key risk factors for kidney cancer. While it cannot be entirely prevented, the risk can be reduced by adopting a healthy lifestyle, maintaining a balanced diet, managing weight and blood pressure, staying hydrated, and avoiding long-term NSAID use. Genetic predisposition also plays a significant role in susceptibility to kidney cancer.

LARYNGEAL CANCER

Description

Laryngeal cancer is most commonly associated with smoking, alcohol consumption and infection by the human papillomavirus (HPV). It is characterized by an overgrowth of cancer cells in the larynx, particularly the glottic structures. This affects the patient's ability to speak, breathe and swallow. Adults are mostly affected by this cancer, and it is more common in men than women.

Risk Factors

- Aged 55 years and older
- More common in men than women
- Smoking
- Excessive alcohol consumption
- Exposure to certain hazardous substances, e.g. asbestos

TYPICAL SYMPTOMS



Persistent sore throat



Difficulty or pain when swallowing



Lump in the neck or throat



Hoarseness or change in voice

2

RESULT Risk Markers Detected



Moderate Risk

Gene: ADH7

Genotype: CC

Gene: Intergenic

Genotype: GG

RECOMMENDED INTERVENTIONS

The survival rate of laryngeal cancer depends largely on its location and the stage at which it is diagnosed. The risk can be reduced by avoiding smoking and excessive alcohol consumption, maintaining a vegetable-rich diet, and treating infections promptly to prevent complications. Additionally, HPV vaccination helps prevent HPV-associated laryngeal cancers.

LUNG CANCER

4

RESULT

Risk Markers Detected

Slightly Lower Risk

Gene: TERT

Genotype: CC

Gene: CLPTM1L

Genotype: CC

Gene: CLPTM1L

Genotype: CC

Gene: CLPTM1L

Genotype: CC

Description

Lung cancer usually develops in the cells lining the bronchi, and also parts of the lung. Later, it may spread to the lymph nodes or other organs in the body, such as the liver or brain. Lung cancers are normally grouped into two main types - small cell lung cancer (which is less common but more aggressive) and non-small cell lung cancer (which is more common but slower-growing). Lung cancer is one of the most common causes of cancer-related deaths.

Risk Factors

- Aged 65 years and older
- Heavy smoker
- Second hand smoke
- Family history of lung cancer

TYPICAL SYMPTOMS



Chest pain that is worse with deep breathing, coughing or laughing



Coughing up blood or dark-colored sputum



Onset of wheezing and shortness of breath



Persistent or worsening cough

RECOMMENDED INTERVENTIONS

Lung cancer is the leading cause of cancer-related deaths, primarily caused by smoking, including exposure to secondhand smoke. Other risk factors include exposure to asbestos, radon gas, and other carcinogens. While lung cancer cannot be completely prevented, the risk can be significantly reduced by quitting smoking, maintaining a healthy lifestyle, testing for radon exposure, and using protective equipment in high-risk workplaces.

MELANOMA

Description

Melanoma is a form of skin cancer which affects the cells responsible for producing melanin in the skin. It is one of the most serious types of skin cancer, and it usually occurs on the outside of the body, especially the skin. However, it can also occur in the eyes and internal structures of the body. Ultraviolet (UV) radiation is thought to play a role in the development of this cancer.

Risk Factors

- Fair skin
- Excessive ultraviolet light exposure
- History of sunburn
- Family history of melanoma
- Have unusual moles

TYPICAL SYMPTOMS



One half of a mole or birthmark does not match the other



Edges are irregular, ragged, notched or blurred



Color is not the same all over



Spot is larger than 6 millimeters across

3

RESULT

Risk Markers Detected



Low Risk

Gene: CDK10

Genotype: AG

Gene: MTAP

Genotype: AG

Gene: PLA2G6

Genotype: AA

RECOMMENDED INTERVENTIONS

The exact cause of melanomas is not determined, but high exposure to ultraviolet (UV) radiation and tanning lamps greatly increase the risk. Individuals should always apply sunscreen on exposed skin to avoid direct sunlight exposure. It is important to check for lesions or moles on the skin, and consult a professional if needed.

MENINGIOMA

2

RESULT

Risk Markers Detected



Moderate Risk

Gene: BRIP1

Genotype: CC

Gene: SOD3

Genotype: CC

Description

Meningioma is a cancer that occurs in the membranes surrounding the brain and spinal cord (meninges). The cancer tumor is often slow growing and it is usually not malignant. It also does not spread to other parts of the body. Meningioma usually develops in individuals aged between 35 and 45 years, and affects twice as many women than men.

Risk Factors

- Most common in adults aged 65 years or older
- Women more likely than men
- Exposure to radiation
- Neurofibromatosis type 2

TYPICAL SYMPTOMS



Language difficulty



Loss of smell



Personality or behavior changes



Seizures

RECOMMENDED INTERVENTIONS

While most cases of meningioma are not preventable, minimizing radiation exposure and identifying genetic predisposition may help assess risk. Avoiding high-voltage areas and limiting exposure to environmental risk factors can also help reduce the likelihood of tumor growth and associated long-term complications.

NASOPHARYNGEAL CANCER

3

RESULT Risk Markers Detected



Moderate Risk

Gene: Intergenic

Genotype: TC

Gene: Intergenic

Genotype: AA

Gene: TNFRSF19

Genotype: AG

Description

Nasopharyngeal cancer is a rare tumor that starts in the upper part of the throat, behind the nose. This cancer is most common in various parts of Asia. Studies have shown that family history, genetic variants, diet and smoking may be a risk factor to developing this cancer. Common symptoms include lump in neck, headache, hearing loss and blurry vision.

Risk Factors

- More common in men than women
- Adults aged between 30 and 50 years
- Commonly affects individuals in parts of China, Southeast Asia and Northern Africa
- Salt-cured foods
- Family history
- Epstein-Barr virus
- Tobacco smoking
- Excessive alcohol consumption

TYPICAL SYMPTOMS



Blocked or stuffy nose (usually 1 side)



Facial pain or numbness



Hearing loss or ringing in ear (especially in 1 ear only)



Lump or swelling in neck (for over 3 weeks)

RECOMMENDED INTERVENTIONS

Nasopharyngeal cancer requires regular follow-ups with the doctor before, during and after treatment. The type of treatment is dependent on the location and stage of tumor, and also on one's overall health. This cancer can be prevented by avoiding cigarette smoking, cutting down on diets high in salt, reducing alcohol intake and avoiding workplace pollutants such as wood dust and formaldehyde.

NON-HODGKIN'S LYMPHOMA

Description

Non-Hodgkin's lymphoma is a cancer of the lymphatic system characterized by the abnormal proliferation and growth of lymphocytes. This can lead to the formation of tumors throughout the body. There are many types of Non-Hodgkin's lymphoma - the most common of which are diffuse large B-cell lymphoma and follicular lymphoma. Common symptoms include bone pain, enlarged lymph nodes and night sweats.

Risk Factors

- Aged over 60 years
- More common in men than women
- Immunocompromised
- Autoimmune diseases
- Family history
- Exposure to chemicals, e.g. insecticide

TYPICAL SYMPTOMS



Abdominal pain or swelling



Chest pain, coughing or trouble breathing



Enlarged lymph nodes



Swollen lymph nodes in your neck, armpits or groin

4

RESULT
Risk Markers
Detected

Low Risk

Gene: FCGR2A Genotype: AG	Gene: Intergenic Genotype: TC	Gene: Intergenic Genotype: TT
Gene: TCF3 Genotype: AG		

RECOMMENDED INTERVENTIONS

Individuals diagnosed with non-Hodgkin's lymphoma commonly do not have any risk factors. However, one can lower the risk of this disease by avoiding overexposure to radiations like x-rays and treating early viral infections to prevent further complications.

ORAL AND THROAT CANCERS

2

RESULT Risk Markers Detected



Moderate Risk

Gene: ADH7

Genotype: CC

Gene: NAA25

Genotype: AA

Description

Oral and throat cancers develop in the mouth, tongue, tonsils and the back of the throat. Cancers found in this region are known as squamous cell carcinoma, where the cancer cells develop at the lining of the mouth of throat. Other related cancers are minor salivary gland cancers. Treatments include chemotherapy, surgery or radiation therapy.

Risk Factors

- Smoking
- Aged 55 years and above
- More common in men than women
- Excessive alcohol consumption
- Betel quid chewing habit
- Human papillomavirus (HPV) infection

TYPICAL SYMPTOMS



**Jaw pain
or stiffness**



**Painful chewing
or swallowing**



**Thickening of skin
or lining of mouth**



**White or reddish
patch inside mouth**

RECOMMENDED INTERVENTIONS

Oral and throat cancer may be avoided by reducing alcohol intake and quitting one's smoking habit. Individuals should also wear their dentures or orthodontic braces correctly, and maintain a healthy diet to ensure they receive proper nutrition.

OSTEOSARCOMA

Description

Osteosarcoma is a type of bone cancer that is commonly found in long bones, such as in the legs, or sometimes, in the arms. However, it can also develop in any bone in the body. This disease can either be inherited or acquired from external factors, such as radiation. It occurs more often in males than females, and those under the age of 25 years.

Risk Factors

- Aged 10 to 30 years
- Taller children
- More common in men than women
- Inherited bone diseases

TYPICAL SYMPTOMS



Incidental fractures caused by low mineralization of bone



Pain and tenderness at affected areas of bones



Tumor formation



Swelling and redness at the site of tumor

2

RESULT

Risk Markers Detected

Low Risk

Gene: ADAMTS6	Gene: GRM4
Genotype: TC	Genotype: TC

RECOMMENDED INTERVENTIONS

Genetic conditions, such as Li-Fraumeni syndrome, play a significant role in osteosarcoma risk. Early genetic screening for high-risk individuals can aid in early detection and management. While osteosarcoma cannot be entirely prevented, minimizing exposure to radiation may help reduce the risk. Genetic testing is also recommended for timely diagnosis and treatment planning.

OVARIAN CANCER (FEMALE)

5

RESULT Risk Markers Detected



Moderate Risk

Gene: ABHD8

Genotype: CC

Gene: HOXD3

Genotype: CC

Gene: Intergenic

Genotype: GG

Gene: Intergenic

Genotype: TC

Gene: RB1

Genotype: GG

Description

Ovarian cancer develops in the ovaries of female individuals, and is one of the most common types of cancer in women. It often goes undetected until it reaches a late stage where the cancer has spread to the pelvis, abdomen and other distant organs, making it more difficult to treat. This cancer usually develops in women who have gone through menopause (normally over 50 years old), but it can also affect younger women.

Risk Factors

- Female only
- Aged over 50 years
- Family history of breast and colorectal cancer
- Never pregnant before or low number of pregnancies
- Use fertility drugs, e.g. clomiphene
- Diet rich in fats

TYPICAL SYMPTOMS



**Abdominal swelling
with weight loss**



**Abdominal or
pelvic pain**



**Trouble eating or
feeling full quickly**



**Urinary symptoms
such as frequent urination**

RECOMMENDED INTERVENTIONS

There is no proven way to prevent ovarian cancer, but one can reduce the risk by maintaining a healthy body weight, practicing a balanced diet, and minimizing smoking and alcohol drinking. Screening for genetic predispositions is also recommended for early detection and treatment.

OVARIAN CANCER (MALE)

Description

Ovarian cancer does not occur in men; however, they may pass on genetic risks to their daughters. If a male individual tests positive for ovarian cancer risk markers, it is advisable for his daughters and granddaughters to check for genetic predispositions, as there is a possibility they may have inherited genetic predispositions from him.

Risk Factors

- Does not exist

r

TYPICAL SYMPTOMS



Men do not develop ovarian cancer, therefore, there are no symptoms.

5

RESULT
Risk Markers
Detected



Moderate Risk

Gene: ABHD8 Genotype: CC	Gene: HOXD3 Genotype: CC	Gene: Intergenic Genotype: GG
Gene: Intergenic Genotype: TC	Gene: RB1 Genotype: GG	

RECOMMENDED INTERVENTIONS

Female descendants of males with genetic predisposition for ovarian cancer should go for more regular screenings for early detection. They should also stop smoking, reduce alcohol intake and maintain a healthy body weight and diet.

PANCREATIC CANCER

Description

Pancreatic cancer develops in the tissues of your pancreas. Several types of growths can begin in the pancreas, which include cancerous and noncancerous tumors. The most common type of pancreatic cancer begins in cells that line the ducts that carry digestive enzymes away from the pancreas. Pancreatic cancer is seldom detected in the early stages as it usually exhibits no symptoms until it has spread to other organs.

Risk Factors

- Risk increases with age
- Obesity
- Family history of pancreatic cancer
- Smoking
- Diabetes
- Chronic pancreatitis
- Family history of genetic syndromes, e.g. BRCA2 gene mutation, Lynch syndrome

TYPICAL SYMPTOMS



**Abdominal
or back pain**



**Dark urine and
colorless stools**



**Digestion problems
or diarrhea**



**Yellow pigmentation
of skin**

2 **RESULT**
Risk Markers
Detected
Low Risk

Gene: HIF1A

Genotype: CC

Gene: SBF2

Genotype: TT

RECOMMENDED INTERVENTIONS

The risk of pancreatic cancer can be reduced by avoiding smoking and excessive alcohol consumption, maintaining a healthy weight, and managing blood sugar levels through a balanced diet. Prompt treatment of underlying conditions like diabetes can help prevent complications that may contribute to pancreatic cancer. Genetic testing is also advisable for individuals with a strong family history to assess their risk.

PROSTATE CANCER (FEMALE)

Description

Although women will not develop prostate cancer, they risk passing down the genes to their sons and grandsons. Should a female individual tests positive for prostate cancer risk, it is recommended that her male descendants check for genetic predispositions to this cancer.

Risk Factors

- Does not exist

TYPICAL SYMPTOMS



Women do not develop prostate cancer, therefore, there are no symptoms.

5

RESULT

Risk Markers Detected



Low Risk

Gene: HNF1B

Genotype: AG

Gene: Intergenic

Genotype: CC

Gene: Intergenic

Genotype: TC

Gene: Intergenic

Genotype: TG

Gene: MSMB

Genotype: TC

RECOMMENDED INTERVENTIONS

As females are only able to pass the genetic risks for prostate cancer to their sons or grandsons, it is important for male descendants to screen for their genetic risks for this cancer as well, besides getting regular checkups.

PROSTATE CANCER (MALE)

Description

Prostate cancer is normally confined to the prostate gland and usually grows relatively slowly. It is one of the most common types of cancer for men. Some prostate cancer grows slowly and may need minimal treatment. However, sometimes it can be more aggressive and may spread quickly to other parts of the body. Prostate cancer that is detected early has a higher chance for successful treatment.

Risk Factors

- Male only
- Aged over 50 years
- Family history
- Obesity
- Smoking

TYPICAL SYMPTOMS



Trouble urinating



**Decreased force
in stream of urine**



**Blood in the
urine or semen**



**Discomfort in
pelvic area**

5

RESULT Risk Markers Detected

Low Risk

Gene: HNF1B

Genotype: AG

Gene: Intergenic

Genotype: CC

Gene: Intergenic

Genotype: TC

Gene: Intergenic

Genotype: TG

Gene: MSMB

Genotype: TC

RECOMMENDED INTERVENTIONS

Various treatments are available for prostate cancer patients such as surgery, radiation therapy and chemotherapy. An individual may also reduce the risk of prostate cancer by maintaining a healthy weight and diet, reducing alcohol intake, treating infections early and stopping smoking.

SQUAMOUS CELL CARCINOMA OF THE SKIN

1

RESULT Risk Markers Detected

Low Risk

Gene: Intergenic

Genotype: AG

Description

Squamous cell carcinoma is the second most common form of skin cancer that develops at the top layer of the skin. It is characterized by accelerated and abnormal growth of squamous cells, which are flat cells located near the surface of the skin. Although it is a slow-growing skin cancer, it can spread to other tissues like the esophagus, mouth, bladder and lung.

Risk Factors

- Unprotected exposure to ultraviolet radiation
- Aged over 50 years
- Immunosuppressed
- Fair-skinned
- History of human papillomavirus (HPV) infection

TYPICAL SYMPTOMS



Firm and red nodule on face, lower lip, ears, hands or arms



New sore or raised area on old scar or ulcer



Red sore or rough patch in mouth



Red, raised patch or wart-like sore around anus or genitals

RECOMMENDED INTERVENTIONS

The risk of squamous cell carcinoma can be lowered by avoiding the sun during the middle of the day, wearing protective clothing and applying sunscreen regularly. If any changes occur on the skin, one should consult a doctor immediately for a professional opinion.

STOMACH CANCER

2

RESULT Risk Markers Detected

Low Risk

Gene: PSCA

Genotype: TC

Gene: PSCA

Genotype: AG

Description

Stomach cancer, or gastric cancer, is caused by abnormal cell growth that can affect any part of the stomach. It tends to develop slowly over many years. Pre-cancerous changes can often occur in the inner lining of the stomach before the cancer develops. It is often asymptomatic in the early stages and can only be discovered when the cancer cells have spread to other tissues and organs.

Risk Factors

- Aged 60 years and above
- More common in men than women
- Family history
- Infection with *Helicobacter pylori* bacteria
- Overweight or obese
- Diet (salt-cured, processed, grilled or charcoal)
- Excessive alcohol consumption
- Tobacco

TYPICAL SYMPTOMS



Sense of fullness in upper abdomen after a small meal



Swelling or build-up of fluid in abdomen



Vague discomfort/pain in abdomen



Vomiting, with or without blood

RECOMMENDED INTERVENTIONS

Preventive measures include treating *Helicobacter pylori* infections, reducing salt intake, and consuming a diet rich in fresh fruits and vegetables. Limiting alcohol consumption and avoiding tobacco products further lower the risk. Additionally, reducing the intake of pickled and smoked foods, such as salted meats and fish, is advisable. While anti-inflammatory drugs like ibuprofen may help reduce stomach cancer risk, they can also increase the risk of gastrointestinal bleeding.

TESTICULAR GERM-CELL TUMOR (FEMALE)

Description

Women are incapable of developing testicular cancer, but they risk passing down the genes to their sons and grandsons. Should a female individual tests positive for testicular cancer risk, it is recommended that her male descendants check for genetic predispositions to this cancer.

Risk Factors

- Does not exist

TYPICAL SYMPTOMS



Women do not develop testicular germ-cell tumor, therefore, there are no symptoms.

4

RESULT

Risk Markers
Detected



Slightly Lower Risk

Gene: KITLG

Genotype: TT

Gene: Intergenic

Genotype: TC

Gene: DMRT1

Genotype: AA

Gene: DMRT1

Genotype: TC

RECOMMENDED INTERVENTIONS

Women with genetic markers for testicular cancer should encourage their male descendants to check for genetic predisposition.

TESTICULAR GERM-CELL TUMOR (MALE)

4

RESULT Risk Markers Detected



Slightly Lower Risk

Gene: DMRT1

Genotype: AA

Gene: DMRT1

Genotype: TC

Gene: Intergenic

Genotype: TC

Gene: KITLG

Genotype: TT

Description

Testicular cancer occurs in the testes, which are located in the scrotum, beneath the penis. This cancer commonly affects males from ages 15 to 44 years old, but may occur at any other age as well. It is highly treatable, especially if discovered early. Family history is the strongest risk factor for testicular cancer. Symptoms may include discomfort in a testicle or back pain.

Risk Factors

- Men only
- Aged 20 to 35 years
- Undescended testes
- Family history of testicular cancer
- Human immunodeficiency virus (HIV) infection

TYPICAL SYMPTOMS



Enlarged testicle



Pain in testicle



**Pain in the lower
abdomen or back**



**Enlarged and
painful breasts**

RECOMMENDED INTERVENTIONS

The prognosis and treatment options for testicular germ-cell tumor depend on the current stage of the cancer and the size of the tumor. Early detection is crucial, especially for those who have a genetic predisposition or a family history of testicular cancer.

THYROID CANCER

2

RESULT Risk Markers Detected

Low Risk

Gene: DIRC3

Genotype: CC

Gene: MIR146A

Genotype: CC

Description

Thyroid cancer occurs in the thyroid gland, a butterfly-shaped gland located at the base of the neck. It may not show early symptoms, but as it grows, it will cause pain and swelling around the neck area. The common form of this cancer is papilloma, which is more common in women than men. Thyroid cancer has a higher genetic risk than other cancers.

Risk Factors

- More common in women
- Aged between 20 and 55 years
- Exposure to high levels of radiation
- Family history of thyroid cancer
- Familial adenomatous polyposis and acromegaly

TYPICAL SYMPTOMS



Lump or swollen glands in front of neck



Hoarseness or voice changes that remains after few weeks



Pain in front of neck, sometimes until ears



Persistent cough that is not because of a cold

RECOMMENDED INTERVENTIONS

There are no proven ways to prevent thyroid cancer, but minimizing radiation exposure and consuming an adequate amount of iodine in daily diet may lower the risk. If there is a family history of thyroid cancer, it can be prevented or treated early by regular screening and examination.

Neurological disorders affect the brain and nerves throughout the body. Structural, biochemical or electrical abnormalities in the brain, spinal cord or nerves can result in a range of symptoms, where some symptoms may be similar. Therefore, knowing whether you have certain genetic predispositions can help a doctor to determine whether further investigation is necessary. This can then be followed up with personalized clinical management or treatment strategies.



Neuro

ALZHEIMER'S DISEASE

1

RESULT Risk Markers Detected

Normal Risk

Gene: CR1

Genotype: AG

Description

Alzheimer's disease is caused by the abnormal accumulation of proteins in the brain. It is the most common cause of dementia. It occurs primarily in individuals aged 65 years or older. It is characterized by a progressive loss of memory and can result in severe degradation of the quality of life. Brain shrinkage is common in patients suffering from Alzheimer's disease.

Risk Factors

- Aged 65 years and above
- More common in women than men
- Family (first degree) history of Alzheimer's disease
- Down syndrome
- Head trauma

TYPICAL SYMPTOMS



Memory loss



Takes longer to complete normal tasks



Frequently getting lost



Poor judgement

RECOMMENDED INTERVENTIONS

While there is no cure for Alzheimer's disease, risks of developing it can be reduced through increased physical activity, treating hypertension, having a healthy diet and identifying and treating depression early.

BRAIN ANEURYSM

4

RESULT

Risk Markers Detected

Low Risk

Gene: CDKN2B-AS1

Genotype: TC

Gene: CDKN2B-AS1

Genotype: TC

Gene: EDNRA

Genotype: GG

Gene: Intergenic

Genotype: AG

Description

A brain aneurysm occurs when a blood vessel in the brain becomes excessively dilated. This is usually caused when an artery loses its elasticity, causing it to expand excessively. This condition results in the walls of the blood vessel becoming thin and then rupture, resulting in bleeding in and around the brain - this is typically referred to as a subarachnoid hemorrhage.

Risk Factors

- Aged 35 to 60 years
- More common in women than men
- Family history
- High blood pressure
- Smoking
- Cocaine use

TYPICAL SYMPTOMS



Double vision



Eye pain



Neck pain



Numbness/
tingling on face

RECOMMENDED INTERVENTIONS

Brain aneurysms occur as a result of underlying conditions such as hypertension. Keeping these conditions under check helps mitigate the risks of developing an aneurysm. There are known genetic links to the development of brain aneurysms. Screening can help identify these factors.

DEPRESSION

Description

Depression is a common but serious mood disorder that can negatively affect how individuals think, feel and behave. It can cause feelings of sadness and loss of interest in previously enjoyed activities, which can impair social, occupational and educational functioning. Depression is considered a serious medical condition that can worsen if proper treatment is not received

Risk Factors

- Genetics
- Life events
- Stress
- Chronic pain
- Poor nutrition
- Sleep disorders

TYPICAL SYMPTOMS



Often feel tired and overwhelmed



Losing interest in previously enjoyable activities



Having emotional reaction to everything



Experiencing changes in appetite

0

RESULT

Risk Markers
Detected



Normal Risk

RECOMMENDED INTERVENTIONS

You may consult talk to a doctor or therapist. Therapy can help you change negative thoughts, and medication can help balance brain chemicals. Exercise, a healthy diet, and enough sleep can improve mood. Talk to family and friends for support. Meditation and yoga can help manage stress.

EPILEPSY

Description

Epilepsy is a condition in which patients experience a disruption or sudden surge in the electrical activity of the brain, resulting in seizures. These seizures can manifest in involuntary actions, suddenly loss of awareness or violent fits. Twenty percent of individuals suffer from epilepsy. The risk for developing epilepsy increases when a genetic history is present; however, anyone can develop epilepsy.

Risk Factors

- Affects individuals of all ages
- Rare conditions, such as tuberous sclerosis and neurofibromatosis
- Trauma
- Stroke

TYPICAL SYMPTOMS



Temporary confusion



Uncontrollable jerking movements of the limbs



Loss of consciousness or awareness



Psychological symptoms such as anxiety or fear

4

RESULT

Risk Markers Detected



Normal Risk

Gene: ADCY9

Genotype: TC

Gene: C6orf7

Genotype: CC

Gene: Intergenic

Genotype: CC

Gene: LOC101927

Genotype: TC

RECOMMENDED INTERVENTIONS

To help reduce epileptic seizures, a doctor may prescribe antiepileptic drugs (AEDs). Other potential options include a special diet (high fat and low carbohydrate), surgery or vagus nerve stimulation. The aim is to reduce the side effects of epilepsy so that an individual can lead a more normal life.

MIGRAINE HEADACHE

0

RESULT
Risk Markers
Detected

Normal Risk

Description

A migraine is a headache which is usually felt as a throbbing pain or pulsing sensation on one side of the head. Migraine attacks can last for hours to days, and vary from one individual to another. Some individuals experience migraines occasionally, while others get them a few times each month. Although the exact causes of migraine are not fully known, both genetics and the environment may play a role.

Risk Factors

- Abnormal brain activity temporarily affecting nerve signals
- Genetics
- Age factors
- Anxiety
- Depression
- Bipolar disorder
- Sleep disorders

TYPICAL SYMPTOMS



Ringing in the ears



Increased sensitivity to light, sound and smell



Pounding headache



Nausea

RECOMMENDED INTERVENTIONS

There is no cure for migraine headaches. Other than common pain relief medicines, you can also try to employ relaxation techniques, obtain sufficient sleep, drink plenty of water and get regular exercise.

PARKINSON'S DISEASE

1

RESULT

Risk Markers Detected

Normal Risk

Gene: PARK2

Genotype: GG

Description

Parkinson's disease is a brain disorder that causes coordination and balance problems. Individuals diagnosed with Parkinson's disease have an increased risk of developing dementia, which is the impaired ability of common intellectual functions. Symptoms of Parkinson's disease are caused by decreased dopamine levels - causing unusual brain activity that leads to slowing of movements.

Risk Factors

- Aged over 60 years
- Hereditary
- Men at higher risk than women

TYPICAL SYMPTOMS



Impaired balance and coordination



Involuntary body shaking (tremor)



Slowness of movement



Stiffness of arms, legs and trunk

RECOMMENDED INTERVENTIONS

As Parkinson's disease is an inherited disorder, it cannot be prevented entirely. Some lifelong habits may be able to help reduce the risks, such as by avoiding toxins and head trauma. One should also maintain a healthy diet and do regular physical exercise to lower the complications.

STROKE

Description

A stroke occurs when a blood vessel that transports oxygen and nutrients to the brain is either blocked by a blood clot or ruptures. When this happens, part of the brain is not able to obtain the blood and oxygen that it needs, causing brain cells to die. This loss of brain function may cause disability or even death. Prompt treatment is crucial to prevent the brain cells from dying.

Risk Factors

- Middle-aged and older adults
- More common in men than women
- Hypertension
- Diabetes
- Smoking
- Obesity

TYPICAL SYMPTOMS



Loss of touch sensation or feel pain in legs and hands



Clumsiness and loss of balance when walking



Trouble speaking and understanding



Confusion or memory loss

3

RESULT

Risk Markers Detected

Low Risk

Gene: CDKN2B-AS1

Genotype: AG

Gene: CDKN2B-AS1

Genotype: AG

Gene: PRKCH

Genotype: TC

RECOMMENDED INTERVENTIONS

Several lifestyle changes can be made to help prevent a stroke, such as having a healthy diet, exercising regularly, and avoiding excessive tobacco and alcohol intake. Any underlying medical conditions such as high blood pressure or atherosclerosis should be treated to prevent further complications.

Imagine your DNA as a cookbook that guides how your body functions. Some people have "special ingredients" in their recipes that make them more prone to allergies. Genetic screening is like checking the cookbook to spot these unique ingredients and determine if you might be sensitive to certain allergens.



Allergy Profile

ALLERGY TO CATS

1

RESULT Risk Markers Detected



Slightly Lower Risk

Gene: Intergenic

Genotype: GG

Description

Individuals highly sensitive to cats may develop symptoms such as respiratory issues or skin reactions within minutes of contact with a cat or exposure to a cat's environment. Genetics indicate that a predisposition to cat allergies can be inherited, with sensitization occurring at any point in life.

Risk Factors

- Oversensitive immune system
- Genetic predisposition

TYPICAL SYMPTOMS



Itchy eyes



Runny nose



Skin rashes



Sneezing/coughing

RECOMMENDED INTERVENTIONS

Avoid contact with cats if sensitive to allergens. If unavoidable, use HEPA filters in frequently occupied rooms. Seek medical advice if symptoms occur.

ALLERGY TO DUST MITES

1

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: TGFB1

Genotype: AA

Description

Dust mite allergy is an immune system reaction to microscopic organisms commonly found in household dust. When genetically predisposed individuals inhale dust mite waste, their immune system may overreact, producing antibodies against these normally harmless substances. This response can lead to respiratory symptoms, skin irritation, and other allergic reactions. Though rare, dust mite allergy can run in families.

Risk Factors

- Oversensitive immune system
- Genetic predisposition

TYPICAL SYMPTOMS



Coughing



Itchy skin



Runny nose



Sneezing

RECOMMENDED INTERVENTIONS

Regularly clean household items such as mattresses, pillows, furniture, carpets, and curtains to reduce dust mites. Doctors may recommend medications to alleviate allergy symptoms.

ALLERGY TO GRASS

4

RESULT

Risk Markers Detected

Low Risk

Gene: ABL2

Genotype: AG

Gene: DNAH5

Genotype: CC

Gene: Intergenic

Genotype: GG

Gene: LOC101927

Genotype: AA

Description

Grass allergy is a seasonal allergic reaction triggered by pollen from various grass species. The immune system mistakenly identifies grass pollen as harmful, causing an allergic response when inhaled. Symptoms may include sneezing, runny nose, and itchy or watery eyes.

Risk Factors

- Oversensitive immune system
- Genetic predisposition

TYPICAL SYMPTOMS



Coughing



Itchy, watery eye



Runny nose



Sneezing

RECOMMENDED INTERVENTIONS

Minimize exposure by staying indoors on high-pollen days, keeping windows closed, and showering after outdoor activities. Medications such as antihistamines, nasal sprays, and decongestants can help. Consult a doctor for severe symptoms.

ALLERGY TO METAL (NICKEL)

Description

Nickel allergy is a common allergic reaction caused by skin contact with nickel, a metal found in jewelry, coins, zippers and electronics. It is a form of contact dermatitis, where the immune system overreacts upon exposure. Once developed, nickel allergy is typically lifelong, with repeated exposure potentially worsening symptoms.

Risk Factors

- Oversensitive immune system
- Genetic predisposition

TYPICAL SYMPTOMS



Red rash



Dry, cracked or scaly skin



Bumps and blisters



itchy skin

2

RESULT

Risk Markers Detected

Low Risk

Gene: PELI1

Genotype: TC

Gene: PGAM1P5, N

Genotype: AG

RECOMMENDED INTERVENTIONS

Avoid contact with nickel-containing items by selecting nickel-free jewelry, clothing accessories, and household products. Use barrier creams and test metal objects for nickel. Apply topical corticosteroids or take antihistamines for symptom relief. Seek medical attention for severe reactions.

ALLERGY TO MILK

Description

Milk allergy is an immune system reaction to proteins found in cow's milk, such as casein and whey. It is one of the most common food allergies, especially in infants and young children, though some may outgrow it. Unlike lactose intolerance, which is a digestive issue, milk allergy involves the immune system and can cause severe reactions, including anaphylaxis.

Risk Factors

- Oversensitivity immune system
- Genetic predisposition
- Atopic dermatitis (eczema)

TYPICAL SYMPTOMS



Diarrhea



Eczema



Runny nose



Anaphylaxis

1

RESULT

Risk Markers
Detected



Slightly Lower Risk

Gene: IL-10

Genotype: TT

RECOMMENDED INTERVENTIONS

Avoid milk and dairy products by reading food labels and choosing plant-based alternatives. Antihistamines can help with mild reactions, while severe reactions may require an epinephrine auto-injector. Consult a doctor for personalized allergy management.

ALLERGY TO PEANUTS

Description

Peanut allergy occurs when the immune system mistakenly identifies peanut proteins as harmful. Direct or indirect contact with peanuts triggers the immune system to release symptom-causing chemicals into the bloodstream, leading to allergic reactions that can range from mild to severe, including anaphylaxis. The prevalence of peanut allergy is increasing, particularly among children.

Risk Factors

- Oversensitive immune system
- Genetic predisposition
- Exposure to peanut proteins

TYPICAL SYMPTOMS



Anaphylaxis



Coughing and shortness of breath



Diarrhea



Vomiting

RESULT

Risk Markers Detected



Low Risk

RECOMMENDED INTERVENTIONS

Completely avoid peanuts and peanut-containing foods. A doctor can confirm peanut allergy through blood or skin prick tests. Inform others about the allergy and emergency response procedures.

ALLERGY TO SHRIMP

0

RESULT
Risk Markers
Detected

Low Risk

Description

Shrimp allergy is a common food allergy caused by an immune system reaction to proteins found in shrimp, particularly tropomyosin. It is classified as a shellfish allergy, which is often lifelong and can be severe. Reactions may occur even from minimal exposures, such as ingestion, skin contact, or inhalation of shrimp particles during cooking. In severe cases, it can lead to anaphylaxis.

Risk Factors

- From childhood
- Family history
- Existing allergies (crabs, lobsters)

TYPICAL SYMPTOMS



Itchy face, lips, tongue, throat



Diarrhea



Runny nose



Anaphylaxis

RECOMMENDED INTERVENTIONS

Avoid shrimp and cross-contaminated foods, read food labels carefully, and exercise caution when dining out. Use antihistamines for mild reactions; severe reactions require an epinephrine auto-injector. Carry emergency medication and consult a doctor for management strategies.

ALLERGY TO WHEAT

Description

Wheat allergy is an immune system response to proteins found in wheat, including gluten and gliadin. It is more common in children but can affect adults. Unlike celiac disease (which involves the digestive system) or gluten sensitivity (which does not involve an immune response but can cause similar symptoms), wheat allergy is an immune-mediated condition that can cause immediate or delayed reactions after consuming wheat or wheat-containing products. Symptoms can range from mild to severe, with anaphylaxis possible in extreme cases.

Risk Factors

- From childhood
- Family history
- Other allergies (grains)

TYPICAL SYMPTOMS



Diarrhea



Itchy skin



Runny nose



Anaphylaxis

2

RESULT

Risk Markers Detected

Low Risk

Gene: IL18

Genotype: TG

Gene: Intergenic

Genotype: TC

RECOMMENDED INTERVENTIONS

Strictly avoid wheat-containing foods by reading labels and choosing wheat-free alternatives like rice or corn. Use antihistamines for mild symptoms and an epinephrine auto-injector for severe reactions. Inform others about the allergy and consult a doctor for management advice.

The early years of a child's life are among the most precious for parents. It is vital to detect the risks of any illnesses or conditions, which may hinder or drastically impact the physical and mental development of a child, as early as possible. For parents with children who are suffering from an unidentified inherited illness or condition, genetic screening may help to prevent years of unnecessary testing and worrying about the unknown.



Childhood - Onset Diseases

ADOLESCENT IDIOPATHIC SCOLIOSIS

Description

Scoliosis is characterized by a sideways curvature of the spine. It occurs during one's growth spurt before puberty. Mild scoliosis does not impose serious medical issues, but severe forms may limit physical movement and cause pain. The diagnosis of scoliosis is made via physical examination, X-rays, CT scans, and MRI scans. Severe cases require surgery to straighten the spine.

Risk Factors

- Children aged 10 to 18 years
- More common in girls than boys
- Family history

TYPICAL SYMPTOMS



**Head not
centered with body**



**Uneven shoulders
and leg length**



**Constant leaning
to one side**



**Prominent
shoulder blade(s)**

3

RESULT Risk Markers Detected



Low Risk

Gene: Intergenic

Genotype: TC

Gene: Intergenic

Genotype: AG

Gene: LBX1-AS1

Genotype: AG

RECOMMENDED INTERVENTIONS

Individuals with scoliosis are encouraged to keep a straight posture while walking or sitting and be more actively involved in sports, especially swimming.

ALLERGIC RHINITIS

Description

Allergic rhinitis is a fairly common disease which affects 1 in 5 people worldwide. It occurs when the body's immune system overreacts to the presence of one or more allergens. Those suffering from allergic rhinitis usually experience symptoms which include itchy nose or throat, coughing, sneezing and watery eyes. However, unlike the common cold, these symptoms are never accompanied by a fever.

Risk Factors

- From birth
- Family history of atopy
- Have other allergies, e.g. food or eczema
- Second-hand smoke

TYPICAL SYMPTOMS



Sneezing



Red, itchy and watery eyes



Sore throat and dry cough



Decreased sense of smell or taste

2

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: IL23R

Genotype: TG

Gene: LTA

Genotype: CC

RECOMMENDED INTERVENTIONS

Allergic rhinitis can be treated with antihistamines. However, it is advisable to prevent bouts of allergic rhinitis by minimizing exposure to allergens. This can be done by using air purifiers or air conditioners with HEPA filters, cleaning your bedding on a regular basis, using anti-allergen covers, and minimizing carpets and upholstery in your home.

ASTHMA

3

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: GSDMB

Genotype: TT

Gene: NOTCH4

Genotype: TC

Gene: ORMDL3

Genotype: CC

Description

Asthma is a common disease in which air exchange is compromised due to inflammation in the airways. When inflammation occurs, the mucosa that line the inner walls of the airways swell and produce excessive mucus, and the muscles of the bronchi also contract. This causes difficulty in breathing and can be life threatening. Asthma attacks are usually triggered by stimuli such as dust, smoke or stress.

Risk Factors

- Childhood onset
- History of asthma in sibling
- Food allergies
- Exposure to cigarette smoke
- Respiratory infections
- Exposure to environmental toxins
- Pollution

TYPICAL SYMPTOMS



Chest congestion or tightness



Frequent and intermittent coughing



Shortness of breath



Wheezing/whistling sound when exhaling

RECOMMENDED INTERVENTIONS

Asthma can be managed with proper medical intervention. Inhalation of second-hand smoke by asthma sufferers can trigger attacks. It is recommended to minimize exposure to pollutants in the environment.

ATTENTION-DEFICIT HYPERACTIVITY DISORDER (ADHD)

Description

Attention-deficit hyperactivity disorder, or more commonly known by its abbreviation ADHD, is a disorder affecting the brain. It is known to cause issues relating to paying attention, sitting still and behaving properly. ADHD is the most commonly diagnosed mental disorder in children. It occurs more commonly in children, particularly boys, and can continue into their teens and adulthood.

Risk Factors

- Mother of child who smokes cigarettes, drinks alcohol, or use drugs during pregnancy
- Exposure to environmental toxins during pregnancy, or when child is at a young age
- Low birth weight
- Brain injuries

TYPICAL SYMPTOMS



**Easily distracted
and forget things**



**Frequently
switch activities**



**Difficulty
paying attention**



**Fidgeting and
squirming when seated**

3

RESULT

Risk Markers
Detected



Slightly Lower Risk

Gene: DBH	Gene: SLC6A2	Gene: CLOCK
Genotype: TT	Genotype: GG	Genotype: AA

RECOMMENDED INTERVENTIONS

There is no known way to prevent ADHD. However, knowing your genetic predisposition for this condition allows for effective, early diagnosis and management.

CELIAC DISEASE

Description

Celiac disease is a genetic disease that is characterized by sensitivity to gluten products. Gluten, a protein found in wheat, barley and rye, causes a localized immune response in the small intestine. Prolonged inflammation from the reaction causes damage to the small intestine and affects nutrient absorption. It is common for people with celiac disease to have diarrhea, feel fatigued and bloated, and have anemia.

Risk Factors

- From childhood
- Family (first degree) history of celiac disease
- Type 1 diabetes
- Thyroid disease
- Autoimmune diseases

TYPICAL SYMPTOMS



Abdominal pain



Intermittent diarrhea



Weakness and fatigue



Weight loss

3

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: Intergenic

Genotype: TC

Gene: Intergenic

Genotype: GG

Gene: LPP

Genotype: AA

RECOMMENDED INTERVENTIONS

The severity of celiac disease can be reduced by avoiding foods with gluten. These are usually products containing wheat, rye or barley. Having a balanced diet is key to maintaining health with celiac disease; regular exercise and, in some cases, supplementing can help.

CHILDHOOD EPILEPSY

Description

Epilepsy is a condition in which individuals experience seizures caused by uncontrolled or abnormal neuron firing. This causes disruptions in normal brain functioning. While seizures can be due to high fever, low blood sugar and other factors, when a child experiences two or more seizures with no known cause, it is typically diagnosed as epilepsy. Certain types of genetic mutations are associated with a higher risk of developing epilepsy.

Risk Factors

- Brain injury
- Family history of epilepsy
- High fever during childhood
- Infections (eg., meningitis, encephalitis)
- Premature birth or neonatal hypoxia
- Prenatal exposure to alcohol or toxins

TYPICAL SYMPTOMS



Change in mood



Change in
perception of sound,
smell, taste and feel



Jerking of parts
of the body



Seizures

4

RESULT Risk Markers Detected



Slightly Lower Risk

Gene: ADCY9

Genotype: TC

Gene: C6orf7

Genotype: CC

Gene: Intergenic

Genotype: CC

Gene: LOC101927

Genotype: TC

RECOMMENDED INTERVENTIONS

Epilepsy can be hereditary. Genetic screening can help identify individuals with these predispositions. However, risk of epilepsy can be mitigated by avoiding head trauma, early intervention for infections and treating underlying conditions.

CROHN'S DISEASE

Description

Crohn's disease is a chronic condition affecting the digestive tract. It typically occurs in the small intestine and colon. The chronic inflammation of the bowel can lead to malnutrition, severe diarrhea, fatigue and weight loss. Severe cases can be debilitating and have many medical complications. Those suffering from Crohn's disease are at a higher risk for certain types of cancers that can include intestinal and colon cancer.

Risk Factors

- Men slightly at higher risk than women
- Family (first degree) history of Crohn's disease
- Smoker

TYPICAL SYMPTOMS



Abdominal pain and cramping



Blood in the stool



Diarrhea



Intestinal ulcers

5

RESULT

Risk Markers Detected

Low Risk

Gene: CDKAL1

Genotype: TC

Gene: IL23R

Genotype: TC

Gene: IRGM

Genotype: AG

Gene: Intergenic

Genotype: GG

Gene: LOC101927

Genotype: GG

RECOMMENDED INTERVENTIONS

While there is little that can be done to prevent Crohn's disease at this time, knowing the genetic risks allow the disease to be detected via further and regular tests. This can then allow treatment to start earlier. Regular monitoring, such as colonoscopy and blood tests, is recommended for early detection in high-risk individuals.

DYSLEXIA

4

RESULT

Risk Markers Detected

Low Risk

Gene: DYX1C1

Genotype: CC

Gene: TDP2

Genotype: AA

Gene: DCDC2

Genotype: AG

Gene: DCDC2

Genotype: AG

Description

Dyslexia is a learning disorder that affects an individual's ability to speak, spell, read and write. Those with dyslexia are unable to link speech sounds and their relationship with words and letters. It affects the regions of the brain that process language. However, dyslexic individuals are not intellectually stunted. They are often hard workers and are able to excel with proper intervention.

Risk Factors

- From infancy
- Family history of dyslexia or other learning disabilities
- Premature birth or low birth weight
- Mother of child exposed to nicotine, drugs, alcohol or infection during pregnancy

TYPICAL SYMPTOMS



Inability to read or slow progress in reading



Inability to write or has many spelling mistakes



Short attention span



Slow speech development with limited vocabulary

RECOMMENDED INTERVENTIONS

Dyslexia is genetically inherited. There is no way to prevent or cure dyslexia. However, with genetic screening, it is possible to identify the risks of being dyslexic. Early diagnosis can then be made with further tests, and early interventions can start. Fetal exposures to nicotine, drugs, alcohol or infections that alter brain development are linked to the development of dyslexia.

EARLY SPEECH DEVELOPMENT

4

RESULT Risk Markers Detected

Typical Development

Gene: CNTNAP2

Genotype: AG

Gene: CNTNAP2

Genotype: AG

Gene: CNTNAP2

Genotype: TC

Gene: CNTNAP2

Genotype: AC

Description

Speech development varies between children and is influenced by both genetic and environmental factors. Most children develop functional speech by age 4. Specific language impairment (SLI) is a neurodevelopmental disorder characterized by delayed language mastery without intellectual or hearing deficits. SLI may coexist with conditions like ADHD or autism but is a distinct diagnosis.

Risk Factors

- From infancy
- More common in men than women

TYPICAL SYMPTOMS



Slow speech development with limited vocabulary



Does not respond to sing along with others



Does not appear to listen or be concerned with words



Does not make common gestures

RECOMMENDED INTERVENTIONS

Parents can support speech development by engaging in responsive communication (e.g., describing daily activities, responding to babbling). Interactive play (e.g., peek-a-boo, singing) and letting the child lead conversations are encouraged. Early intervention by a speech-language pathologist is recommended if delays persist beyond 2–3 years.

INVASIVE PNEUMOCOCCAL DISEASE

Description

Invasive pneumococcal disease (IPD) is caused by the bacterium *Streptococcus pneumoniae*, which invades sterile sites like the blood (bacteremia), brain (meningitis), or lungs (pneumonia), or the middle ear (otitis media). Severe cases may lead to sepsis, hearing loss, or neurological damage. High-risk groups include children under 2, adults over 65, and immunocompromised individuals.

Risk Factors

- Children aged <2 years (highest incidence)
- Adults aged ≥65 years
- Chronic conditions
- Immunodeficiency (e.g.: HIV, chemotherapy)
- Smoking or alcohol abuse

TYPICAL SYMPTOMS



**Chest pain
and muscle aches**



**Cough
and nausea**



Fever and chills



**Shortness
of breath**

3

RESULT Risk Markers Detected



Slightly Lower Risk

Gene: NFKBIA	Gene: NFKBIA	Gene: NFKBIZ
Genotype: GG	Genotype: TT	Genotype: AC

RECOMMENDED INTERVENTIONS

Since pneumonia commonly affects children, parents should take the right precautions to prevent their children from facing serious complications. For instance, children should receive their seasonal flu shot and get vaccinated. Good hygiene and having an active lifestyle can also help mitigate the risk of pneumonia.

JUVENILE IDIOPATHIC ARTHRITIS

5

RESULT Risk Markers Detected



Moderate Risk

Gene: REEP3

Genotype: CG

Gene: ERAP1

Genotype: TC

Gene: IL20

Genotype: TC

Gene: JMJD1C

Genotype: TC

Gene: TIMMDC1

Genotype: AG

Description

Juvenile idiopathic arthritis is a form of arthritis commonly seen in teens and children. It is an autoimmune disease in which the body's immune cells attack the tissue between the joints, resulting in chronic and prolonged inflammation. This causes discomfort, pain and tenderness in the joints, making mobility and movement more difficult. Prolonged inflammation of this condition will eventually damage the surrounding cartilage and bones.

Risk Factors

- Children aged below 16 years
- More common in girls than boys

TYPICAL SYMPTOMS



**Inflammation
of the joints**



Joint stiffness



**Restriction of
joint movements**



Skin rash

RECOMMENDED INTERVENTIONS

Juvenile idiopathic arthritis is influenced by both genetic and environmental factors. By watching out for symptoms like joint swelling and stiffness in a child, parents can treat the infections early before the symptoms escalate. A healthy, balanced diet aids in reducing the symptoms of this arthritis.

KAWASAKI DISEASE

Description

Kawasaki disease is an acute inflammation of the connective tissues of the body. Vessels such as lymph nodes, arteries, veins and capillaries become inflamed, thus affecting their normal functions. It occurs primarily in children younger than 5 years old. It is thought to have a genetic component. If untreated, this disease can lead to the development of aneurysms and heart dysfunctions.

Risk Factors

- Children aged below 5 years
- More common in boys than girls
- Asian

TYPICAL SYMPTOMS



High fever lasting more than 5 days



Red rashes



Swollen lymph nodes



Very red lips, tongue and eyes

2

RESULT Risk Markers Detected

Low Risk

Gene: Intergenic

Genotype: TC

Gene: BTBD1

Genotype: AG

RECOMMENDED INTERVENTIONS

Kawasaki disease is strongly linked to genetic factors and family history. Individuals with the genetic predisposition for this disease should consume a high anti-inflammatory food diet as they are more vulnerable to inflammations. If a child has persistent high fever, he or she should be taken to the doctor for prompt treatment to prevent lasting damage.

LONG QT SYNDROME

4

RESULT Risk Markers Detected

Low Risk

Gene: Intergenic

Genotype: GC

Gene: Intergenic

Genotype: TC

Gene: Intergenic

Genotype: TC

Gene: NOS1AP

Genotype: GG

Description

Long QT syndrome is a heart rhythm condition where the heart beats chaotically, resulting in a long QT wave in an electrocardiogram (ECG) that records electrical signals from the heart. It is a congenital heart condition which can result in sudden fainting episodes, arrhythmias, heart attack and sudden death. This syndrome may be caused by certain medications, medical conditions or mineral imbalances.

Risk Factors

- At birth or later in life
- Inherited disorder
- More common in women than men

TYPICAL SYMPTOMS



Irregular heartbeat and palpitations



Fainting



Lightheadedness



Seizures

RECOMMENDED INTERVENTIONS

Long QT syndrome may be caused by genetic mutation or other medical conditions. Regardless, children with genetic predisposition for long QT syndrome should take extra precautions by avoiding strenuous exercise, as rapid heartbeat may lead to fainting. They should always stay hydrated to maintain a balanced electrolyte level. Medications may be prescribed to prevent dangerous heartbeat episodes.

MENINGOCOCCAL DISEASE

Description

Meningococcal disease is caused by the *Neisseria meningitidis* bacteria, also known as meningococcus. The bacteria spread through respiratory and throat secretions. Infections can have deadly complications, including infections of the lining of the brain and spinal cord (meningitis), infections in the bloodstream (septicemia or bacteremia) or clotting of blood vessels (disseminated intravascular coagulation). Getting immediate medical attention is very important when infected.

Risk Factors

- Children aged 2 to 10 years
- Teens and young adults
- Immunosuppressed

TYPICAL SYMPTOMS



High fever



Severe headache



Vomiting or nausea with headache



Seizures

2

RESULT
Risk Markers
Detected

Low Risk

Gene: CFH

Genotype: TG

Gene: CFH

Genotype: TC

RECOMMENDED INTERVENTIONS

As meningococcal disease poses a great risk among children and babies, it is important for them to be diagnosed early for treatment in order to prevent serious complications. All children should be vaccinated against meningococcal bacteria to prevent severe symptoms. Good hygiene habits also help reduce the risk of bacteria transmission.

OBESITY (CHILDHOOD ONSET)

Description

Childhood onset obesity is the occurrence of obesity in children and adolescents. It is characterized by a body mass index (BMI) of more than 30. Obese children commonly have a high body fat content and typically present with central adiposity. Children with obesity are at greater risk to develop metabolic syndrome. This also increases their risk of developing cardiovascular disease, diabetes and renal disease.

Risk Factors

- From childhood
- Genetic predispositions
- Sedentary lifestyle
- High-fat diet

TYPICAL SYMPTOMS



At 2 years of age,
BMI exceeding 19.3



At 4 years of age,
BMI exceeding 17.8



At 9 years of age,
BMI exceeding 21.0



At 13 years of age,
BMI exceeding 25.1

4

RESULT

Risk Markers Detected

Low Risk

Gene: BDKRB2

Genotype: TC

Gene: GNPDA2

Genotype: AG

Gene: Intergenic

Genotype: CC

Gene: LOC107985

Genotype: CC

RECOMMENDED INTERVENTIONS

Childhood onset obesity poses a greater risk for other major diseases. Some of the ways to reduce it is by providing the right mix of food in the child's diet plan, following regular mealtimes, eating in moderate quantity and checking for genetic predispositions, for early and effective treatment.

OBSESSIVE-COMPULSIVE DISORDER (OCD)

Description

Obsessive-Compulsive Disorder (OCD) is a psychological disorder characterized by recurring, unwanted thoughts, ideas or obsessions which compel individuals to do something repetitively. These behaviors can severely interfere with their daily lives. Not performing these repetitive compulsions can cause great distress and can further disrupt their daily lives.

Risk Factors

- Children aged 10 to 14 years
- More common in men than women
- Family history of the disorder

TYPICAL SYMPTOMS



Avoidance of situations that can trigger obsessions



Fear of being contaminated from touching objects



Hair loss as a result of hair pulling



Intense stress when objects are not orderly

2

RESULT

Risk Markers Detected

Low Risk

Gene: ANKK1

Genotype: AG

Gene: SLC1A1

Genotype: TT

RECOMMENDED INTERVENTIONS

Obsessive-Compulsive Disorder (OCD) symptoms can be managed through a combination of treatments, including medicine and therapy. Some treatments include psychotherapy and relaxation exercises like meditation and yoga. Psychiatric drugs can be prescribed to control the feeling of obsession and compulsion.

PATHOLOGICAL MYOPIA

Description

Pathological myopia is the leading cause of visual impairment worldwide. This condition can cause serious near-sightedness, which can lead to blindness. It is a highly progressive disease and is not reversible. This disease is caused by a major alteration of the eye shape. Hence, it is crucial for early examination among children to prevent permanent vision loss.

Risk Factors

- Younger age
- Asian
- Family history of myopia

TYPICAL SYMPTOMS



Blurred vision



Excessive blinking



Eye strain



Light sensitivity

3

RESULT

Risk Markers
Detected



Slightly Lower Risk

Gene: MIPEP

Genotype: AG

Gene: PAX6

Genotype: CC

Gene: TGFB1

Genotype: CC

RECOMMENDED INTERVENTIONS

There are different treatment options for pathological myopia that may help improve one's vision quality and eyesight. Those diagnosed with pathological myopia can consider prescription eyeglasses, contact lenses or photodynamic therapy.

RESTLESS LEGS SYNDROME

Description

Restless leg syndrome is a condition where the individual has an uncontrollable urge to move their limbs due to uncomfortable sensations, which include tickling, aching and crawling feelings. In addition, the sensations usually begin after having a rest, e.g. when lying down for long periods of time. This condition can be relieved with medications such as dopamine agonists and opioids.

Risk Factors

- Iron deficiency (primary risk)
- Family history

TYPICAL SYMPTOMS



Overwhelming urge to move legs



Unpleasant crawling or creeping sensation in the feet, calves and thighs



Increase in intensity of symptoms at rest or during sleep



Sensation is often worse in evening or night

2

RESULT

Risk Markers Detected

Low Risk

Gene: PTPRD

Genotype: AG

Gene: PTPRD

Genotype: AG

RECOMMENDED INTERVENTIONS

Restless leg syndrome can be prevented by simple self-care and lifestyle changes. One should maintain a balanced diet to avoid iron or vitamin deficiencies which can worsen the condition. Additionally, genetic screening is recommended to support early diagnosis and effective treatment.

TEETHING

4

RESULT

Risk Markers Detected

Delayed Development

Gene: BMP4

Genotype: AG

Gene: IGF2BP1

Genotype: AG

Gene: Intergenic

Genotype: GG

Gene: Intergenic

Genotype: GG

Description

Teething is the process where an infant's teeth break through the gums. Symptoms of teething include swollen gums and the baby's tendency to place fingers into the mouth. Abnormalities in teething may cause missing or extra teeth, leading to difficulty in chewing. Therefore, early and proper dental hygiene helps babies maintain healthy teeth and gums.

Risk Factors

- No sex-based risk - affects all infants equally
- Premature birth may delay teething

TYPICAL SYMPTOMS



Swollen and tender gums



Chew on hard things



Generate lots of saliva



Changes in eating or sleeping patterns

RECOMMENDED INTERVENTIONS

Parents can help soothe their baby's sore gums by keeping it cool with a teething ring or use a wet gauze to rub the baby's gum, as the pressure helps ease the discomfort. Try to avoid using over-the-counter remedies or any teething medications for the baby's safety. If the child has not developed any teeth by 15 months old, parents should consult a professional healthcare practitioner.

TOURETTE SYNDROME

Description

Tourette syndrome (TS) causes people to have 'tics' which are sudden twitches, movements or sounds that cannot be controlled. This disease usually has a genetic link and runs in the family. Types of tics and the frequency which it occurs changes over time. TS commonly occurs with disorders like attention deficit hyperactivity disorder (ADHD) and dyslexia. Those with this syndrome are able to live a normal life, as it can be controlled by medication or behavioral therapies.

Risk Factors

- Onset typically between 5-7 years
- Male:Female ratio ~4:1

TYPICAL SYMPTOMS



Simple movement ticks
(blinking, grimacing, twitching)



Complex movement ticks
(bending, jumping, self-injuring acts)



Simple vocal ticks
(barking, coughing, squealing)



Complex vocal ticks
(Uttering, repeating words)

2

RESULT

Risk Markers Detected

Low Risk

Gene: POLR3B

Genotype: TT

Gene: COL27A1

Genotype: AG

RECOMMENDED INTERVENTIONS

Treatment for Tourette syndrome is focused on controlling tics which may disrupt daily activities and functioning. Various medications and therapies can be introduced to reduce the symptoms of the tics.

ULCERATIVE COLITIS

1

RESULT

Risk Markers Detected



Slightly Lower Risk

Gene: LOC101927

Genotype: GG

Description

Ulcerative colitis is an inflammatory bowel disease that causes inflammation in the digestive tract, including the rectum and colon. Small ulcers first develop, leading to internal bleeding. This disease affects both males and females and is highly linked to genetics. Ulcerative colitis can sometimes lead to life-threatening complications. Although there is no known cure, treatment can reduce symptoms of the disease.

Risk Factors

- Younger age
- Family history of inflammatory bowel disease

TYPICAL SYMPTOMS



Abdominal pain and discomfort



Diarrhea/blood in stool



Skin lumps and mouth sores



Nausea and vomiting

RECOMMENDED INTERVENTIONS

Symptoms of ulcerative colitis can be controlled by avoiding food that causes allergy, consuming small amounts of food at a time, and minimizing dairy products or high fiber food such as beans and nuts. Early treatment of bowel infection lowers the risk of this infections.



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